

MISCELLANEOUS COURSES

OPERATIONAL SAFETY

TRAINING MANUAL
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MISCELLANEOUS COURSES

OPERATIONAL SAFETY

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1. OBJECTIVES

At end of this presentation, a production operator (or going to be) will be able to understand and analyse the main risk encountered on an Oil & Gas site.

All risks, all precautions to undergo, all the 'tricks' to know cannot be enumerated here but at least after having followed what is treated in the present manual, the attendee would be ready for :

- Localise, anticipate the potential sources of Fire on a site
- Localise, anticipate the potential sources of Accident on a site
- Interpret the critical situations and conditions in exploitation and operation (excluding Process)
- Take the necessary references in existing Company Standards Files
- Interpret, analyse, apply rules established in/by 'SIMOPS, HSE, and all other Safety Documents within the Group
- Follow, analyse, apply the logics of ESD et F&G systems on a site
- Apply, make it applied and followed all the different Work Permits t issue on a production site
- Decide between Production and Maintenance Works what is most imperative
- Work and supervise / organise works respecting and applying all the necessary safety measures for all the types of interventions, maintenance and operations
- Use the adapted individual safety kit for routine and non routine operations
- Operate the Fire Fighting equipment's
- Make Safety of Personnel as First Priority
- Place what is not 'Safety of Personnel' in Second Priority

2. INTRODUCTION

Controlling safety is a central concern for the Total group. Our different activities generate industrial risks of all types, relating to the development of reservoirs and the production of hydrocarbons.

Safety primarily concerns the protection of individuals at their work stations and in their day-to-day operations.

Each of our professions has inherent risks. This is why we have established some rules on the basis of feedback. These rules are presented in this course. They cover the most frequently faced situations on Exploration & Production sites.

The compliance with such rules will therefore significantly contribute to preventing accidents in this field. Safety concerns all of us in our day-to-day activities, therefore we must implement these rules and we owe it for our families.

The commitment of each individual to the objective of improving safety is a decisive factor for progress and will enable our safety performance to be enhanced.

These rules will contribute to the development of a stronger HSE culture within Exploration & Production.

Most works executed by our services include routine tasks which can however represent a certain level of danger.

In fact, this manual could be as thick as we want but if the individual does not have the common sense and realism, this same manual will be useless…..

2.1. DEATH

In 2006, 3 deaths occurred in Exploration & Production. The causes were:

- Vehicle accident: 1
- Impact due to hitting an object: 1
- Sickness: 1

2.2. ACCIDENTS WITH LEAVE

In 2006, 110 LTI occurred in Exploration & Production (listed in the SYNERGI base).

The causes were:

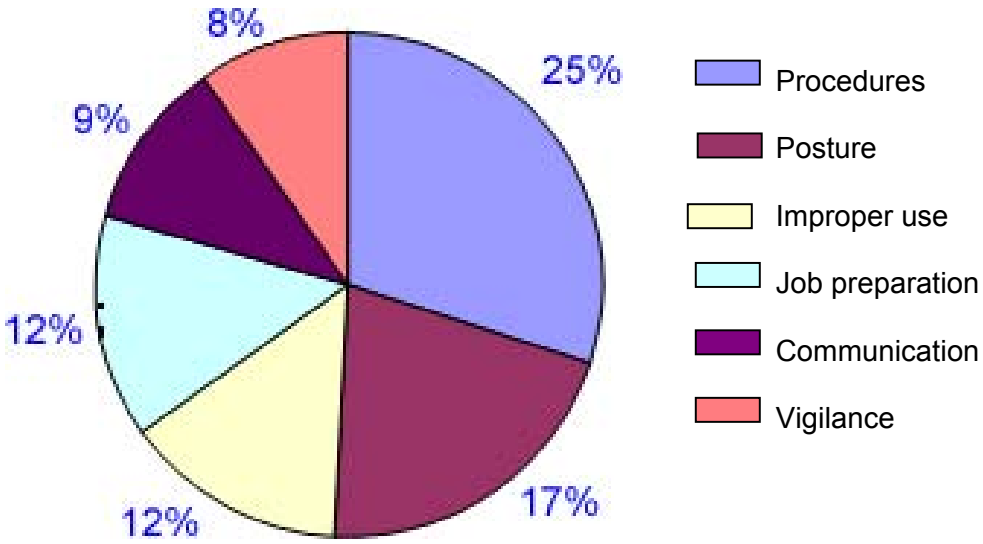


Figure1: LTI causes in 2006

LTI 2006 - Total LTI on SYNERGI Base = 110			
Rank	%	Causes	Details
1	25%	Procedures	Failed to comply with procedure or instruction Non-respect of Total HSE rules Ignorance of HSE rules Exceptional/routine violation of safety rules Non-respect of laws and regulations Failing to do regular safety tour Ignoring warnings Working without authorization
2	17%	Posture	Adopting unsafe work position, posture, placement
3	12%	Improper use	Using inadequate / faulty tool, equipment, materials Using tool, equipment, materials improperly
4	12%	Job preparation	Failure to evaluate risk prior to critical job Failing to provide pre-job briefing Insufficient listing of precautions Inappropriate work planning

LTI 2006 - Total LTI on SYNERGI Base = 110			
Rank	%	Causes	Details
5	9%	Communication	Communication failure between team-mates Failing to warn, inform Inadequate communication systems Management-personnel communication failure
6	8%	Vigilance	Lack of safety awareness Lack of attention, of vigilance

Table1: Accidents with leave

2.3. FIRE, EXPLOSIONS

In 2006, 5 fires or explosions occurred in Exploration & Production (listed in the SYNERGI base).

The causes were:

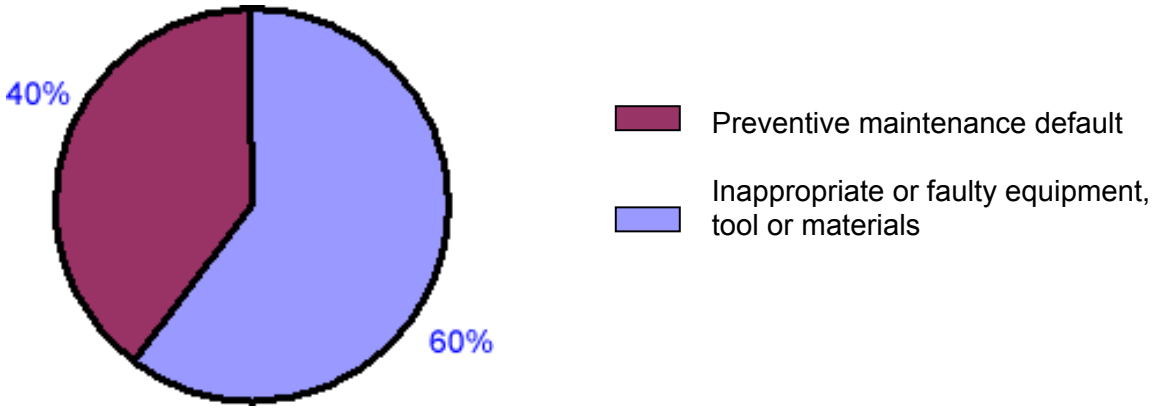


Figure2: Fire and in 2006

2.4. HIGH POTENTIAL INCIDENTS

In 2006, 232 HIGH POTENTIAL incidents occurred in Exploration & Production (listed in the SYNERGI base).

The causes were:

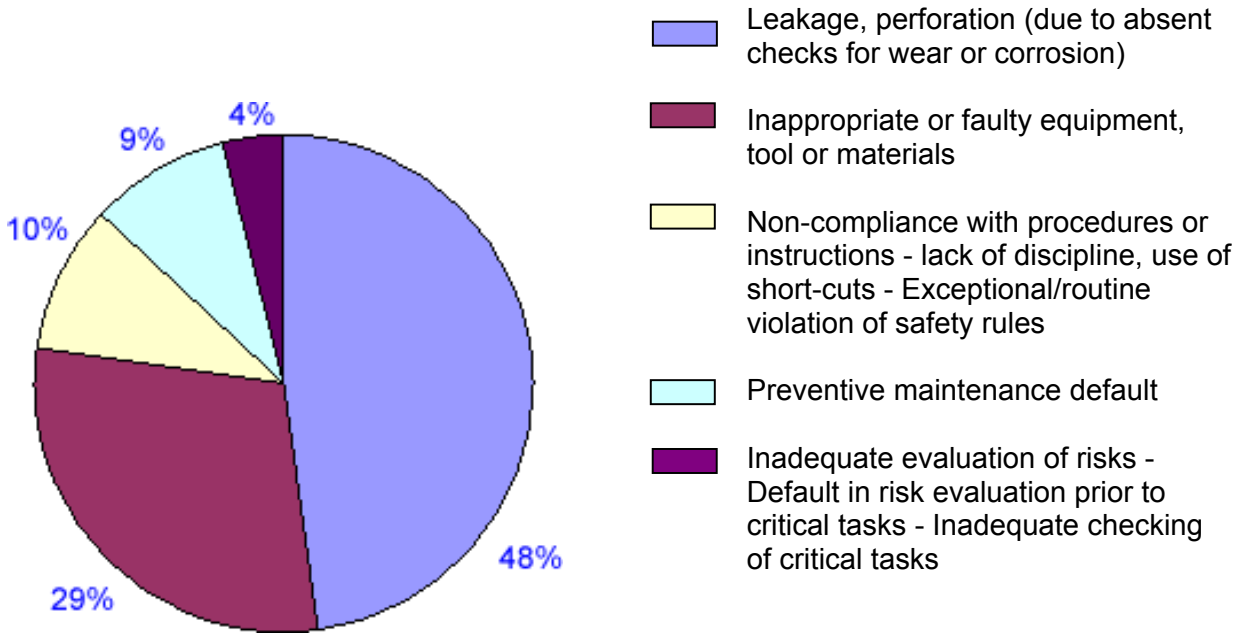


Figure3: High potential incidents

2.5. LEAKS

In 2006, 364 leakages occurred in Exploration & Production (listed in the SYNERGI base).

The causes were:

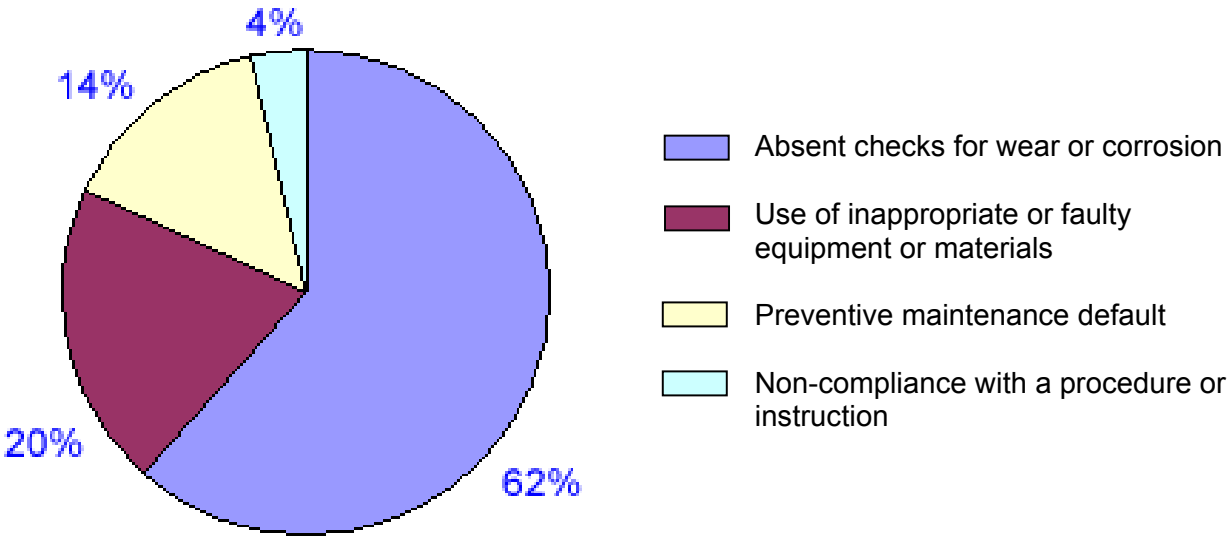


Figure4: Leaks

3. ESSENTIAL SAFETY NOTIONS FOR OPERATORS

These safety notions are applicable to all installations and must be fully understood and integrated in the work of any operator. They are particularly defined in E&P reference documents.

We will consider elementary notions for operators in the following paragraphs.

3.1. TRANSMISSION OF INSTRUCTIONS

Why? : Teams working in different day/night or rollover/rest shifts must ensure that instructions are transmitted to the next shift to guarantee the long-term application of the process and the safety of all those concerned, by indicating the situation of the installations when changing shift.

How? : Using the different instruction logs in the control room and the supervisor's office:

- Knowledge of the process situation: to ensure the effective tracking of events occurring during a shift, they must be noted in a log: the shift log. This log will mention the time of the event and a detailed description. This log will also include the list of Work Permits, Work Slips opened during the shift and a shift-end installation status (well and main unit status).

- Knowledge of current operations (Work Slip, Work Permits): a list is printed in the control room in the morning for the rapid identification all operations subject to Work Permits and Work Slips. This list must be updated with specification of all operations started, completed and under way.

- Knowledge of inhibitions: in case of specific operating conditions, it may be necessary to inhibit/mask alarms/safety for the process on the DCS. Each mask/inhibition will be recorded in a specific log and carefully monitored. It is essential to not leave any active mask/inhibition at shift-end.

- Knowledge of consignments: for traditional operating requirements (maintenance) or specific operations, some equipment may be consigned. These consignments will be noted in a log.

- Knowledge of downgraded situations: some equipment may have a downgraded operating mode for safety purposes. All downgraded situations will be recorded in a log indicating the compensatory measures taken to remedy the situations.

The operator also has access to several documents included in the HSE folder for the correct transmission of instructions. A copy of this folder must be kept in the control room.

The HSE folder for the installation will be created and updated. It includes:

- the validation table for the HSE folder from the Management,
- the register of key personnel,
- the list of documents describing the installation and emergency management,
- the full folders concerning exemptions accorded,
- the list of corrective action under way, started on the basis of audit recommendations (e.g.: Opersafe audit, etc.),
- the downgraded situation tracking table,
- the long-term inhibition tracking table,
- the log of verifications, inspections, tests, certifications, inspections, measures and other regular operations in terms of HSE,
- the log of regular safety exercises.

The folder is: kept in a specific location, generally in the control room, and may be accessed by any member of personnel, organized per platform or per installation, for simpler access in case of simultaneous operations (SIMOPS).

No.	Description	Classification	Standard content and requirements
1	The validation table for the HSE folder from the Management.		The checking and the validation of the HSE folder at each inspection of sites and installations is the responsibility of the hierarchy.
2	The register of key personnel.		Name, position, responsibilities, certificate validity (if applicable).
3	The list of documents describing the installation and emergency management.	General plans and process diagram	Plot plans, P&ID, PFD, …
		Safety logic diagrams	AU logic diagrams, Fire and gas (F&G) logic diagrams, …
		Layout drawings for safety systems	Location of the AU, AU push button, fire fighting equipment, emergency exits, rescue and evacuation equipment for the installation, location of hazardous areas, etc.…
		Safety concept	Installation safety concept.
		Evaluation of *risks*	Log for major risks, risk evaluation folder for industrial hygiene (RAF), evaluation of security *risks*.
		Study of environmental impact	EIE or the summary of EIE commitments.
		Emergency management system	Contingency plan, site response plan, specific response plan.
			The date of the Last revised and the location of the master document must be mentioned on the list.
4	The full folders concerning exemptions accorded.		Applicant (division, department, etc.) rule reference document (CR, GS, etc.), reference document for the application and the exemption accorded, date of creation.

No.	Description	Classification	Standard content and requirements
5	The list of corrective action underway, started on the basis of audit recommendations.		Type and date of the audit, object (e.g.: HSE policy, SIMOPS, etc.), summary of recommendations, reference document (audit sheet, etc.), action, action coordinator, deadline.
6	The downgraded situation tracking table.		Description of equipment, description of the *downgraded situation*, consequence, compensatory measure, corrective action, date of detection, current status, reference document (evaluation of *risks*, action plan, etc.).
7	The long-term inhibition tracking table.		The location of the short-term inhibition tracking table and insulation logs must be mentioned on the list. .
8	The log of verifications, inspections, tests, certifications, inspections, measures and other regular operations executed in terms of HSE.	Safety devices relating to the emergency stop system .	o Fire & Gas detection. o AU active instrumentation, e.g.: PSHH. o Emergency stop devices in processes (e.g.: ESDV, BDV, SDV valves) and wellheads (e.g. : DHSV, SSV valves). o High Integrity Protection System (HIPS): o Ultimate safety/protection devices, e.g.: PSV.
		Alarm device, with light and audio alarm.	o Emergency stop, e.g.: PA/GA. o Anti-collision, e.g.. : radar beacon.
		Fire fighting systems.	o Fixed fire fighting equipment and related equipment, e.g.: deluge system, fire pump, flush hydrant, water/foam gun. o Fixed fire fighting equipment for deluge system, e.g.: Inergen, CO2. o Mobile fire fighting equipment, e.g.: mobile extinguisher, fire nozzle.
		Personal protective equipment.	Respirators, anti-fire blankets, life buoys, life jackets, etc.
		Rescue and evacuation equipment.	Life boat, life raft, including the list of equipment carried and their validity dates, etc.

No.	Description	Classification	Standard content and requirements
		Floating units.	Classification certificates for floating units.
		Status of open/closed locked valves for the P&ID.	Discrepancies for the P&ID must be assessed and the reason for the discrepancy mentioned in the log.
		Regular measures/studies: o Concerning health hazards o Concerning environmental hazards.	o Noise, radioactivity, asbestos, dangerous substances, etc. o Discharge of effluents and atmospheric emissions, flows and analysis results, pouring of drilling excavation and oil on excavations, waste log and tracking forms, study reports on environmental tracking, checking of air quality, surveillance wells for underground water, etc.
		Hygiene of indispensable products.	The content of the first aid kit (e.g.: expiry date for drugs in the kit), water quality for the safety shower, eye-rinser, etc.
9	Log of regular safety exercises and the anti-pollution combat.		Report on safety exercises, report on anti-pollution combat exercises, etc.

Table2: HSE folder for the installation

3.2. KNOWLEDGE OF DOWNGRADED SITUATIONS

Any situation in which the risk level is temporarily increased from that of a normal situation must be managed as a 'downgraded situation' according to the following principles:

- the situation is identified and formally notified to Management,
- induced risks are identified and analysed,
- compensatory measures to reduce the risks are defined and approved,
- application and regular checking of the measures,
- a list of 'downgraded situations' will be updated for the site on a daily basis and made available for the personnel concerned.
- **Deterioration or loss of functionality for an important installation in terms of safety.**
- **The new situation does not allow for the operation of the installation as originally intended or "as at modification". »**

Downgraded situations: a few examples

Operation control:

- Inhibitions lasting more than 24 hours or affecting several shifts
- Errors in logic diagrams

Incident/anomaly reports:

- Delays in reporting/action to be implemented following incidents

Compatibility of personnel:

- No appropriate training provided for personnel for this task
- No appropriate training provided for personnel concerning the control of major risks
- Personnel did not comply with the safety recommendations
- Communication's systems are not operational
- Non-familiarity with evacuation plans

Inhibitions:

- Voluntary and deliberate modification rendering a given functionality inoperative for an installation.

Compensatory measures:

- Measures enabling the existence of an acceptable safety level for a given downgraded situation.

Corrective measures:

- Measures enabling the suppression of the downgraded situation

Any abnormal situation consisting of a temporary increase in the inherent risk level is considered as a downgraded situation and is therefore recorded in the specific log.

This refers to:

- abnormal situations concerning the safety barriers for the installations – including assessment and protective devices – whether this refers to dynamic safety barriers (e.g.: valves…) or static safety barriers (i.e. any equipment containing a section, such as seals, tubes, etc.),
- loss of containment (e.g. corrosion,…),
- abnormal situations for the main structural elements, such as the deterioration of part of the structure or the non-availability of measuring devices for structural integrity (probes, strain gauges, etc.),
- abnormal operation of installations, particularly those with an impact on the environment or not satisfying environmental objectives,
- abnormal organization and qualifications (e.g.: no supervision, absence of competent personnel, etc.).

All downgraded situations will be subject to the exhaustive evaluation of risks leading to the definition and implementation of corrective and compensatory action, as follows:

- all risks induced by the downgraded situation are identified,
- the appropriate compensatory measures will be immediately identified and implemented with the approval of the operational Management of the RSES or the entity, depending on the level of residual risk,
- the status of the downgraded situation and the effectiveness of compensatory measures will be constantly revised,

- corrective measures (final measures) will be identified and implemented as rapidly as possible.
- the downgraded situation log will be updated and displayed.

The **downgraded situation log** must be visible and all personnel concerned must be aware of its presence. It will mention:

- Name of the issuer
- The date of observation
- The identification of the system/equipment concerned
- A description of the situation
- The priority level
- A description of compensatory measures
- A definition of corrective measures
- Assignment of responsibilities for tracking the situation
- Date of recovered normal operations

When controlling installations, each operator must always be aware of the list of downgraded situations. The operator must also actively participate in detecting some abnormal situations which may, if extended, become downgraded situations, and report these situations to hierarchy.

3.3. KNOWLEDGE OF INHIBITIONS AND CONSIGNMENTS

- **Knowledge of inhibitions**: in case of specific operating conditions, it may be necessary to inhibit/mask alarms/safety for the process on the DCS. Each mask/inhibition will be recorded in a specific log and carefully monitored. It is essential to not leave any active mask/inhibition at shift-end.

- **Knowledge of consignments**: for traditional operating requirements (maintenance) or specific operations, some equipment may be consigned. These consignments will be noted in a log.

Works requiring the isolation of sources of energy (hydraulic, mechanic, electric, process, etc.) will only be undertaken if the following conditions are satisfied:

- Prior identification of all tasks requiring this isolation, for a strict check of operations,

- Definition of the isolation and discharge method for the energy stored,

- Discharge of energy,

- Use of a locking and tagging system with integrated safety item, i.e., authorization of over-consignment when several separate tasks exist,

- Verification of isolation and regular checking of effectiveness,

- Recording of the start and end of isolation in a specific log.

This isolation rule dictates the process isolation, mechanical isolation and electric isolation, which may be managed separately per entity organization.

In view of risk control, all isolation is subject to risk analysis, is formalized and consigned in specific logs as follows:

- authorization of the person in charge at the appropriate level – particularly the Operating Manager – to execute the consignment, with particular attention to situations which may lead to a downgraded situation,

- formalization of the isolation in a specific certificate and recording in the appropriate log,

- updating of the corresponding tracking table and recording in the appropriate log.

If only one isolation system is used for various operations, the following requirements must be satisfied:

- implementation of multiple isolation to ensure that no isolation is removed prior to all the official closing of all of the corresponding permits. Coordination is the responsibility of the isolation Manager, and a master document will be used for all teams working on the same isolation system.

All of the requirements mentioned in the previous paragraph also apply to inhibition, with the following additional requirements: an inhibition relating to a safety system (e.g.: gas detector, top-top pressure switch (PSHH),...) will be considered as a downgraded situation if not removed within 1 day. Two separate logs are required for inhibitions: the long-term log and the short-term log. Inhibitions which are not removed within 1 month are transferred from the short-term log to the long-term log: the long-term inhibition tracking table is updated and included in the HSE folder for the installation.

The inhibition caused by a sudden detection or other instrument problem must also be formalized in the form of a specific certificate, but will only be consigned in the appropriate log if the inhibition has not been removed at the completion of work by a team.

Standard content of an isolation/inhibition certificate and an isolation/inhibition tracking table:

Isolation and inhibition certificates

The standard content of an isolation/inhibition certificate is as follows:

- Description of the equipment or system.
- List of related Work Permits and identification.
- Name of the isolation/inhibition Manager.
- Names of the individuals installing the padlock or other locking device at the consignment point, and status of the isolation (isolation required/not-required)
- List of documents (plans and schematics, check-lists*, etc.) enclosed with the certificate.
- Authorization to proceed with isolation/inhibition (e.g.: signature of the operational Manager or other) if required.

Note: Specifically check the list of the positions of open locked/closed locked valves (i.e. the status of the valves as indicated in the P&ID, status before and after isolation)

Isolation and inhibition tracking table

The standard content of an isolation/inhibition tracking table is as follows:

- Description of the equipment or system (e.g.: name, location, tag, etc.)
- Type of isolation/inhibition (e.g.: inhibition/isolation for works, for a process problem, etc.)
- Reason for and consequence of isolation/inhibition.
- Name of the isolation/inhibition Manager and the isolation/inhibition actor.
- Consignment - LO/TO (Lock Out / Tag Out – Locking/Tagging)
- The appropriate Work Permit number, if applicable.
- History: start date and time, planned end date, and end date and time.

3.4. KNOWLEDGE OF SIMOPS

All works or simultaneous Operations in Production and Construction and Production and Drilling (SIMOPS) are likely to increase the level of risk. These SIMOPS must not start prior to:

- a preliminary inspection of installations by the authorized representatives,
- an identification, evaluation and complete analysis of risks,
- the application of all recommendations on the basis of this risk analysis,
- the definition of responsibilities in SIMOPS and the nomination of a RSES (Site Safety Environment Manager),
- the creation of specific SIMOPS meetings,
- the presentation of a duly approved SIMOPS file,
- obtaining of formal authorization to proceed with operations, as issued by the Operations director (or Technical director) following a site inspection.

The operator must pay particular attention to these situations as they frequently generate the presence of extra personnel on installations and increased risks due to the simultaneous execution of operations by day and night.

Safety systems are often installed in addition to the fixed systems for the installation, and it is important to be aware of their location and functioning.

3.4.1. The general safety dossier

SIMOPS requires prior start-up a SIMOPS general safety dossier issued by the RSES to all the responsible persons involved in the operations.

This dossier becomes the General Safety Dossier of the installation for the duration of SIMOPS.

Prior to moving in of the heavy marine units or the rig, the SIMOPS General Safety Dossier shall essentially include:

- Any existing statutory texts relative to the design and operation of the relevant installation
- The General Safety-Environment Standing Instructions
- The Contingency Plans

- The Work Permit System procedure
- The Subsidiary Standing Instructions for conducting SIMOPS
- The ESD block diagram of the installation
- The specific SIMOPS procedure including:
- The detailed technical documents of the installation, updated for SIMOPS, and in particular the layout of ESD manual push button, life-saving and evacuation means on the installation, and depressurization systems
- SIMOPS Decision Matrix
- The risk assessment study
- The SIMOPS organization charts
- The detailed operations program
- The report on the visit of the installation prior to SIMOPS
- The acceptance report on works to be completed on the facilities **prior to rig installation** as defined during the visit prior to SIMOPS
- The specific procedure detailing tests on safety equipment and systems with inhibition of effects, and the precautions to be taken with regard to inhibition (especially deactivation of inhibition on completion of testing)
- Any specific memos or documents concerning safety and the environment during SIMOPS
- The fire-fighting dossier including the platform and rig fire-fighting equipment and plans.

When the rig is installed and before well operations starts, the following main additions are made to the SIMOPS General Safety Dossier:

- The rig acceptance report
- The acceptance report on works to be completed on the facilities **prior to start well operations**, as defined during the visit prior to SIMOPS
- The minutes of the kick off meeting
- Any updates made to the installation's technical documents mentioned above.

3.4.2. The Technical Safety Dossier

The RSES is the custodian of this dossier which becomes Technical Safety Dossier of the installation for the duration of SIMOPS. The purpose of this dossier is to record the checks performed and the provisions adopted in respect of safety.

In addition to the requirement detailed in CR EXP 008 as follows,

- The record of checks performed on equipment subject to compulsory inspection
- Where applicable, the record of inspections performed by the authorities
- The follow-up sheets, dated and signed, for periodic testing of safety equipment and systems
- The record of periodic safety drills
- The production Log-Book kept in the control room for shift personnel
- The closed out permits to work and isolation certificates
- The minutes of daily meetings or the logbook recording decisions taken there
- The down graded situations log book or inhibit/override register.

The Technical Safety Dossier also comprises:

- The facilities hand-over file
- The RSES Log-Book
- The statements signed and dated by the RSES, the Operating Authorities, declaring that they are familiar with the specific SIMOPS procedures in the General Safety Dossier
- The follow-up sheets, dated and signed, for periodic checks and tests on safety equipment and systems specific to SIMOPS.

The technical safety dossier is verified and signed, at each visit to the installation performed by line management and in particular by the Operations Manager.

3.5. CLASSIFIED AREAS

3.5.1. Definitions

Firstly, it must be reiterated that this type of area is defined in order to install the appropriate electric equipment and **not to select the locations for which a welding permit procedure would not be required**.

The **classification of hazardous areas** takes into consideration events which are "liable to occur during normal or abnormal plant operating conditions"

Areas are classified as follows:

- **Area 0**: area in which an explosive gaseous/or dust-impregnated atmosphere is constantly present or for long periods.
- **Area 1**: area in which an explosive gaseous/or dust-impregnated atmosphere is likely to be present during normal operations.
- **Area 2**: area in which an explosive gaseous/dust-impregnated atmosphere is not likely to be present during normal operations, or, if this atmosphere was present, it could only remain for a short period.

In addition, to simplify the understanding of this text, **non-hazardous areas** are defined: these are areas in which the probability of the appearance of gas or flammable vapours is marginal independently to operating conditions. These are explosion risk-free areas. (Example: pressurized electric cabin).

3.5.2. Delimitation of areas

The delimitation of areas with explosion risks meets two objectives:

- Limiting the extension of hazard areas
- Installing appropriate electric equipment.

This delimitation must be defined in a written document and completed with detailed plans.

3.5.3. Sources of emissions

These are the points at which flammable substances are emitted into the atmosphere. *

Sources of emissions are classified:

Area 0 sources:

This essentially includes:

- The inside of closed storage tanks
- The inside of closed production or mixing devices.

Area 1 sources:

This essentially includes:

- Open tanks or storage containers
- Open production or mixing devices
- Vents from closed tanks (Separators, desalters, etc)
- Vent orifices for hydraulic guards
- End of articulated arms and flexible loading arms for cisterns and containers
- Loading buffers and drain valves for devices
- Sampling or free draw-off valves
- Pump or compressor packing, etc. if leakage subsists (e.g. functional leakage from a gland)
- Pits or unsealed gutters
- Siphoid sight holes
- Pig trap scrapers
- Well head.
- Area including dust which may lead to a risk of explosion.

Area 2 sources:

This essentially includes:

- Flanges, connections, valves and piping connections
- Gauge glass sights or tubes

- Pump or compressor packing, etc., designed to prevent leakage
- Instruments in fragile materials such as glass, ceramic, graphite
- Breathing orifices for expansion diaphragms
- Retention pond (catchpits).

Remarks:

- Welded piping without flanges or connections is not considered as a source of emissions for area 2.
- A pressurized enclosure may be a source of emissions for area 1 or a significantly extended area 2 (sampling or draw-off valves, safety valves, rupture disks, vent orifices for the expansion diaphragm, etc.).

Provisions required for electric equipment:

- Area 0: Underlying category "i" safety
- Area 1: "Safety" equipment + cut-out at the detection of 25% LEL for gas
- Area 2: "Safety" equipment if spark or hot surface, or sealed equipment otherwise

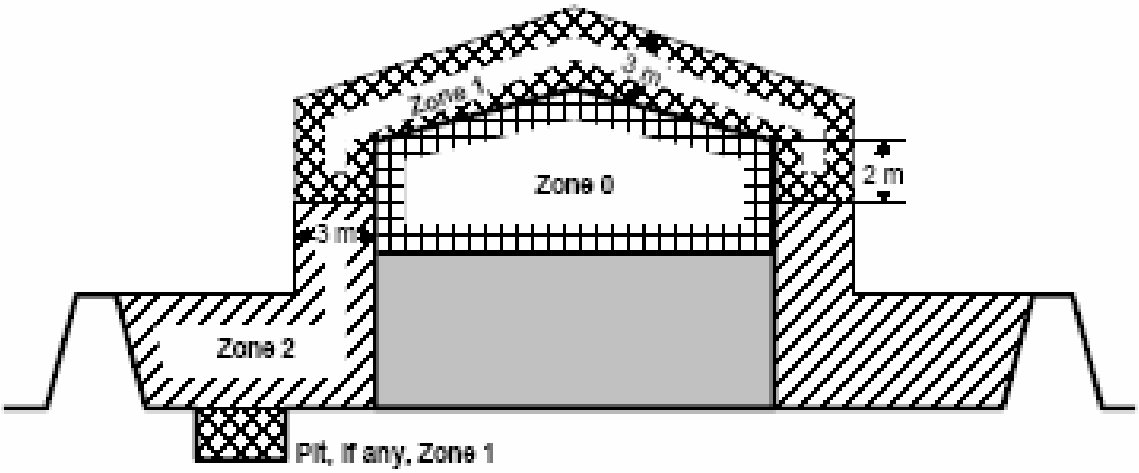


Figure 5: Example of area delimitation for an onshore storage vessel

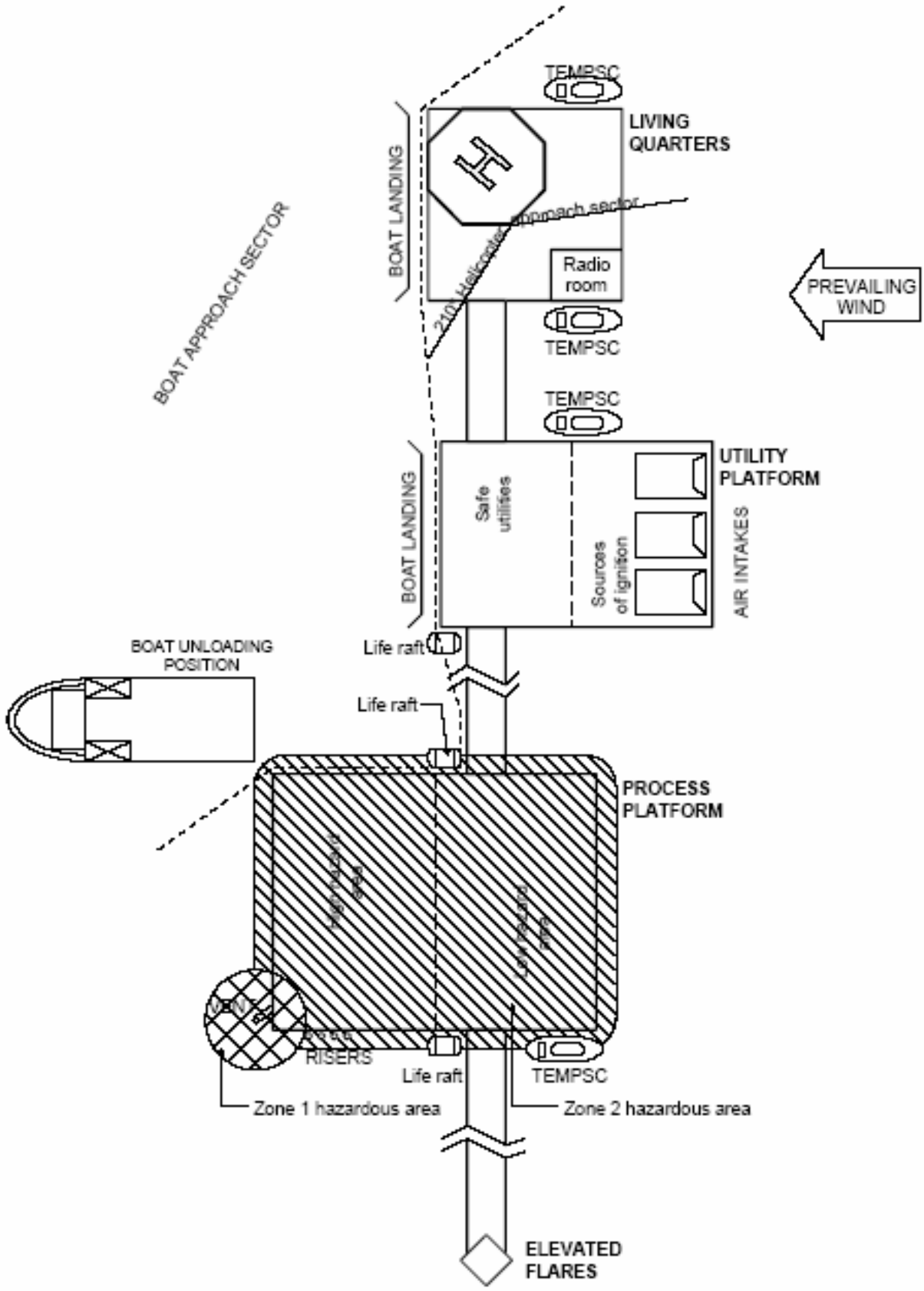


Figure 6: Layout of fixed offshore platforms

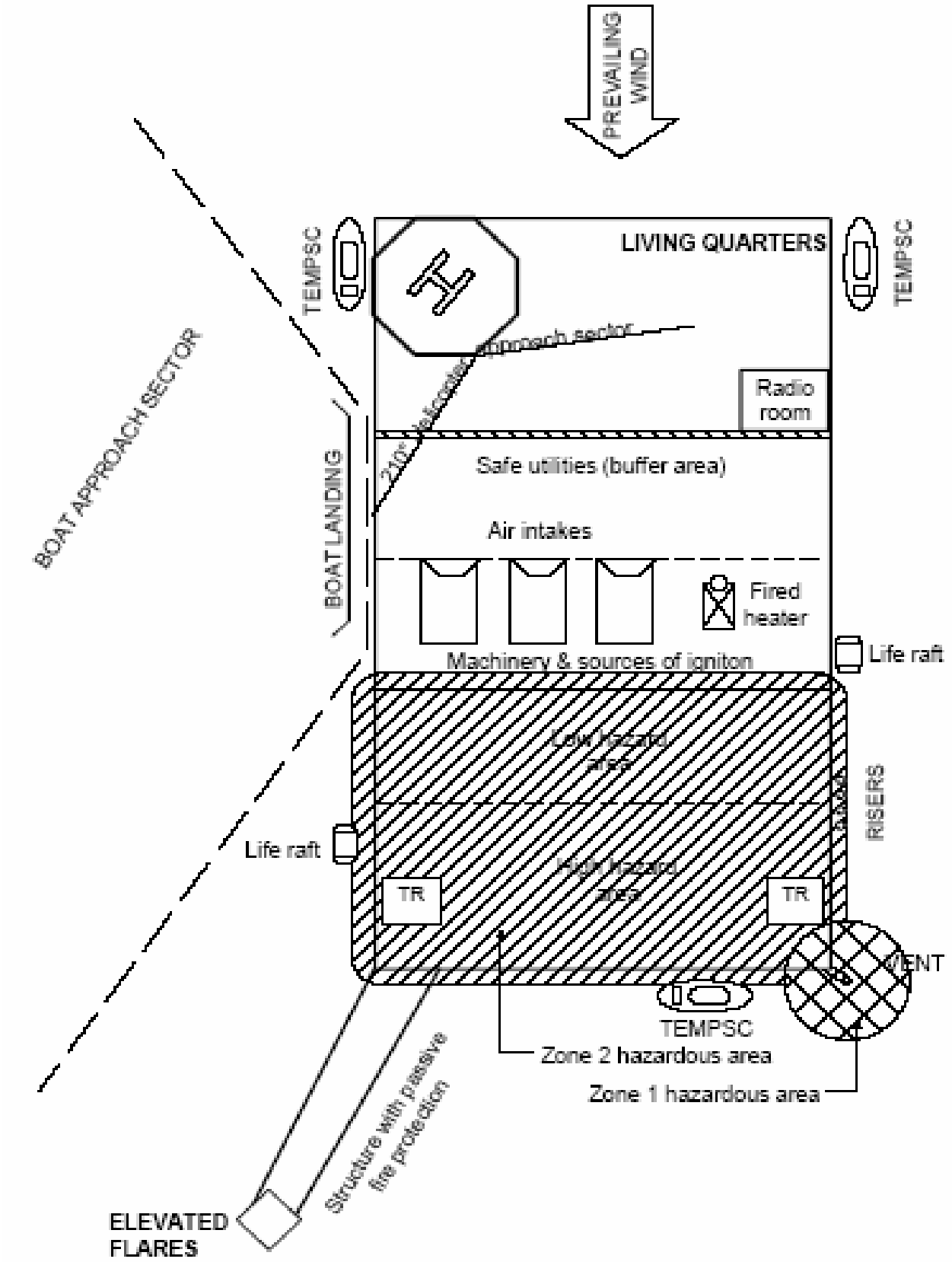


Figure 7: Layout of integrated floating platform

3.6. SAFETY BARRIER LOGIC

3.6.1. Emergency Shut-Down (ESD)

ESD system is here used as a generic term and consists in fact of process shutdown (SD) and emergency shutdown (ESD) functions.

A safety shutdown system contains different levels (process, emergency, fire & gas and if required others), each of them consisting of a set of safety loops. In general, safety loops consist of field sensors (initiators), logic solvers and final elements (e.g. valves).

Its (emergency) shutdown is associated with other independent safety systems (PSVs, HIPS) and safeguard systems (fire fighting, escape evacuation and rescue, personnel protection systems, etc.) to reduce the industrial risk of the installation.

The main purposes of ESD systems are as follows:

- To protect personnel, e.g. smoke and gas detection in the HVAC intakes of Living Quarters,
- To limit the loss of containment, by isolating hydrocarbon production, processing and storage equipment,
- To execute automatically a set of remedial actions, upon manual or automatic triggering,
- To prevent ignition by elimination of potential sources of ignition,
- To reduce flammable or toxic inventory by depressurization through the EDP system, when appropriate.

3.6.2. Architecture of the Shutdown system

It is essential to distinguish three functionally different safety shutdown systems:

Functional system	Abbrev.	Function
Process Safety System	PSS	Trips and associated shutdown actions + local (equipment / package) F&G
Emergency Shutdown System	ESD	Emergency shutdown actions
Fire and Gas System	F&G	Outdoor and indoor general fire and gas related ESD actions

Table 3: Architecture of the Shutdown System

The **PSS** controls all causes/actions pertaining to SD-3 shutdowns (i.e. individual equipment), including fire and gas at local (equipment/package) level. In this respect the PSS can include a F&G sub-system, generally provided with the equipment/package and by its VENDOR, and distinct from the main F&G system mentioned below.

The **ESD** system manages all process-related inputs and outputs relative to ESD-0 (whole facility, if applicable), or ESD-1 (fire zone) or SD-2 (process unit) shutdowns. It is also fed by signals from the main F&G system (see below).

The main **F&G** system deals with fire and gas detection outdoor and indoor (e.g. technical room, control room, etc.), where they may consequentially affect more than just one specific equipment. It generates the corresponding ESD-1 actions, except those related to process that are undertaken by the ESD system. The F&G system thus provides input to the ESD system. The F&G system does not generate SD-2 shutdown actions.

Besides the above mentioned three safety shutdown systems there are two additional instrumented systems, whereby one is optional.

Functional system	Abbrev.	Function
Process Control System	PCS	Controls and associated alarms
Ultimate Safety System	USS	Back up of ESD actions

Figure 8: Additional instrumented systems

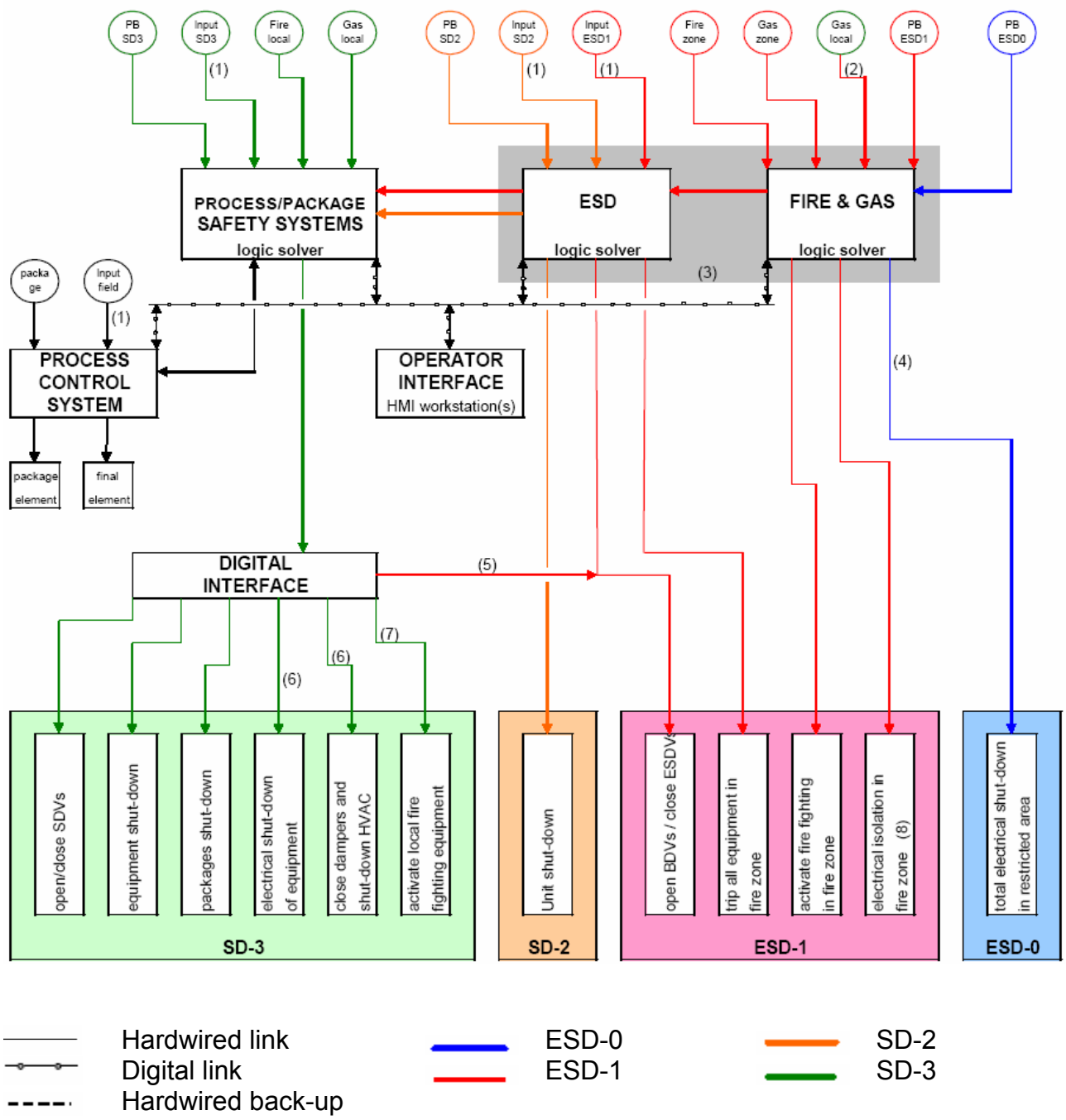


Figure 9: Typical shutdown system architecture

1: input = field sensors or initiators
2: gas detection in package ventilation/combustion air duct, if compatible
3: ESD and F&G logic functions may be housed in the same logic solver
4: main power supply and all battery outgoers
5: PSS action on ESDVs, if necessary
6: if local gas detection is activated
7: if local fire detection is activated
8: except vital consumers and controls

3.6.3. Definition of the shutdown matrix

It is a common practice within COMPANY to define a maximum of four typical shutdown levels with decreasing criticality, numbered 0 to 3 and affecting:

- all installations within a single restricted area (level-0) = ESD-0
- a given fire zone within the installation (level-1) = ESD-1
- a given unit within a given fire zone (level-2) = SD-2
- an individual equipment or package within a given unit (level-3) = SD-3

Level-0 and level-1 shall be called ESD levels because they involve either fire/gas detection in unconfined environment (hence a situation subject to possible escalation) or manual emergency action.

Level-2 and level-3 shall be called SD levels because they correspond either to a mere process upset or to confined fire/gas detection (sufficiently well contained) not threatening immediately the safety of the personnel and of the installation.

The safety shutdown system of an installation, consisting in a set of safety loops and devices, comprises different sub-systems organized as complementary barriers to the Process Control System, as represented in the following schematic.

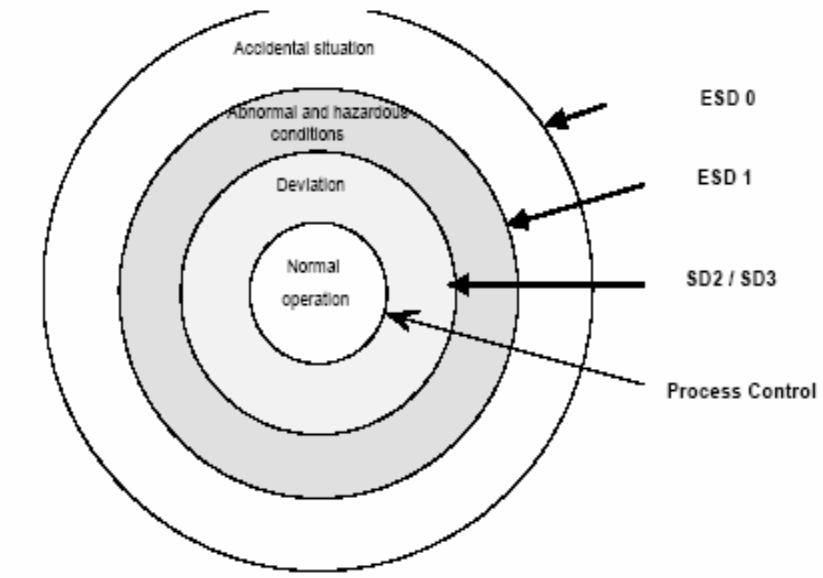


Figure 10: Schematic of safety shutdown system operation

For each installation an ESD/SD logic shall be defined and represented in an ESD/SD logic diagram. This logic is based on the hierarchy of ESD and SD levels, the level N activating the level N+1. The ESD/SD logic diagram shows the top-down hierarchy of ESD and SD levels, all their causes and actions in the form of a block logic diagram.

Each level is subdivided into several safety bars (up to one bar per equipment). The number of safety bars varies with the type of installation, the number of fire zones and their location, the number of independent units in each fire zone and other characteristics. Each case is specific and the following development is intended to provide guidelines and simplified examples.

The ESD/SD logic diagram shall cover all the facilities of a petroleum installation. The causes and actions shall be described at a functional level (type and location of detection, closure/opening of valve, shutdown of equipment, etc. …).

Differences onshore/offshore

The fundamentals driving shutdown logic design are always the same, however the environment (onshore versus offshore) leads to three main differences:

- The **ESD-0** level shall be applicable for permanently manned offshore installations, unless statutory requirements do not impose to do so and a risk assessment (size, lay-out and manning criteria) demonstrates the non-necessity of ESD-0.
    - In all other cases, not permanently manned offshore installations and all onshore plants (regardless of size), the number of shutdown levels may be limited to three, starting from **ESD-1** level. The wordings “muster & evacuation of personnel” and “muster” denote voluntary procedures involving personnel but are not to be considered as ESD levels.
- For all offshore installations (permanently and not permanently manned) **Emergency De-Pressurization** (EDP) shall be (if installed) automatic upon activation of ESD-1 level. This requirement is not compulsory for onshore facilities and EDP strategy shall be duly addressed in the SAFETY CONCEPT.
- **De-energizing** including battery powered systems, but with the exception of emergency devices (emergency lighting, navigation aids, etc.) and equipment suitable for operation in Zone 1 hazardous area, can be achieved on permanently manned offshore installation through activation of ESD-0. Onshore this functionality does not have to exist and shall then be compensated by the implementation of a specific push button for each fire zone that shall perform total de-energizing, including controls (24 VDC), with **possible** exception for emergency post-lube pumps, machinery helper, etc. **and only if** they are suitable for operation in Zone 1 hazardous area.

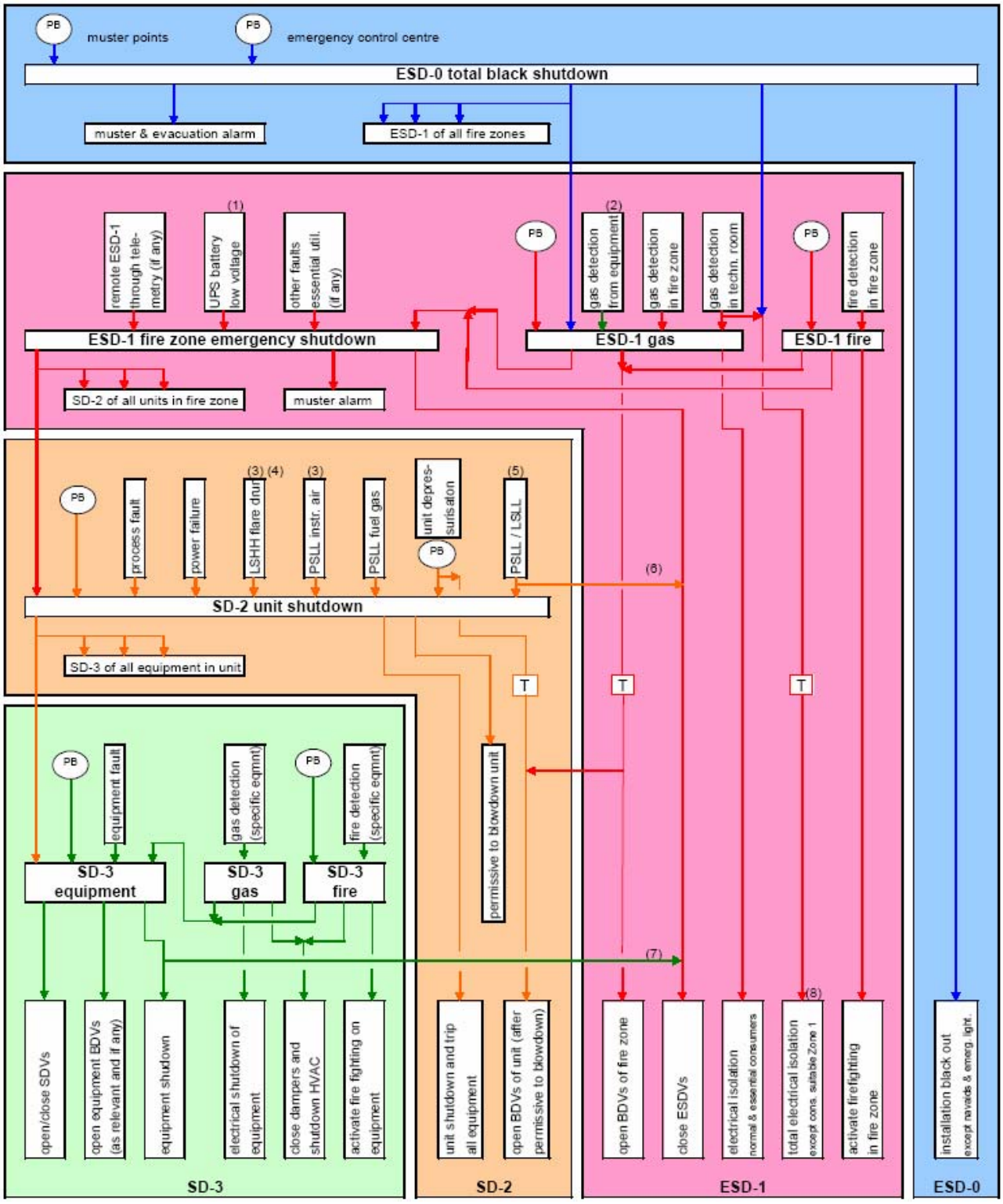


Figure 11: Typical shutdown logic diagram (offshore processing facility)

1 : to avoid uncontrolled sequence of ESDV/BDVs closing/opening
2 : unprocessed gas detection signal from equipment to ESD-1 if required
3 : also to other units if common
4 : as an alternative, LSHH flare drum could also initiate an ESD-1 (risk assessment)
5 : list to be assessed on a case by case basis
6 : close ESDVs if no SDVs upstream PSLL/LSLL used as leak detection device
7 : closing of fuel gas ESDVs serving the concerned equipment
8 : emergency/vital systems remaining powered; telecom, PAGA and post lube (if any)

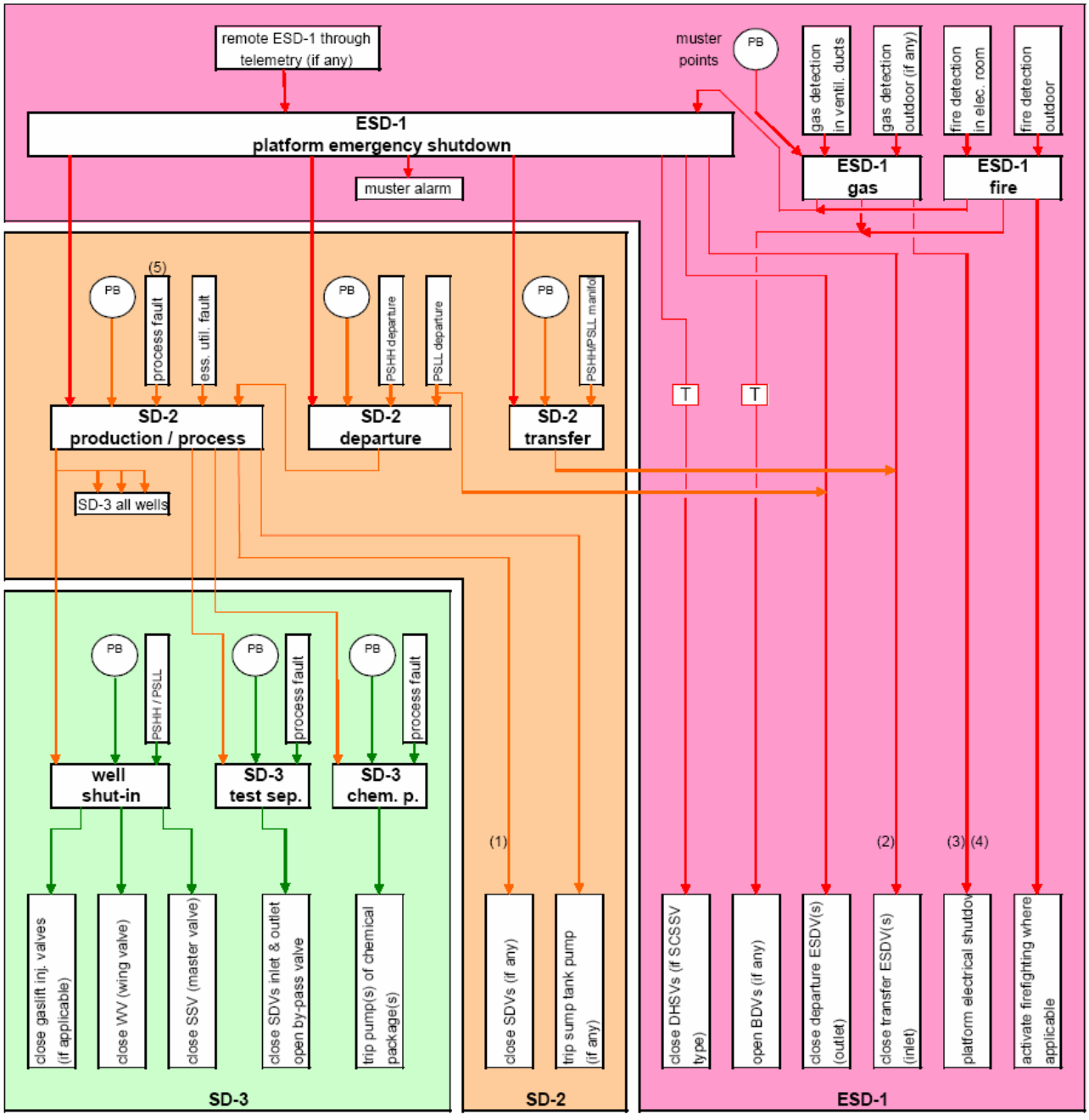


Figure 12: Typical shutdown logic diagram (wellhead & riser platform with test separator)

1: downstream of production manifold where connecting with transfer manifold
2: assuming transfer manifold ties-in upstream of platform outlet ESDV
3: emergency & vital systems remaining powered: navaids, emergency lighting, general alarm, telecom and public address (if any)
4: shutdown crane engine if diesel powered
5: as alternative and based on risk assessment, LSHH flare drum can also initiate an ESD-1

3.6.4. ESD-0 (total black shutdown)

This is the highest level of ESD, intended to make an installation safe before evacuation.

This level concerns the restricted area of a petroleum installation.

There shall be one ESD-0 for each restricted area.

Although very rare, within the property boundaries of the same site two or more completely independent installations may be present, i.e. each installation runs independently with different sources of power and controls and are at sufficient distance, creating thus several (non-overlapping) restricted areas. Each restricted area has its own ESD-0 instead of a common site ESD-0.

3.6.4.1. Causes ESD-0

It is in general, manually initiated, only once the voluntary decision has been taken by the person in charge (when in manual decision), i.e. RSES (French abbreviation for Responsable Sécurité Environnement de Site, translated in English as Site Safety Environment Manager), to evacuate the installation.

Exceptionally it is automatically initiated. This is only the case when the ESD and F&G systems have to be de-energized due to presence of a flammable atmosphere in the building where the ESD and F&G systems are located (generally in the CCR). Whenever possible, an installation should be designed to avoid the need for automatic ESD-0 initiation.

As far as practicable, buildings containing the ESD and F&G systems (I/O cabinets, racks, power supplies and PLCs) should be located outside the restricted area of the installation. If so the initiation of ESD-0 shall only be manual.

If not practicable, the probability of a spurious ESD-0 on false gas detection in the CCR shall be minimized by implementing action when 2 different detections are actuated simultaneously in air inlets and air locks and gas detectors located downstream of the HVAC inlet shutter (fire dampers) closing first the dampers before initiating ESD-0.

3.6.4.2. Actions ESD-0

ESD-1 of all fire zones within the restricted area.

Shutdown of all process and utility systems, with depressurization, for all fire zones in the restricted area.

- ESD-0 does not stop diesel engine driven firewater pumps if they were already started up automatically (selector on automatic mode and signal from Fire & Gas system, or PSLL ring main).

- Some post-lubrication pumps may need to be kept in service to prevent damage of major rotating equipment. To prevent major financial loss in the event of an ESD-0, this equipment may be kept in service. It shall however be shutdown after a pre-set time, i.e. the run-down of the machine, and this shall be duly addressed in the SAFETY CONCEPT.

Shutdown of all potential sources of hazard and ignition including essential and emergency loads, except navigational aids (marine and aviation) and emergency lighting.

Shutdown of all potential sources of hazard and ignition is achieved without delay.

Shutdown after a pre-set time (normally not exceeding 1 hour) of the critical communications within the installation (public address) and with external parties (radio, satellite).

Audible alarm and visual signals for personnel to muster and prepare for evacuation.

All the equipment and their associated power supply systems, staying operational after an ESD-0, shall be certified for Zone 1 hazardous area and shall have their own dedicated uninterruptible power supply (UPS).

3.6.5. ESD-1 (fire zone emergency shutdown)

There is one ESD-1 for each fire zone within the restricted area and it is the highest level of shutdown which allows the presence of personnel on site.

In general all hydrocarbon flows within the fire zone shall be stopped and hydrocarbon inventories blocked-in and possibly released upon an ESD-1.

As fire and gas detection leads to different actions, the ESD-1 shall be further split into ESD-1/F for the particular fire case, ESD-1/G for the particular gas detection case and the subsequent generic ESD-1 fire zone.

3.6.5.1. Causes ESD-1

ESD-0 within the restricted area.

Manual initiation through push button (based on a probable or actual, catastrophic situation).

A signal from the installation F&G system: (1) (2)

- Outdoor (or in a not totally enclosed area) flammable gas detection in the fire zone,

- Gas detection in the HVAC inlets of technical rooms located in the fire zone,

- Gas detection in the air inlets of fired equipment located in the fire zone,
- Outdoor fire detection in the fire zone.

Detection of inevitable loss of a utility which is essential for the safety of the installation:

- FSLL or PSLL flare purge gas,
- UPS low voltage (loss of power supply to ESD and F&G systems),
- Other utility failures, as advised by a specific study.

Fire detection inside a technical room does not result in an ESD-1, as the local fire fighting and HVAC isolations are handled by the F&G system.

Fire detection in an electrical room does not result in an ESD-1, except in remote and not permanently manned premises where intervention is not quickly possible.

3.6.5.2. Actions ESD-1

SD-2 of all units, process and utility systems, within the fire zone

Close all ESDV's, fuel (except diesel) supply lines to the fired equipment shall therefore be fitted with an ESDV.

Close the SCSSV (Surface Controlled Subsurface Safety Valve) of the wells located within the fire zone.

- SSVs (Surface Safety Valves) of the wells are closed on the SD-3 level (via the SD-2 level) and SCSSVs and SSVs are regarded as ESDVs.

Main power supply (and power generation if located in the fire zone) shutdown (electrical isolation), thereby shutdown of all motors in the fire zone.

Shutdown of the large electrical motors (redundant with main power supply shutdown).

- Considering that essential utilities are suitable for operation in Zone 1 hazardous area, the shutdown of non-essential utilities with a time delay, where applicable, may be acceptable.
- Upon confirmed fire and/or gas detection, automatic emergency de-pressurization (EDP) offshore, and optional onshore. Open all the BDVs (Blow-Down Valves) in the fire zone with a pre-set time delay (30 s to 1 min.). If de-pressurization is not automatically initiated upon ESD-1/F and/or ESD-1/G, a push button located in the CCR initiates ESD-1/F and/or ESD-1/G and opens all BDVs with a pre-set time delay.

- Initiate the SD-2 of the hydrocarbon facilities located outside the ESD-1 fire zone, which send hydrocarbons to the ESD-1 fire zone.
- In case of gas detection, shutdown of all potential sources of hazard and ignition (except running firewater pumps, see (1) in section 4.2.3.2) in the fire zone and except controls and emergency or vital equipment on individual battery systems and suitable for Zone 1.
- In case of fire detection, activation of fire-fighting means in the fire zone.
- Audible alarm and visual signals for personnel to escape from fire zone and to muster.

3.6.6. SD-2 (unit shutdown)

There is one SD-2 for each independent functional unit.

Hydrocarbon production and process facilities within a fire zone are shutdown upon an SD-2. It does however not necessarily shutdown the fuel gas system. Upon production/process shutdown and if fuel gas is still required for power generation or flare purge gas, then the fuel gas source shall be independent from production (e.g. fuel gas from an import or export pipeline) and this source shall not be interrupted on SD-2.

There is no F&G input at SD-2 level. F&G initiates either ESD-1 (outdoor detection) or SD-3 (specific to an equipment or package).

3.6.6.1. Causes SD-2

ESD-1 of the fire zone to which the unit belongs.

ESD-1 of another fire zone from which the concerned unit fire zone receives hydrocarbons.

Manual initiation through push button (based on a probable or actual unit failure).

Process fault or failure that requires the automatic shutdown of the unit and would have inevitably resulted in a complete shutdown of the production/process unit by cascade.

Detection of inevitable loss of a utility, which is essential for production/process in the unit :

- LSHH in the flare KO drum(s) connected to the unit,
- PSLL instrument air,

- Leak detection (PSLL, LSLL, etc.) on process systems studied on a case by case basis,
- Main power very low voltage,
- Loss of normal power.

3.6.6.2. Actions SD-2

SD-3 of all equipment within the unit (process or utility systems)

SD-3 of all hydrocarbon processing and production equipment within the unit, close the associated SDVs and shutdown of associated motors.

To avoid cascaded shutdown, shutdown of some non-hydrocarbon treatment facilities, which are directly linked to production/process but not required when production/process is stopped (e.g. chemical injection into production/process hydrocarbon flow)

Send a signal (e.g. by telemetry) to close remotely operated choke valves of the wells outside the SD-2 fire zone, which send hydrocarbons to the concerned SD-2 fire zone.

May close the ESDVs located at the battery limits of a process train or process platform (fire zone).

Although the installation battery limit ESDVs (i.e. import/export pipeline ESDVs) are the ultimate safety barriers of the installation and only closed upon ESD-1, these ESDVs shall be closed upon their corresponding pipeline leak detection PSLL.

Permissive to perform manually emergency depressurization if relevant to concerned unit.

3.6.7. SD-3 (equipment shutdown)

There is one SD-3 for each process or utility equipment within a unit. The objectives of an SD-3 shutdown are to put the equipment in a safe position and to provide the operator the opportunity to prevent escalation to a higher (SD-2 or ESD-1) shutdown level.

In some cases, equipment can have different SD-3 sequences depending on the tripping fault.

Where fire and gas detection lead to particular and different actions, SD-3 of an equipment shall be further split into SD-3/F for the particular fire case, SD-3/G for the particular gas detection case, and the subsequent generic SD-3 equipment.

The SD-3 logic is mainly processed into the PSS system (process equipment) but in some cases into the ESD system (utility equipment).

3.6.7.1. Causes SD-3

SD-2 of the unit.

Manual initiation through push button (based on a probable or actual equipment failure).

For prime movers and machinery, manual initiation (push button) from a local panel.

Trip of a process or utility operating parameter (excursion outside operating limits).

Fire or gas detection inside a non-fired equipment enclosure.

For fired equipment, a signal from the installation Fire & Gas system.

- Flammable gas or fire detection inside the enclosure of the fired equipment (e.g. gas engine or turbine) shall trigger an SD-3 of the fired equipment package and close the ESDV of the fuel supply to the package. For the latter, the fire and gas detection signal shall also be processed by the installation F&G system, which sends a signal to the ESD system.

- Fire-fighting and associated isolations (air intake, exhaust and electrical) inside a fired equipment package shall normally be handled by its own internal package F&G system.

3.6.7.2. Actions SD-3

Close SDVs or open SDVs (for diverting purposes) through PSS system.

Close some specific ESDVs (e.g. fuel supply to packages) through ESD system.

Close the SSV (Surface Safety Valve) of the wells located within the fire zone.

- SCSSVs (Surface Controlled Subsurface Safety Valves) of the wells are closed through the ESD-1 level and SCSSVs and SSVs are regarded as ESDVs.

Stop motors.

Initiate package shutdown, e.g. compressor package.

Shutdown of production or utility equipment, with either (if relevant) automatic depressurization or (if required) unlatching of a “permissive to depressurize” lock allowing thus manual emergency depressurization.

In case of gas detection inside an enclosure (from an internal gas source), shutdown of all potential sources of hazard and ignition within the enclosure (including essential loads) except emergency or vital equipment on individual battery system and suitable for Zone 1.

In case of fire detection inside an enclosure, activation of fire-fighting means in the equipment enclosure and closure of dampers (as relevant).

3.6.8. Fire and Gas system versus ESD system

The F&G manages all inputs provided by fire and/or gas detectors, performs the corresponding logic treatment and generates the relevant outputs. The F&G deals only with safety actions of the highest level, i.e. ESD-0 and ESD-1. Fire and gas detection and logic relating to packages shall be achieved locally by a system provided by the package VENDOR.

Outputs from the F&G system shall be either directly to equipment (e.g. electrical isolation, activation of fire-fighting means, etc.) or else shall feed the ESD system that performs the process related actions (e.g. close ESDVs, open BDVs, etc.).

The F&G and ESD systems shall always be functionally independent, even if these two functions are performed by a common equipment. This option is sound providing the F&G reliability is not impacted and also if the software managing ESD and F&G are treated as two independent functional entities and the links between ESD and F&G are clearly identified and documented.

3.6.9. Shutdown devices, protection and other requirements

3.6.9.1. Process safety valve definitions

ESDV: Emergency Shut-Down Valve

BDV: Blow-Down Valve

SDV: Shut-Down Valve.

Other on/off motorized valves (XVs) and Hand Valves (HVs) cannot be considered as safety valves, neither ESDVs nor SDVs.

It is possible that an ESDV or SDV is controlled simultaneously by the ESD system and by the PSS system. In this case two solenoid valves shall be mounted in series, one connected by dedicated hard wire to the ESD system, the other connected to the PSS system.

3.6.9.2. Wellhead safety valve definitions

DHSV: Down-Hole Safety Valves (SCSSVs) shall be considered as ESDVs.

SSV: Surface Safety Valves (automatic upper master valves) shall be considered as ESDVs.

- SSVs shall always close before SCSSVs to avoid pressure differential across the SCSSV.

WV: Wing Valves (automatic wing valves) shall be used. They shall be considered as SDVs.

- WVs shall always close before SSVs to avoid pressure differential across the SSV.
- WVs may be remotely controlled if their control circuit is fitted with a specific solenoid independent from the safety trip circuits,
- Remote WV re-opening through telemetry is authorized only if the concerned well was closed voluntarily and in absence of fault (F&G or PSHH/PSLL).

Gas-lift or gas re-injection isolating valves are considered as SDVs.

Chokes, even motorized, cannot be considered as safety valves, neither ESDVs nor SDVs as they are Pressure/Flow control valves only.

3.6.9.3. Emergency Push buttons

Push buttons shall be installed as follows:

Location	Offshore Platform	Drilling / WO rig	Onshore Plant
Emergency control centre	ESD-0	ESD-0 (1)	
Muster points / temporary refuge	ESD-0	ESD-0	
Driller's console		ESD-1 SD-2	
Control room (CCR)	ESD-0 (2) ESD-1 SD-2 SD-3	ESD-1 SD-2 SD-3	ESD-1 SD-2 SD-3
Unit local panels (3)	SD-2 SD-3	SD-2 SD-3	SD-2 SD-3
Outdoor	ESD-1 (4)		ESD-1 (4)

Table 4: Installation of emergency push buttons

(1): Relates to drilling rig shutdown at an ESD-0 level (no ESD-0 level on a wellhead platform) - SIMOPS dossier to define the relevant actions
(2): Push buttons in CCR only for remote facility controlled from CCR
(3): Outdoor panel close to equipment or unit
(4): ESD-1 push buttons can be provided outdoor at convenient locations, if imposed by site specifics (not base case)

Push buttons shall be properly located, tagged and illuminated by essential lighting. They shall be physically protected against spurious activation and fitted with a specific unlocking tool to return to normal position.

In case the activation of a shutdown push button unlatches a "permissive to EDP" signal, the corresponding EDP push button shall be located close by.

	Wellheads			Process		
	DHSV	SSV	WV	ESDV	BDV	SDV
Local reset after ESD-0 or ESD-1	Yes	Yes	Yes (1)	Yes	Yes (2)	No
Open from CCR	No	No	No (1)	No	Yes (5)	(3)
Close from CCR	Yes	Yes	Yes	Yes	No	(3)
Open / Close local command	Yes	Yes	Yes	Yes	Yes	Yes
Open / Close status display in CCR	Yes	Yes	Yes	Yes	Yes	Yes
Partial stroking facilities	No	No	No	Yes (4)	No	Yes (4)
ESD signal test facilities	Yes	Yes	Yes	Yes (4)	Yes	Yes (4)

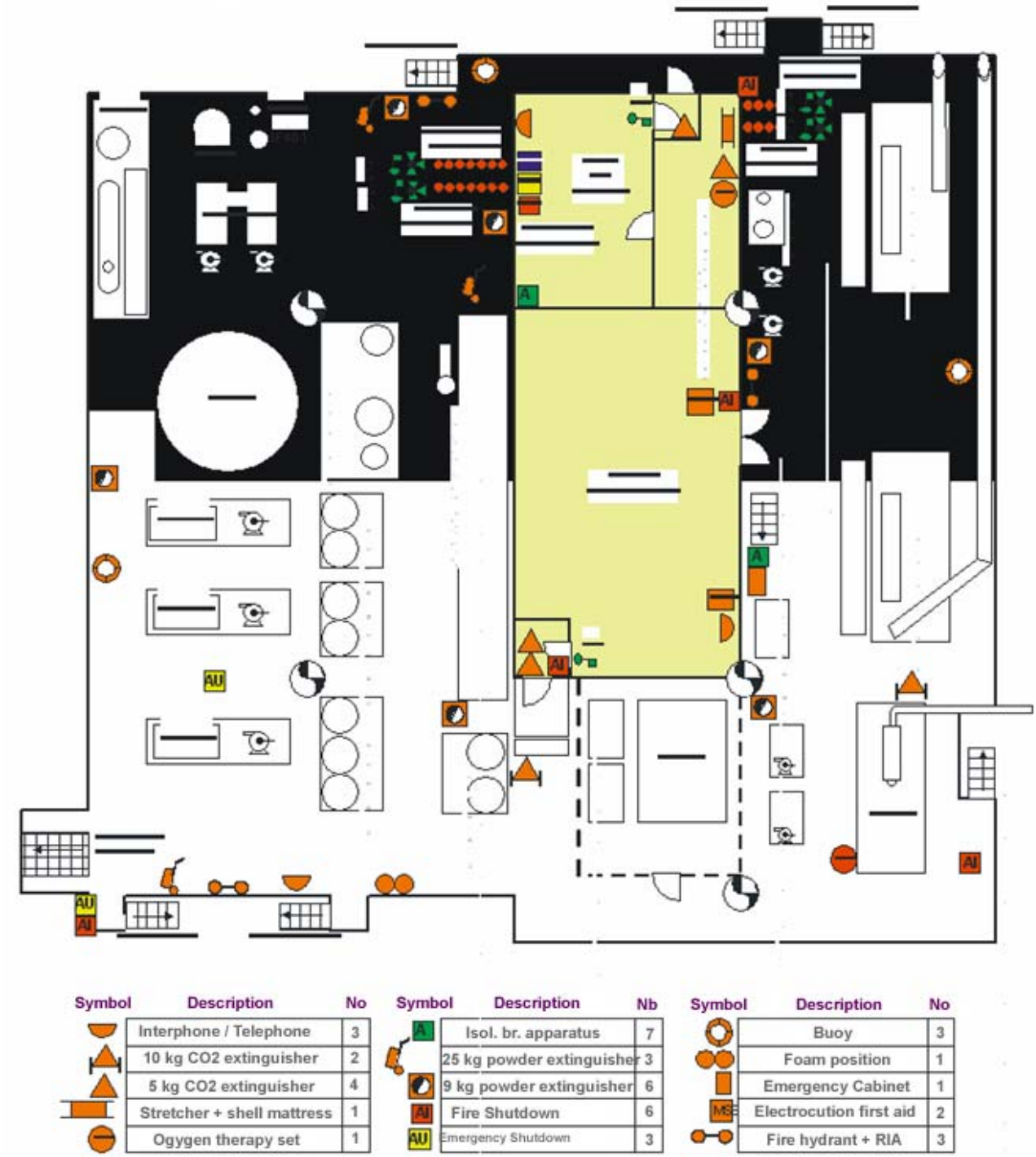
Table 5: Functional requirements

(1): Except if WV was voluntarily closed from CCR
(2): Automatic reset upon reset of ESD level may be envisaged from CCR
(3): As required by Process and Field Operations
(4): Recommended for the ESDVs and SDVs, that cannot be tested during scheduled equipment shutdown
(5): Interlocked with "permissive to EDP" signal

3.7. LOCATION OF EMERGENCY PUSH BUTTONS

Several emergency push buttons are installed on the various installation levels and in buildings. These push-buttons may have different functions according to their location (e.g.: ESD push-button; F&G push-button) and the operator must be aware of their location and function.

In addition, during phases of specific works (e.g.: SIMOPS), it is essential to notify all non-installation personnel working on the site to take extra care and not hit theses push-buttons with equipment.



Symbol	Description	No
	Interphone / Telephone	3
	10 kg CO2 extinguisher	2
	5 kg CO2 extinguisher	4
	Stretcher + shell mattress	1
	Ogygen therapy set	1

Symbol	Description	Nb
A	Isol. br. apparatus	7
	25 kg powder extinguisher	3
	9 kg powder extinguisher	6
AI	Fire Shutdown	6
AU	Emergency Shutdown	3

Symbol	Description	No
	Buoy	3
	Foam position	1
	Emergency Cabinet	1
MS	Electrocution first aid	2
	Fire hydrant + RIA	3

Figure 13: Example of the location of emergency push buttons

3.8. PADLOCKED VALVES

Some operating conditions require the padlocking of valves in the open or closed position according to needs:

- on the production systems during normal operating (e.g.: flare system)
- on the systems during maintenance or works (e.g.: suction/ref pump in maintenance)

The padlocking of these valves is subject to a very specific procedure drafted by each subsidiary. The implementation philosophy is standard:

- determination of the number and location of valves to be padlocked
- numbering of valves and declaration on the Work Permit
- padlocking of valves by the operator using the locking and display system applied in the subsidiary.
- reporting of information in the control room and recording in the shift log
- removal of the padlock after the operation
- reporting of information in the control room and recording in the shift log

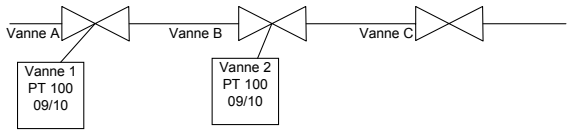
Figure 14: Example of a Work Permit appendix for valve padlocking

PERMIS 100 (09/10)
Demande la consignation des vannes A et B

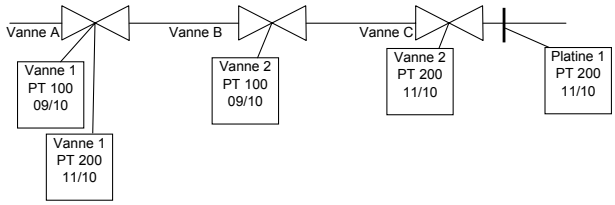
PERMIS 200 (11/10)
Demande la consignation des vannes A et C + l'installation d'une platine en aval de la vanne C

PERMIS 300 (12/10)
Demande la consignation des vannes A, B et C + l'installation d'une platine en amont de la vanne B et une platine en aval de la vanne C.

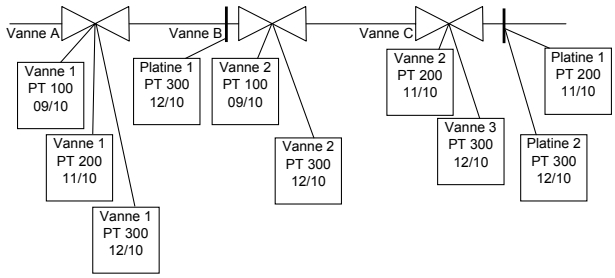
Après ouverture du permis 100, la situation sur le site est donc la suivante:



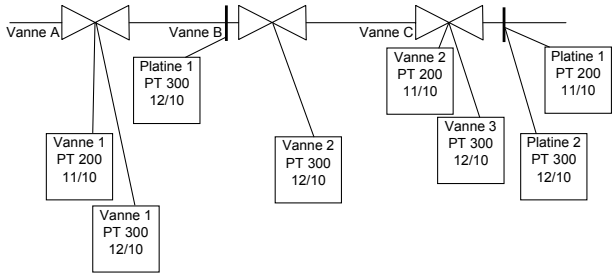
Après ouverture des permis 100 et 200, la situation sur le site est donc la suivante:



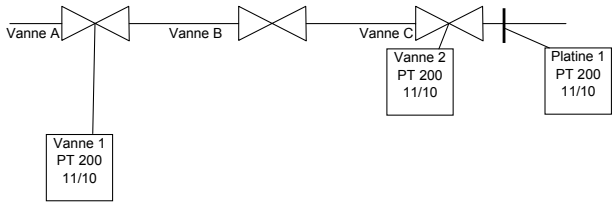
Après ouverture des permis 100, 200 et 300, la situation sur le site est donc la suivante:



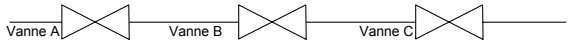
Après fermeture du permis 100, la situation sur le site est donc la suivante:



Après fermeture du permis 300, la situation sur le site est donc la suivante:



Après fermeture du permis 200, la situation sur le site est donc la suivante:



A few days ago, we received a case containing red "consigned valve", green "consigned valve" and green "plate" tags.

These tags must be used from now on, as defined below, for each consignment executed.

- One **red "consigned valve"** tag must be placed on **each valve** which is consigned as **closed**,
- One **green "consigned valve"** tag must be placed on **each valve** which is consigned as **open**,
- One **green "plate"** tag must be placed on **plate installed**.

The following must be indicated on each tag:
- The valve or plate number as mentioned in the consignment table (if applicable),
- The main permit number for the consignment,
- The date of consignment.

To this end, and to avoid "dirtying" the tags, the items of information mentioned above will be noted on selotape, which will be removed when the tags are definitively removed.

If **XX** (e.g. 2) different permits (not covered by a coverage permit) require the consignment of one same valve and/or the installation of a plate at the same location, then:
- XX (e.g. 2) "CONSIGNED VALVE" AND/OR "PLATE" TAGS MUST BE PLACED.

These tags will be placed at start of validity for a permit and removed when the corresponding permit expires.

This applies from receipt of the memo.

Figure 15: Extract from the internal memo concerning the use of consignment tags

3.9. Work Permit PROCEDURE

All works carried out on the site, other than "routine" tasks (defined in a controlled and widely distributed list), require a Work Permit:

- the permit issuer will ensure that all tasks and works are clearly specified and that operational risks are analysed,
- if required, specific additional permits (confined spaces, isolation of systems with energy supplies, digs, etc.) will be established by an authorized individual prior to executing works,
- to execute several separate tasks on one equipment, the manager must establish a link between all permits and documents involved,
- the site works supervisor will ensure that all prior conditions for the permit are satisfied before starting and during works,
- the Work Permit system formalizes the return to normal operations.

The Work Permit applied for entity operations must comply with the principles described in CR EP HSE 036.

3.9.1. Basic content of a Work Permit

The Work Permit procedure uses specific forms which include or formalize:

- a precise description of the task. This includes the identification of the area, unit or equipment concerned, and the resources required (team or provider, list of specific tools and equipment to be used), planned duration of work, etc.;
- the hazards identified for HSE. This concerns all risks relating to the intervention, the working area and adjoining installations, and any potential interference with other works or operations;
- precautionary and intervention measures (reduction and rescue) considered necessary and the checking of their effective implementation at start of work or at each change of shift;
- the references of all pertinent documents: reports from preparatory meetings, studies of risks and safety analyses, procedures, schematics, related permits, related inhibition/isolation certificates, etc.;
- the signature of the personnel involved in the preparation, consolidation and approval of the permit, with mention of the period of validity of the permit;

- (re)validation at each change of shift by personnel responsible for the execution and supervision of work;
- the provision of the installation or equipment required for the current shift prior to starting work and return to operations after works;
- measures as required to ensure the correct acceptance of work and the inspection of effective implementation;
- the official closing of the Work Permit.

3.9.2. Different types of Work Permits

The Work Permit procedure is based on different forms which are simple to distinguish.

Three types of forms are available:

- a standard form (the most frequently used form) known as the **Cold Work Permit**
- a specific form covering most hot works, known as the **Hot Work Permit.** The Hot Work Permit form must clearly differentiate hot works involving "**Bare flames**" from hot works with "**no bare flames**"
- a specific form covering all working in confined spaces, known as the **Confined Area Work Permit.**

Depending on the organizational context and the type of operations, entities may envisage developing and implementing other forms:

- specific Work Permit forms accompanied by specific check-lists for the identification of hazards/assessment of risks, to be used for certain types of work, instead of the "standard" Cold Work Permit forms
- a simplified form known as the **Work Slip**, to be used on an exceptional basis.

3.9.3. Field of application

Generally, all works require the use of one or several permits. The appropriate forms are selected on the basis of the type of the various tasks to be carried out, as described below.

3.9.3.1. Use of a Cold Work Permit

The Cold Work Permit **form** automatically applies to all types of works other than those covered by a more specific form.

3.9.3.2. Use of a Hot Work Permit

A **Hot Work Permit** is required instead of (or possibly in addition to) the Cold Work Permit if the work involves real or potential ignition sources, in particular:

- sources of bare flame, producing sparks or heat (blowpipe cutting, welding, grinding, etc.), unless they are used in workshops or other areas specifically designed for this purpose, and are not near to hazardous areas. The "Bare flame" box will be checked on the corresponding Hot Work Permit
- other sources of potential ignition (manual tools, equipment with no or no longer intrinsic safety etc.) located or used in or near to a hazardous area and for which isolation is not possible. The "No bare flames" box will be checked on the corresponding Hot Work Permit.

3.9.3.3. Use of a Confined Area Work Permit

A Confined Area Work Permit is required in addition to the Cold Work Permit or the Hot Work Permit if works involve activities in a pressurized space or tank, or require the presence of personnel in a confined area – i.e. an area with limited natural ventilation and where a hazardous atmosphere is present or could arise.

3.9.3.4. Use of other permits

For any other forms which may be produced and used in the entity, the Work Permit procedure will also specify the scope in an identical manner.

3.9.3.5. Exceptions – Works subject to Work Slips

Some works and tasks may only require a Work Slip, subject to the satisfaction of the following conditions:

- they must be executed on a regular basis, e.g. several times annually;
- the study of applicable risks clearly indicates that no unusual hazards or high risks are involved;
- they are subject to detailed procedures including the exhaustive identification of all potential hazards and applicable precautions, and these procedures have been carefully tested and declared satisfactory;
- they are executed by fully trained and experienced employees or personnel of service providers who are part of permanent site personnel;
- they are mentioned on the “list of works requiring a Work Slip" as duly approved by operational Management for the entity.

These permits apply, for example, to recurrent preventive maintenance tasks.

3.9.4. SIMOPS Work Permit system

With simultaneous operations, the Work Permit system is adapted when preparing the SIMOPS folder, in order to account for the highest risk level.

This particularly refers to:

- the lists of exceptions (works requiring Work Slips, works on the basis of verbal instructions) are revised and modified as necessary;
- interference between the different works is re-assessed according to SIMOPS conditions and any additional limitation or stricter rule – if considered necessary – will be clearly identified;
- the approval process for Work Permits is revised to comply with changes to company structure and the RSES responsibilities as defined for the duration of the SIMOPS. It must be checked that the RSES in charge of SIMOPS approves all permits and the associated daily log covering all works executed in the SIMOPS area.

3.9.5. Key personnel

The Work Permit procedure will identify each function or position playing a key role in the Work Permit procedure per site or activity area, and in particular:

- the **RSES** (or delegate), as defined in the CR EP HSE 035 rule;
- the **Operational Manager**, i.e. the person bearing the technical responsibility for the operations carried out and the installations of the entity on the operation site, if applicable;
- the **Permit Control Manager**, i.e. the person responsible for consolidating the preparation of permits, on the basis of the existing structure on the operation site (generally the permit coordinator, the Operational Manager or a designated representative, etc.);
- the **HSE Representative**, i.e. the person responsible for checking HSE elements in the various activities executed on the site (generally the HSE consultant or supervisor);
- the **Shift Leader**, i.e. the person responsible for the global and permanent monitoring and supervising of equipment located and works executed in a given area, for the entire duration of the shift;
- the **Applicant**, i.e. the person submitting the Work Permit application; in general the maintenance people or method engineer.
- the **Intervention Supervisor**, i.e. the person responsible for the execution of works (generally a leader or supervisor for the profession, the representative designated by a provider, etc.). Whenever possible, the Intervention Supervisor will be the person requesting the Work Permit, which will guarantee complete commitment to the preparation of works.

It is important to ensure that the following functions are never assigned to the same person:

- the RSES and the Operational Manager, and
- The Operational Manager, the Permit Control Manager and the HSE Representative.

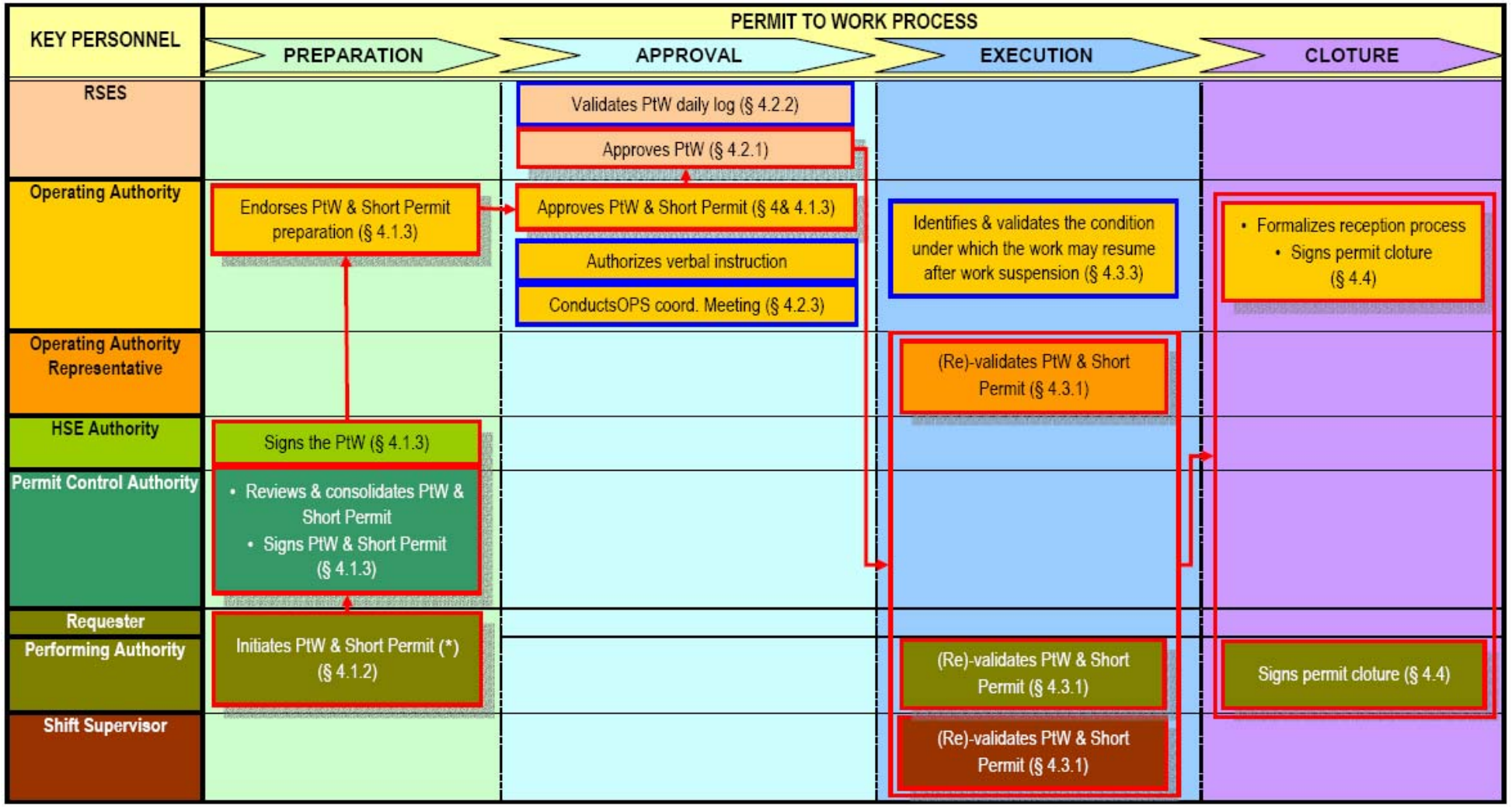


Figure 16: Roles of key personnel in the Work Permit process

(*) If possible, the Intervention supervisor will request the Work Permit

3.9.6. Work Permit process

The Work Permit process is governed by the following key principles:

- a permit will expire a maximum of 14 days after the date of approval or 7 days after the start of works. A Hot Work Permit for "bare flames" will expire a maximum of 7 days after the date of issue or 2 days after the start of works, whichever is earlier.
- a permit is subject to approval with adequate notice, i.e. a minimum of 24 hours prior to the start date for works;
- a permit is issued for one single job, i.e. a set of interdependent tasks which must be accomplished by one single shift in one single area. Consequently, the following situations cannot be subject to a common permit: tasks with no direct relation, jobs executed in 2 separate locations, activities by 2 different providers, etc.;
- each permit bears the references of all permits, documents and related inhibition/isolation certificates. In the same way, all inhibition/isolation certificates bear the references of all permits requiring these inhibitions or isolations. According to the requirements of the CR EP HSE 031 rule, a multiple inhibition/isolation system is implemented to ensure that no inhibition, no override mechanism, no electric consignment, process or mechanical isolation is deleted before all of the corresponding permits have been officially closed;
- standard or specific permits are approved both by the Operational Manager and the RSES. Work Slips are approved by the Operational Manager.

3.9.7. Permit application

The Work Permit Applicant must:

- describe the works to be executed,
- identify the risks,
- enclose all pertinent documents,
- sign and submit the permit for review and consolidation.

3.9.8. Review and consolidation

The Work Permit will be revised and consolidated during a specific meeting attended by all personnel involved in the preparation process.

The permit control manager must:

- check that all prerequisites have been satisfied (Cf. § 4.1.1) and check all elements included in or enclosed with the permits assessed (Cf. § 4.1.2);
- identify, with the assistance of the HSE Representative, all compensatory measures to be implemented and inform the applicant of any problem requiring specific attention;
- ensure that all inhibitions and isolation required for the execution of works are referenced and listed, and that the plating plan is enclosed with the permit, if applicable.

The preparatory phase will end when the Permit Control Manager and the HSE Representative have both signed the Work Permit and when the Operational Manager has approved the permit, thus validating the preparation.

3.9.9. Approval phase

3.9.9.1. Work Permit approval

The Operational Manager will submit all standard and specific Work Permits to the RSES for approval. The RSES is responsible for:

- checking that the approval procedure of the Work Permit has been satisfied, and particularly that all forms and documents presented have been correctly filled in;
- checking that all HSE elements have been duly accounted for, and particularly that the hazards identified and the compensatory measures recommended fully match the actual risk;
- determining the period of validity and adding other specific instructions if necessary;
- approving the permit.

3.9.9.2. Daily schedule/permit register

A register including all current permits is drafted and updated daily. This register mentions:

- the type of permit: cold work, hot work, work in confined areas, work slips, etc.;
- the working areas concerned;

- the status of each permit, i.e. work completed, under way/deferred to the next day, suspended, etc.

When it is necessary to view the different working areas in order to better assess the potential interference between jobs under way, permits are noted on the layout drawing.

3.9.10. Execution phase

3.9.10.1. (Re) validation at each change of shift

Prior to the start of works or when a new shift starts, all standard or specific permits and Work Slips are (re)validated. The following conditions must be satisfied:

- the approved Work Permit remains valid and all related documents are attached. The Work Permit is noted as "under way" for the shift in the last register review;
- actors are fully informed or notified of the various tasks to be carried out, the related hazards and the action to be taken, via site opening meetings or site meetings. The reports for these meetings will be enclosed with the permit as applicable;
- all resources are available, including equipment, the tools spare parts, consumables, etc, required for works and the supervisory team;,
- the conditions for works and installations are those accounted for or planned in the preparatory phase;
- the correct implementation and effectiveness of all compensatory measures (including inhibitions and isolations) are carefully checked and the corresponding boxes have been checked on the Work Permit;
- the Work Permit is (re)validated by the shift leader, the Representative of the Operational Manager (i.e. the shift responsible for the permanent or occasional supervision of works, as applicable) and the Intervention Supervisor.

3.9.10.2. Permit management during the execution of works

In addition to the original form, at least one copy of each Work Permit will be systematically printed.

While the works are under way, the original permit and related documents will be displayed near to the working area. A copy will be kept in the permit control room (generally the control room, if any, the Shift leader's office, etc.), in the register for current permits.

Should the works be interrupted, the original form will be returned to the shift leader, who will ensure that the document is kept with the copies in the permit control room, until future use.

3.9.10.3. Suspension of works

A permit must be suspended:

- in case of a general alarm or a specific instruction from the shift leader, the operational Manager or the RSES;
- if risk control can no longer be satisfactorily maintained and/or one of the requirements of the permit is no longer satisfied.

Following the suspension of works, the conditions for the continuation of works are identified and validated by the Operational Manager. A minimum of the re-assessment of working conditions and exhaustive revalidation is required at changes in shift.

3.9.10.4. Closing phase

The permit is closed when the works have been completed, the period of validity has expired or the works have been suspended and revalidation is not authorized (in the two latter cases, works can only be continued if a new permit has been requested and issued, according to all stages in the process).

When works have been completed, the acceptance process will be formalized subject to the responsibility of the Operational Manager and include the following, as applicable and appropriate:

- tests, controls and verifications including the visiting and visual inspection of the works area;
- identification of inhibitions and isolations to be maintained and those to be removed before re-commissioning equipment;
- re-start-up instructions and related precautions;
- off and/or on-load operating tests;
- availability or acceptance certificate, possibly mentioning reserves.

The closing of the permit requires the signature of the Intervention supervisor and that of the Operational Manager.

Following closing, the original permit, appendices and other related documents will be archived for a minimum of one year.

3.9.11. Examples of Work Permits

The following pages show a few standard forms as recommended for use in the Work Permit procedure.

We specify that the pre-defined lists of hazards and precautions inserted in forms are given for information only. These lists only cover the most frequent types of incidents and cannot be considered as exhaustive. This must be remembered when identifying hazards or considering the precautions to be taken.

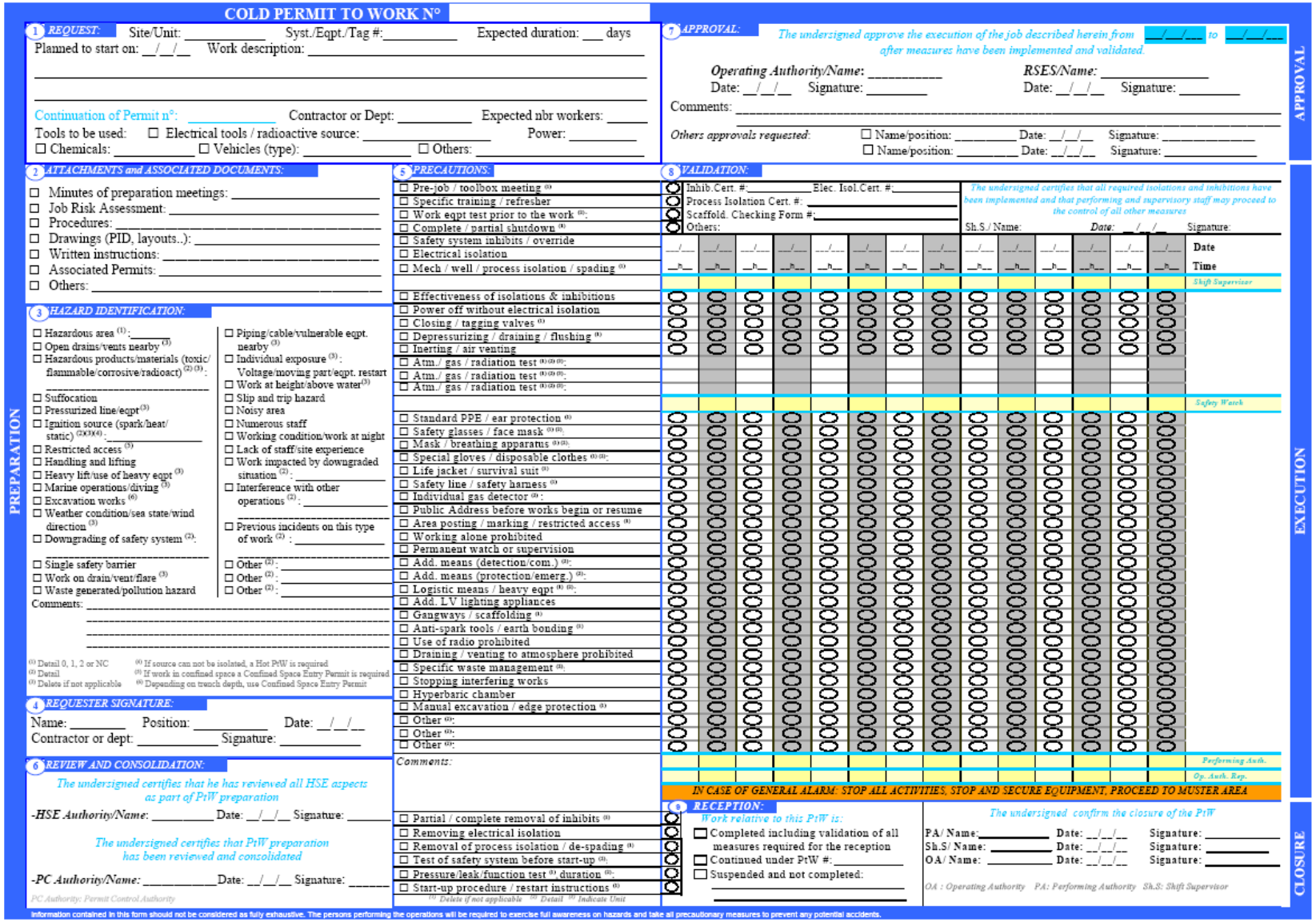


Figure 17: Cold Work Permit

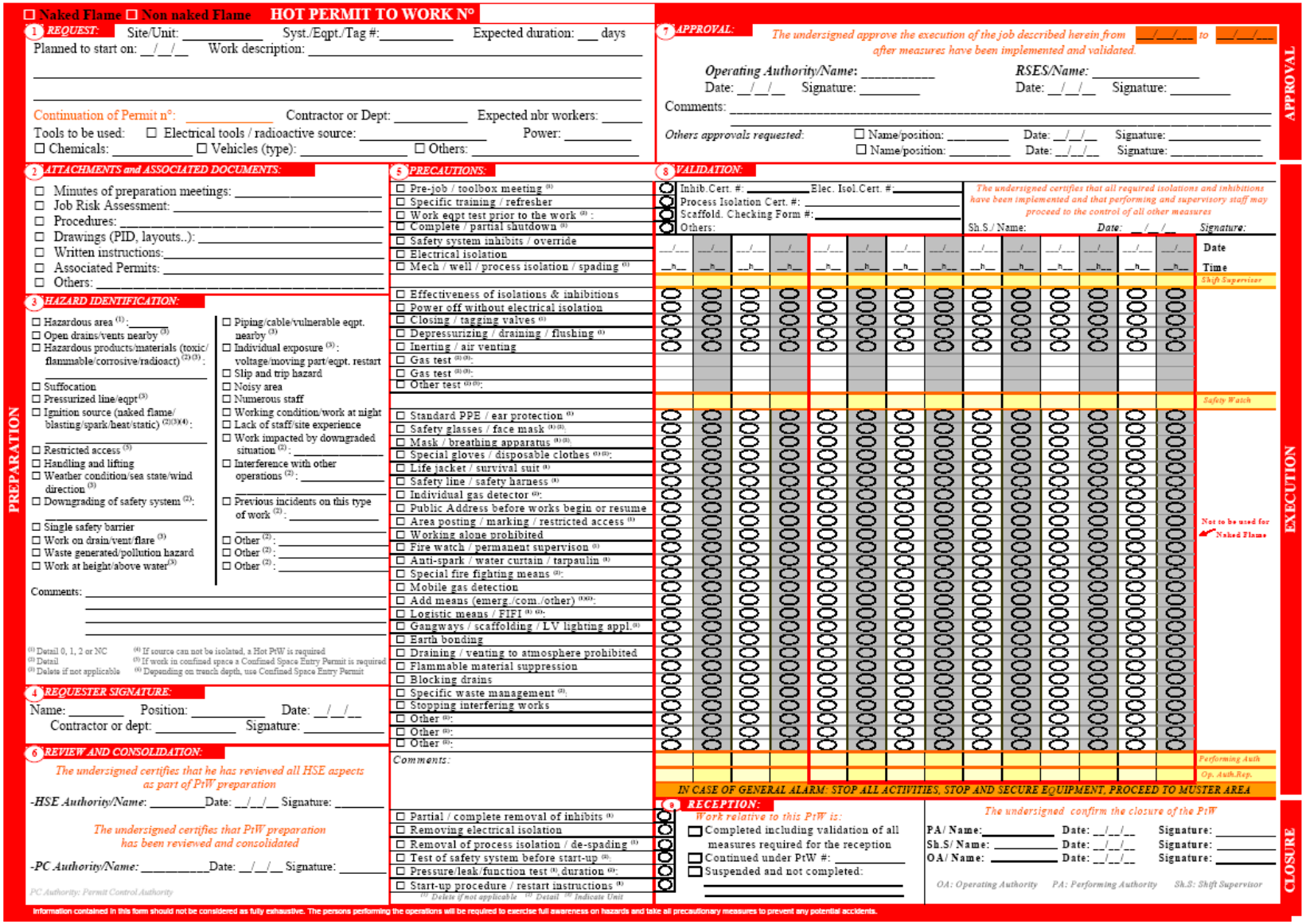


Figure 18: Hot Work Permit

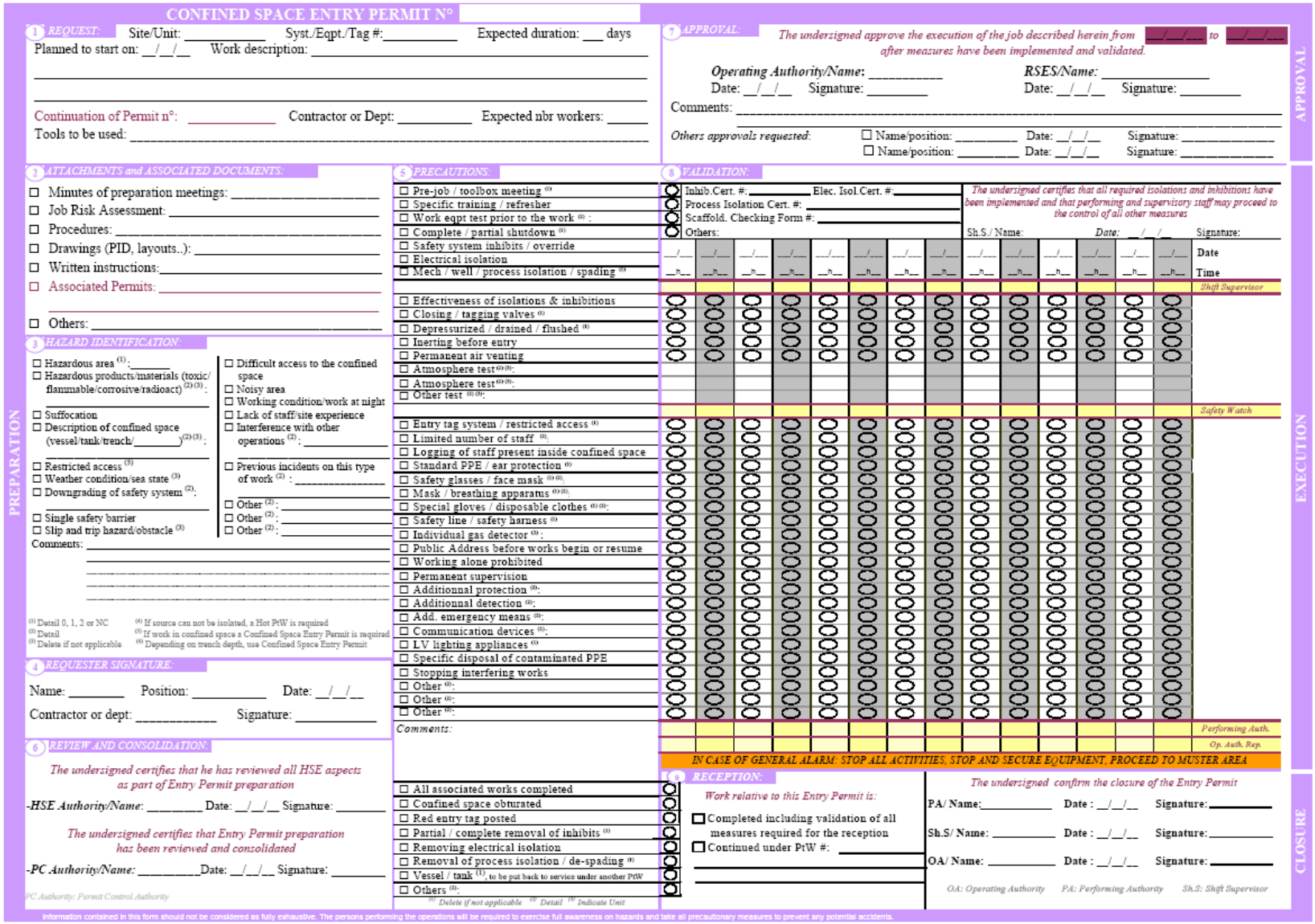
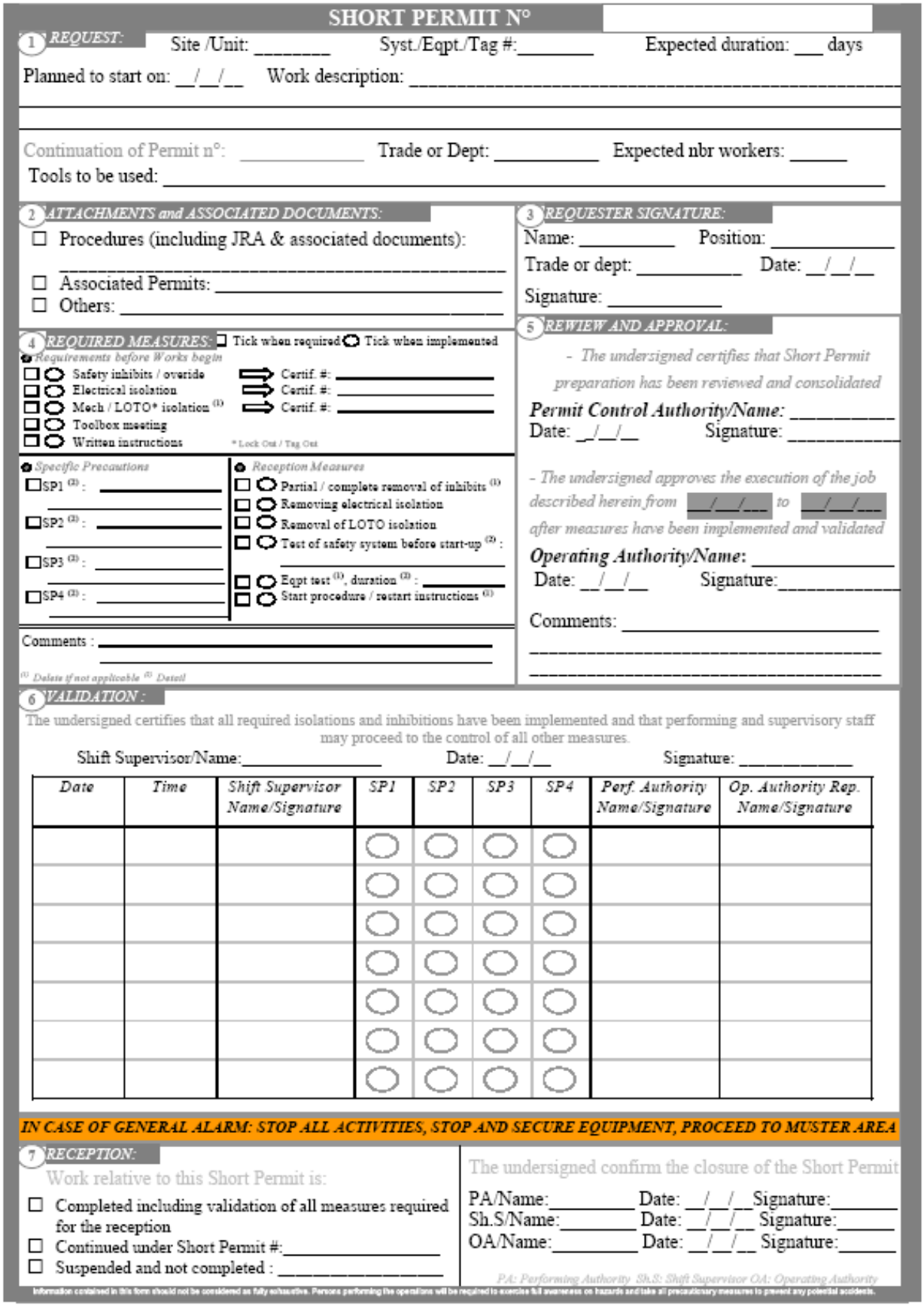


Figure 19: Confined Area Work Permit



SHORT PERMIT N°

1 REQUEST: Site /Unit: _____ Syst./Eqpt./Tag #:_____ Expected duration: ___ days

Planned to start on: __/__/__ Work description: _____

Continuation of Permit n°: _____ Trade or Dept: _____ Expected nbr workers: _____

Tools to be used: _____

2 ATTACHMENTS and ASSOCIATED DOCUMENTS:

☐ Procedures (including JRA & associated documents):

☐ Associated Permits: _____

☐ Others: _____

3 REQUESTER SIGNATURE:

Name: _____ Position: _____

Trade or dept: _____ Date: __/__/__

Signature: _____

4 REQUIRED MEASURES: ☐ Tick when required ○ Tick when implemented

Requirements before Works begin

☐ ○ Safety inhibits / override ⇨ Certif. #: _____

☐ ○ Electrical isolation ⇨ Certif. #: _____

☐ ○ Mech / LOTO* isolation (1) ⇨ Certif. #: _____

☐ ○ Toolbox meeting

☐ ○ Written instructions

* Lock Out / Tag Out

Specific Precautions

☐ SP1 (2) : _____

☐ SP2 (2) : _____

☐ SP3 (2) : _____

☐ SP4 (2) : _____

Reception Measures

☐ ○ Partial / complete removal of inhibits (1)

☐ ○ Removing electrical isolation

☐ ○ Removal of LOTO isolation

☐ ○ Test of safety system before start-up (2) : _____

☐ ○ Eqpt test (1), duration (2) : _____

☐ ○ Start procedure / restart instructions (1)

Comments : _____

(1) Delete if not applicable (2) Detail

5 REWIEW AND APPROVAL:

- The undersigned certifies that Short Permit preparation has been reviewed and consolidated

Permit Control Authority/Name: _____

Date: __/__/__ Signature: _____

- The undersigned approves the execution of the job described herein from __/__/___ to __/__/___ after measures have been implemented and validated

Operating Authority/Name: _____

Date: __/__/__ Signature: _____

Comments: _____

6 VALIDATION :

The undersigned certifies that all required isolations and inhibitions have been implemented and that performing and supervisory staff may proceed to the control of all other measures.

Shift Supervisor/Name: _____ Date: __/__/__ Signature: _____

Date	Time	Shift Supervisor Name/Signature	SP1	SP2	SP3	SP4	Perf. Authority Name/Signature	Op. Authority Rep. Name/Signature
			○	○	○	○		
			○	○	○	○		
			○	○	○	○		
			○	○	○	○		
			○	○	○	○		
			○	○	○	○		
			○	○	○	○		

IN CASE OF GENERAL ALARM: STOP ALL ACTIVITIES, STOP AND SECURE EQUIPMENT, PROCEED TO MUSTER AREA

7 RECEPTION:

Work relative to this Short Permit is:

☐ Completed including validation of all measures required for the reception

☐ Continued under Short Permit #: _____

☐ Suspended and not completed : _____

The undersigned confirm the closure of the Short Permit

PA/Name: _____ Date: __/__/__ Signature: _____

Sh.S/Name: _____ Date: __/__/__ Signature: _____

OA/Name: _____ Date: __/__/__ Signature: _____

PA: Performing Authority Sh.S: Shift Supervisor OA: Operating Authority

Information contained in this form should not be considered as fully exhaustive. Persons performing the operations will be required to exercise full awareness on hazards and take all precautionary measures to prevent any potential accidents.

Figure 20: Short Permit

3.9.12. Special precautions

Some types of works require special precautions as described below.

3.9.12.1. Hot work with a "bare flame"

All entities will take all action possible to eliminate hot work, particularly hot work with a "bare flame", executed simultaneously in one area. The quantity and duration of these works must be kept as low as possible, should this restriction be clearly defined in the entity procedure. Under all circumstances, hot work should be avoided in transitory phases, e.g. when stopping or starting installations.

If hot work cannot be avoided, the safest conditions and times for the execution of works must be defined. The gas detection operator and the fire safety officer must be near to the work area, and fire fighting equipment must be available and ready-for-use throughout the entire duration of works.

3.9.12.2. Working in confined areas

Working in confined areas is only authorized if no other method is possible.

In this case, the following requirements apply:

- identify all potential risks, such as the atmosphere in the confined space, any defects in prevention resources (isolation, breathing equipment, etc.),…
- define the necessary prevention resources: place signs at the entrance to the confined space, restrict entry to authorized individuals only, wear the appropriate personal protective equipment, use adequate lighting, always ensure that the atmosphere has been analysed and results recorded, etc.
- establish a Work Permit according to the procedure described in CR EP HSE 036 including a detailed evacuation and rescue plan for emergency situations and the appropriate isolation certificate,
- regularly, and whenever necessary, analyse the atmosphere and record results, place an attendant near to the working area and ensure that emergency equipment is in place and ready-for-use for the entire duration of works. The attendant must have a direct means of communication with the control room and, if necessary, with workers inside the confined space.

3.9.12.3. Work on live systems

Works requiring the isolation of sources of energy must not be started before the following has been implemented:

- identification of all sources of energy to be isolated, the isolation method and the discharge method for stored energy, including the analysis of risks should the isolation device fail,
- establishment of a detailed plan/schema for the isolation on the updated document (generally the “execution compliant" document). This includes the identification of the isolation devices and, if appropriate, the technical specifications for each isolation device,
- discharge of stored energy,
- checking of the effective isolation of sources of energy (if possible with measurements) prior to each operation and during successive operations.

3.9.12.4. Excavation

It is necessary to assess the situation for the working area when preparing works. This involves the possibility of earthwork and all aspects relating to work in confined areas, and the identification, checking, on-site confirmation and isolation (if required) of all underground hazards (e.g.: pipes, cables, etc.).

The verification of underground hazards will be carried out by the appropriate tradesmen (e.g.: pipeline staff, electric staff, etc.).

3.9.12.5. Overhead work

Prior to starting works, ensure that:

- workers are wearing the appropriate fall-prevention equipment, as defined in CR EP HSE 062. Climbers must hold a certificate issued by the IRATA (Industrial Rope Access Trade Association), the SPRAT (Society of Professional Rope Access Technicians) or an equivalent body,
- scaffolding is systematically considered as a temporary installation and is designed, installed, inspected and certified on a regular basis by competent personnel. The rig up is than approved and tagged.

3.9.12.6. Lifting

The precautions to be taken for lifting operations are indicated in CR EP HSE 043.

3.9.12.7. Diving

Diving operations are governed by CR EP LSO 400. Prior to diving, all risks relating to the diving operations to be carried out – including risks involved in the execution of simultaneous activities on the installations or near-by – are assessed and the necessary precautions defined.

It is also necessary to assess and coordinate the interface between the various actors, e.g. by establishing an interface document and designating, if necessary, a representative of the company, between the diving service provider and the company, or by establishing a communications link between the diving service provider and the ship or barge captain, if the ship or barge is near to the operating site, but is not subject to the responsibility of the diving service provider.

3.9.12.8. SIMOPS

The presence of simultaneous operations (SIMOPS) is likely to increase the level of risk. Consequently, prior to starting SIMOPS work, the following must be carried out:

- an inspection of the installation in order to identify the hazards and define the precautions to be taken as well as preparation tasks to be carried out prior to starting SIMOPS,
- identification and execution of an exhaustive assessment of all risks and the implementation of all recommendations further to the assessment,
- identification of restrictions (which activities are or are not authorized) by drafting an interference matrix between the SIMOPS and defining the precautions to be taken,
- designation of the RSES for SIMOPS, as stipulated in CR EP HSE 035,
- establishment of a SIMOPS folder subject to the responsibility of the RSES designated for the SIMOPS,
- obtaining of the formal approval of operational management for the entity,
- revision and adaptation of all HSE risk control procedures, such as the Work Permit procedure, if required,
- formalization of the transfer of responsibilities between the "Site" RSES and the "SIMOPS" RSES. This concerns the availability of installations/equipment, the HSE folder for the installations concerned, the updated copy of all plans and general schematics mentioned in the folder, the new definition of the regularity of controls/tests required during SIMOPS.

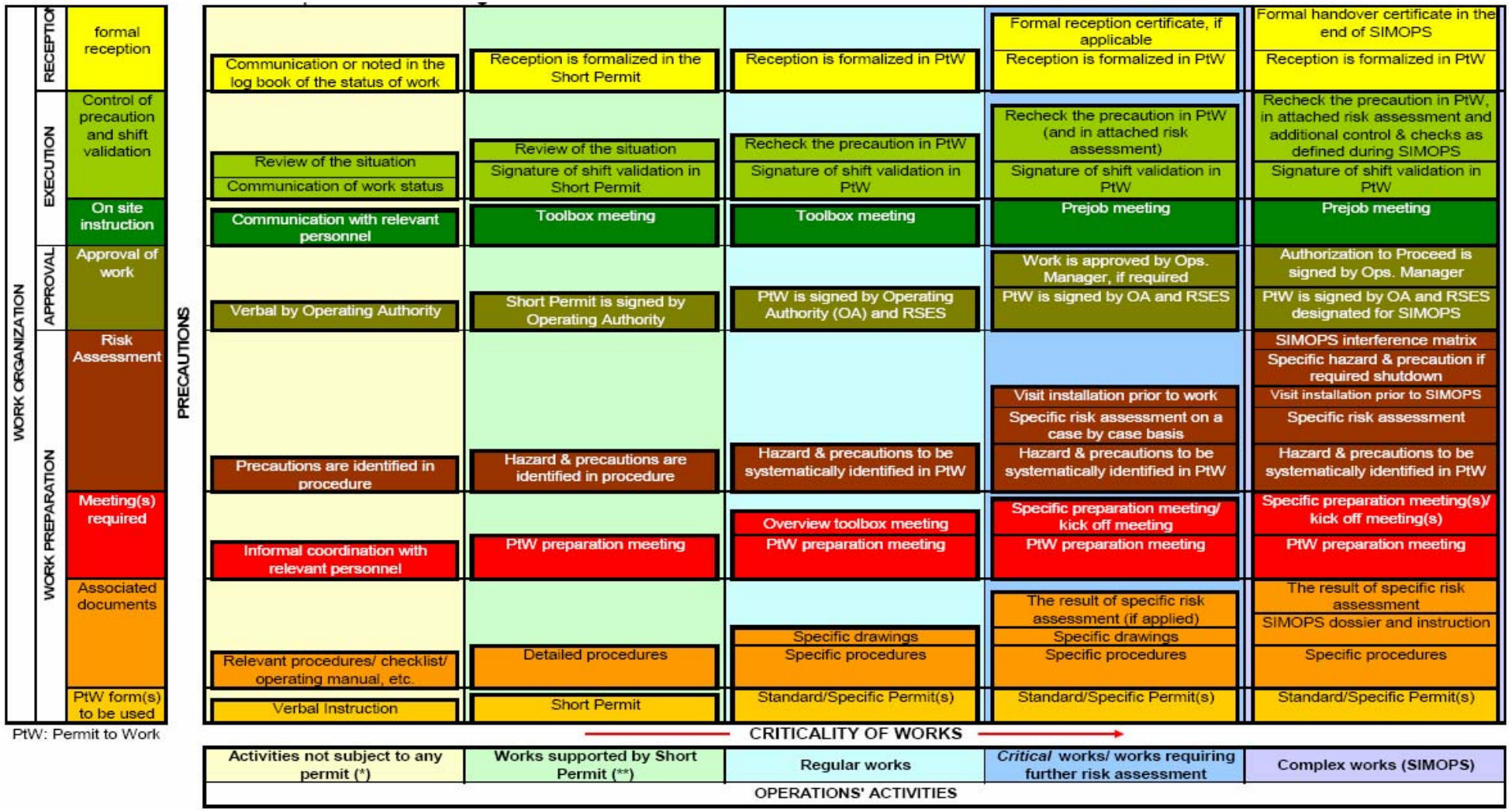


Figure 21: Precautions to be taken in each phase of the organization of works

(*) Conditions according to CR EP HSE 036 Work Permit; E.g.: routine operation. (**) Conditions according to CR EP HSE 036 Work Permit; E.g.: routine maintenance

3.10. INCOMPATIBLE WORKS

The operator may rapidly find himself in hazard due to the multitude of "traditional" product operations and the works carried out on installations if they are concomitant.

To give an example, all hot works carried out on an installation are strictly regulated, or even subject to cancellation, as they could trigger an event if executed simultaneously to operations which may involve the controlled release of gas into the atmosphere.

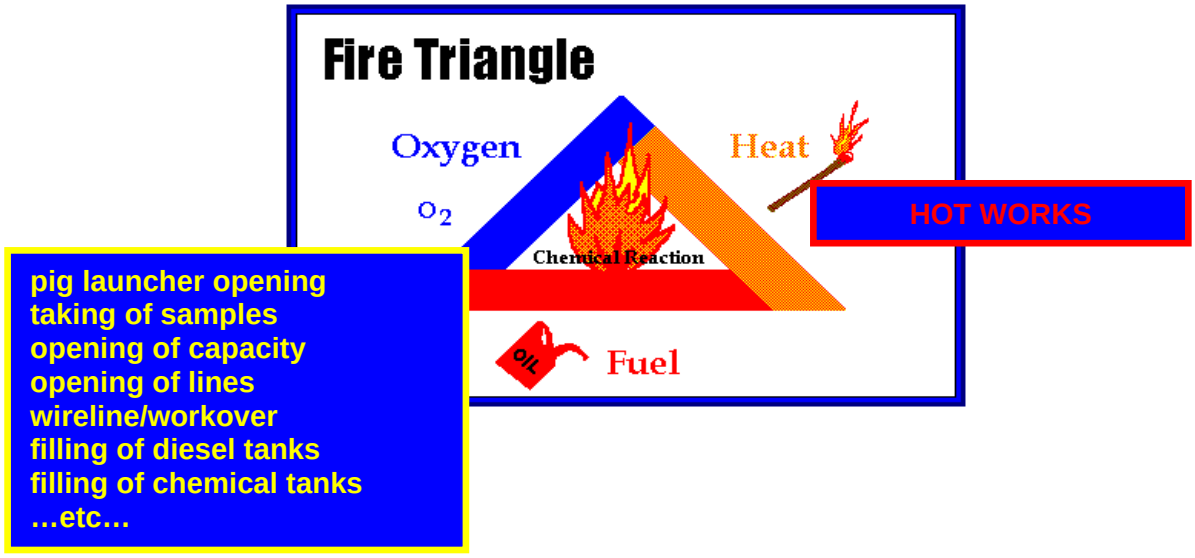


Figure 22: Incompatible works

Other types of works are also incompatible with some operations, e.g.:

- Underwater works and lifting operations
- Underwater works and FIRE/INJECTION water lifting operations.

The Operational Manager must strictly analyse Work Permits and Work Slips in order to schedule works on the basis of daily operations and to avoid any simultaneous works which could endanger personnel and equipment.

The operator must ensure that no other operation can hinder the correct execution of works prior to authorizing works to start, for the entire working area. No doubt may remain. Should this not be the case, works must not be started and hierarchy must immediately be notified.

Attention, hazard may come from a higher or lower level.

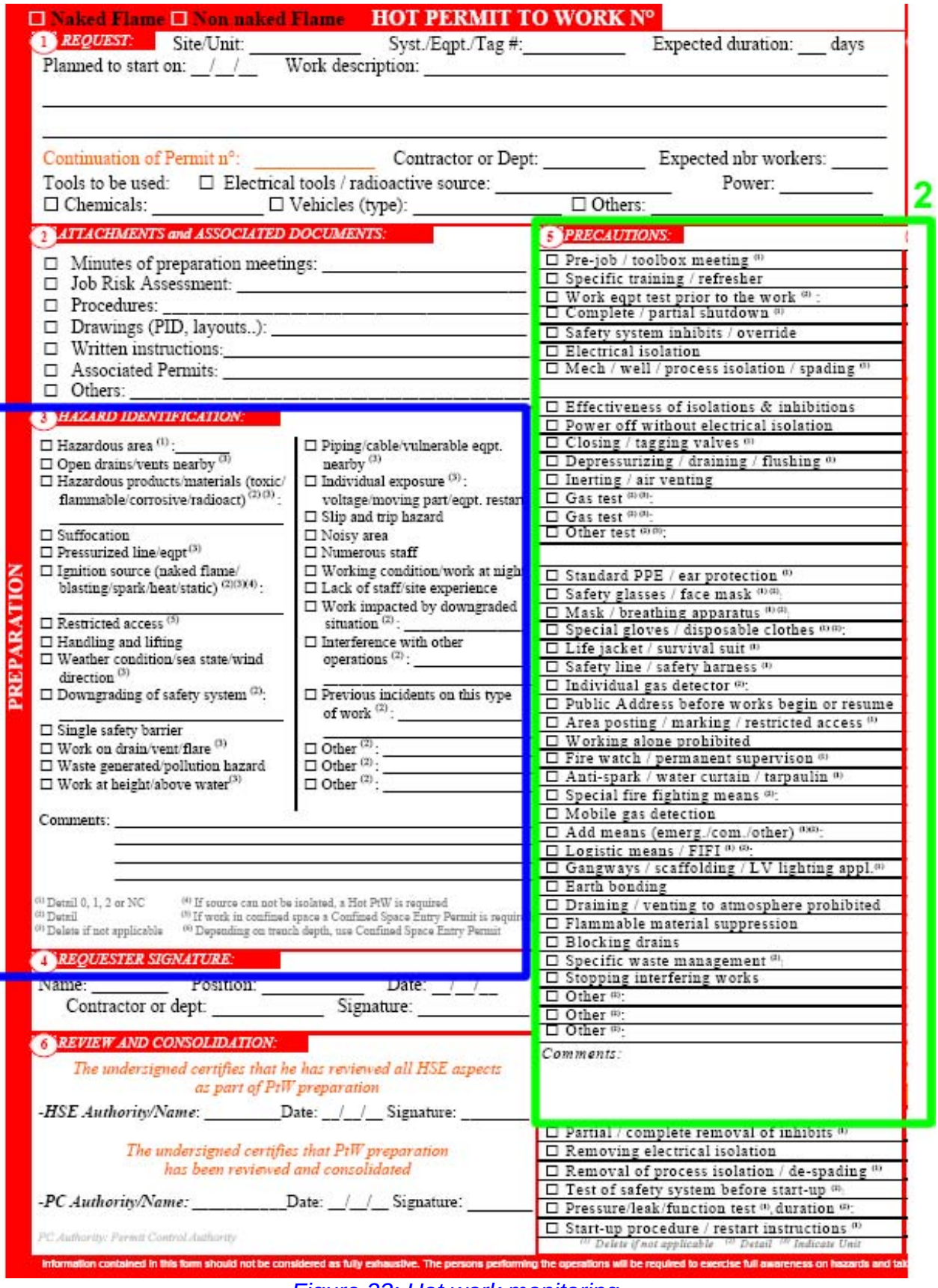
3.11. SURVEILLANCE OF HOT WORKS (ESSENTIAL RULES)

Hot works are known to be a source of many incidents.

The operator in charge of monitoring hot works for the installation must be very strict in terms of:

- the complete reading of the Work Permit and appendices.
- the identification of risks (1): a site analysis must be repeated prior to starting works and hierarchy must be immediately contacted if new risks are present.
- the application of the compensatory measures defined in the Hot Work Permit (2):
  - take the time to correctly install prevention equipment: area protection, shields, flame retardant tarpaulin, etc.;
  - take the time to correctly install detection equipment: gas detectors, detection markers, permanent visibility of the operation;
  - take the time to correctly install protective equipment: extinguishers, battery fire nozzles, FIFI (FIre FIghting) positioning.
- ensuring good and permanent communication:
  - with the control room;
  - with the company responsible for works (ensure that they have fully understood: site safety rules, the work to be carried out, compliance with the Work Permit procedure).

The main essential rule is that the start of hot work will only be authorized if the atmospheric conditions are satisfactory: NO EXPLOSIVE ATMOSPHERE PRESENT.



☐ Naked Flame ☐ Non naked Flame **HOT PERMIT TO WORK N°**

1 *REQUEST:* Site/Unit: _____ Syst./Eqpt./Tag #: _____ Expected duration: ___ days

Planned to start on: __/__/__ Work description: _____

Continuation of Permit n°: _____ Contractor or Dept: _____ Expected nbr workers: _____

Tools to be used: ☐ Electrical tools / radioactive source: _____ Power: _____

☐ Chemicals: _____ ☐ Vehicles (type): _____ ☐ Others: _____

1

2 *ATTACHMENTS and ASSOCIATED DOCUMENTS:*

- ☐ Minutes of preparation meetings: _____
- ☐ Job Risk Assessment: _____
- ☐ Procedures: _____
- ☐ Drawings (PID, layouts..): _____
- ☐ Written instructions: _____
- ☐ Associated Permits: _____
- ☐ Others: _____

PREPARATION

3 *HAZARD IDENTIFICATION:*

- ☐ Hazardous area (1): _____
- ☐ Open drains/vents nearby (3)
- ☐ Hazardous products/materials (toxic/ flammable/corrosive/radioact) (2)(3): _____
- ☐ Suffocation
- ☐ Pressurized line/eqpt (3)
- ☐ Ignition source (naked flame/ blasting/spark/heat/static) (2)(3)(4): _____
- ☐ Restricted access (5)
- ☐ Handling and lifting
- ☐ Weather condition/sea state/wind direction (3)
- ☐ Downgrading of safety system (2): _____
- ☐ Single safety barrier
- ☐ Work on drain/vent/flare (3)
- ☐ Waste generated/pollution hazard
- ☐ Work at height/above water (3)
- ☐ Piping/cable/vulnerable eqpt. nearby (3)
- ☐ Individual exposure (3): voltage/moving part/eqpt. restar
- ☐ Slip and trip hazard
- ☐ Noisy area
- ☐ Numerous staff
- ☐ Working condition/work at nigh
- ☐ Lack of staff/site experience
- ☐ Work impacted by downgraded situation (2): _____
- ☐ Interference with other operations (2): _____
- ☐ Previous incidents on this type of work (2): _____
- ☐ Other (2): _____
- ☐ Other (2): _____
- ☐ Other (2): _____

Comments: _____

(1) Detail 0, 1, 2 or NC
(2) Detail
(3) Delete if not applicable
(4) If source can not be isolated, a Hot PtW is required
(5) If work in confined space a Confined Space Entry Permit is requir
(6) Depending on trench depth, use Confined Space Entry Permit

4 *REQUESTER SIGNATURE:*

Name: _____ Position: _____ Date: __/__/__

Contractor or dept: _____ Signature: _____

6 *REVIEW AND CONSOLIDATION:*

The undersigned certifies that he has reviewed all HSE aspects as part of PtW preparation

-HSE Authority/Name: _____ Date: __/__/__ Signature: _____

The undersigned certifies that PtW preparation has been reviewed and consolidated

-PC Authority/Name: _____ Date: __/__/__ Signature: _____

PC Authority: Permit Control Authority

2

5 *PRECAUTIONS:*

- ☐ Pre-job / toolbox meeting (1)
- ☐ Specific training / refresher
- ☐ Work eqpt test prior to the work (2) :
- ☐ Complete / partial shutdown (1)
- ☐ Safety system inhibits / override
- ☐ Electrical isolation
- ☐ Mech / well / process isolation / spading (1)
- ☐ Effectiveness of isolations & inhibitions
- ☐ Power off without electrical isolation
- ☐ Closing / tagging valves (1)
- ☐ Depressurizing / draining / flushing (1)
- ☐ Inerting / air venting
- ☐ Gas test (1)(2):
- ☐ Gas test (1)(2):
- ☐ Other test (1)(2):
- ☐ Standard PPE / ear protection (1)
- ☐ Safety glasses / face mask (1)(2)
- ☐ Mask / breathing apparatus (1)(2)
- ☐ Special gloves / disposable clothes (1)(2):
- ☐ Life jacket / survival suit (1)
- ☐ Safety line / safety harness (1)
- ☐ Individual gas detector (2):
- ☐ Public Address before works begin or resume
- ☐ Area posting / marking / restricted access (1)
- ☐ Working alone prohibited
- ☐ Fire watch / permanent supervison (1)
- ☐ Anti-spark / water curtain / tarpaulin (1)
- ☐ Special fire fighting means (2):
- ☐ Mobile gas detection
- ☐ Add means (emerg./com./other) (1)(2):
- ☐ Logistic means / FIFI (1) (2):
- ☐ Gangways / scaffolding / LV lighting appl. (1)
- ☐ Earth bonding
- ☐ Draining / venting to atmosphere prohibited
- ☐ Flammable material suppression
- ☐ Blocking drains
- ☐ Specific waste management (2)
- ☐ Stopping interfering works
- ☐ Other (1):
- ☐ Other (1):
- ☐ Other (1):

Comments:

- ☐ Partial / complete removal of inhibits (1)
- ☐ Removing electrical isolation
- ☐ Removal of process isolation / de-spading (1)
- ☐ Test of safety system before start-up (1)
- ☐ Pressure/leak/function test (1), duration (2):
- ☐ Start-up procedure / restart instructions (1)

(1) Delete if not applicable (2) Detail (3) Indicate Unit

Information contained in this form should not be considered as fully exhaustive. The persons performing the operations will be required to exercise full awareness on hazards and tak

Figure 23: Hot work monitoring

3.11.1. Ignition and explosion limits.

The **lower flammable limit** or lower explosive limit (LFL or LEL) for a gas or vapour is the minimum concentration above which propagation of flames occurs.

The **high flammable limit** or high explosive limit (HFL or HEL) for a gas or vapour in the air is the maximum concentration below which propagation of flames occurs.

The **flammability range** (Z.I.) is the hazardous range. This includes all values between the LEL and the HEL (yellow values).

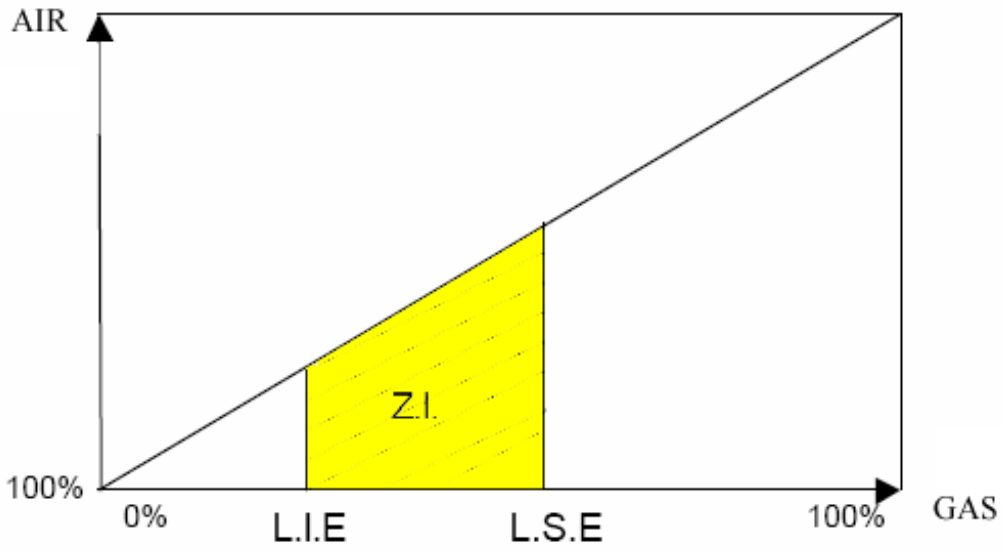


Figure 24: Flammability range

To cause an explosion, three elements are required:

- Oxygen (in the air)
- An flammable substance (fuel) which may be a gas (methane, acetylene), a liquid (petrol, solvent) or a solid (sulphur, sawdust)
- A source of flammability with sufficient energy (electric arc, spark) and/or a rise in temperature

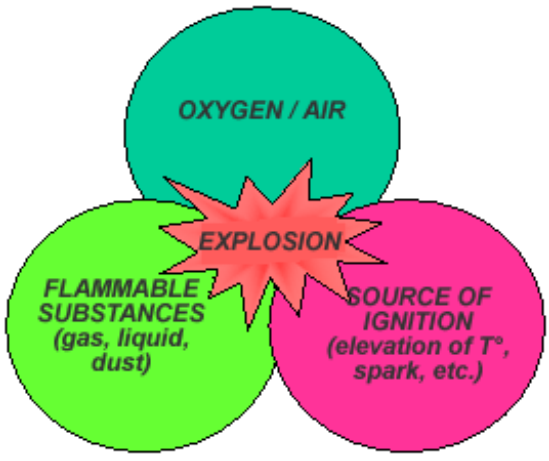


Figure 25: The elements required to cause an explosion

GROUP	NAMES	FORMULA	LEL%	HEL%	DENSITY
Saturated hydrocarbons	Methane Ethane Propane Butane	CH_4 C_2H_2 C_3H_8 C_4H_{10}	5 3.1 2.1 1.86	15 12.45 10.1 8.41	0.6 1 1.6 2.1
Hydrogen	Hydrogen	H2	4	74.2	0.07
Alkene	Ethylene Propylene Butadiene Butylene Pentene	C_2H_4 C_3H_6 C_4H_6 C_4H_8 C_5H_{10}	2.75 2 2 1.98 1.65	28.6 11.1 11.5 9.65 7.7	1 1.5 1.9 1.9 2.4
Aromatics	Benzene Toluene Styrene	C_6H_6 C_7H_8 C_8H_8	1.35 1.27 1.1	6.75 6.75 6.1	2.8 3.1 3.6
Oxides	Carbon Ethylene Propylene	CO C_2H_4O C_3H_6O	12.5 3 2	74.2 80 22	1 1.5 2
Miscellaneous compounds	Ammoniac Turpentine Gas oil Petrol Kerosene White spirit	$NH_3C_{10}H_{16}$	15.5 0.8 6 1.4 0.7 1.5	26.2 13.5 7.5 5 6.5	0.6 4.7 3 - 4 4.5

Table 6: Examples of the explosion limits in an atmospheric environment with oxygen presence of 21%

Influential factors include:

- Pressure: If pressure increases, risk will generally tend to increase (more gas in an equal volume), therefore the LEL% will drop and the flammability range will increase;
- Temperature: An increase in temperature will increase the flammability range;

- Oxygen: an increase in O_2 content will not modify the LEL% for a gas, but will considerably increase the HEL%

Example: Methane CH4

- HEL/ air = 15%
- HEL/O_2 = 61%

3.11.1.1. Examples of the explosion limits.

The **open container**:

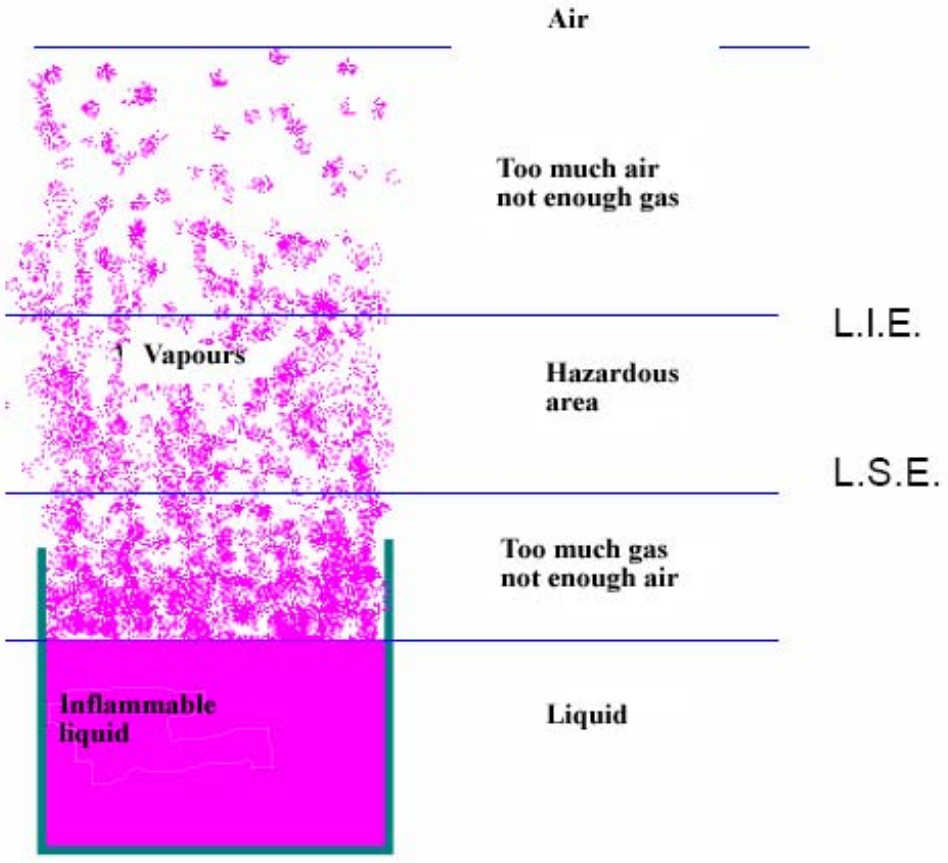


Figure 26: Open container

The **ignition engine**:

It is possible to compare the situation with petrol engines. If not enough petrol vapours are present in the cylinders, the engine will not start. The mixture is too poor as the concentration is less than the LEL (i.e. 1.4% for petrol). On the other hand, the engine will

not start if the petrol vapour content exceeds the HEL (7.6% with petrol). In this case the "engine is flooded" (mixture too rich).

It is important to note that the limits for explosion are normally given for the mixture of gas or vapours with air. The mixture with a combustive gas generally extends the explosion range (specifically, it increases the HEL) and increases the power of the explosion. To continue the comparison with a petrol engine, consider the effect of adding nitrous oxide in the admission phase of racing cars.

Nitrous oxide is a combustive gas and oversupplies the engine, increasing its power.

3.11.2. Explosimeter

Explosimeters are calibrated for a given gas, with specific explosive limits for this gas.



Figure 27: An explosimeter

3.11.2.1. Precautions when using an explosimeter.

The calibration gas must be identified.

The explosimeter must be started in an area free of fuel gas.

The batteries or fuel cells must be checked.

An explosimeter operating on catalytic oxidation does not indicate the presence of fuel gas in an inert environment, and must not be used in an O_2 enriched atmosphere.

An explosimeter operating on catalytic oxidation detects vapours and fuel gas (no dust from oils, carbon or cereals, etc.).

Do not press on the centre of the measuring cell, it could be damaged.

Check that the case orifices in front of the sensors are clean.

Catalyst inhibitors (Chlorinated or sulphur compounds, silicones, lead, tetraethyl, etc.) damaging the explosimeter cell.

Water will damage the explosimeter.

When taking measurements, the geometrical characteristics of the room must be accounted for, e.g. presence of high or low points, gutters,…, where gas may accumulate.

When taking measurements, the type of gas must be accounted for (density).

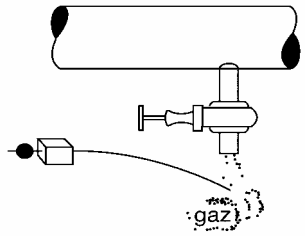
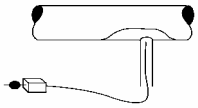
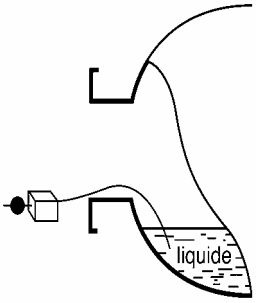
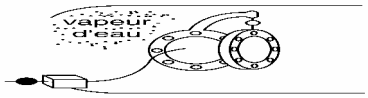
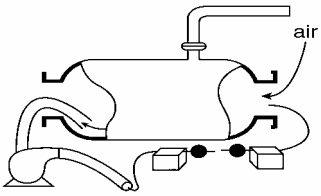
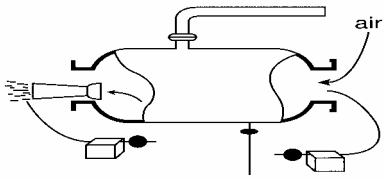
 Significant gas leak = No oxygen = No detection	 No detection in a container bled with a neutral gas
 Aspiration of liquid = Damage to the device	 No detection in the water vapour
 When aspiring or discharging in a venturi = False results	 When aspiring or discharging in a fan = False results

Table 7: Errors not to be committed when measuring with an explosimeter

3.11.3. Open drain systems and siphoids

In order to avoid any possible degassing by the siphoids on the open drain system, one means of compensation is to blank the siphoids:

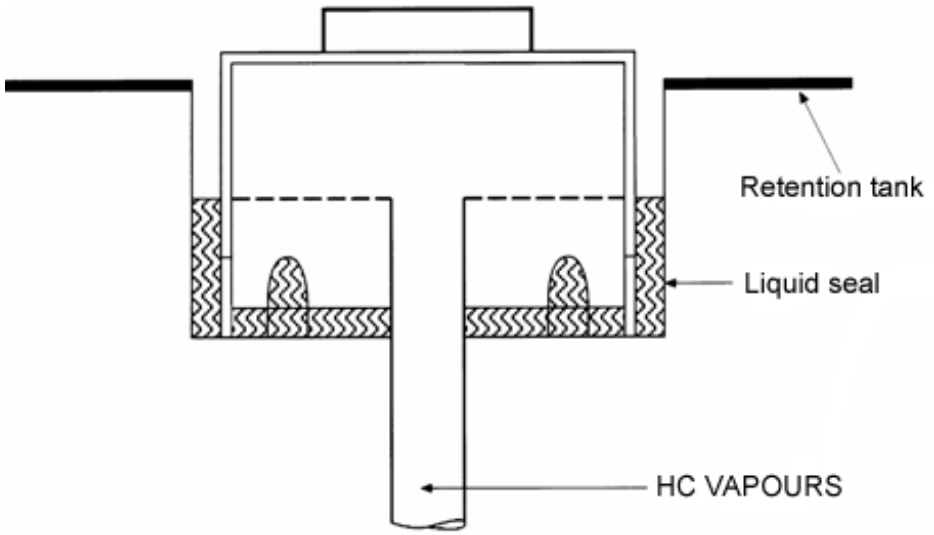


Figure 28: Flow chart

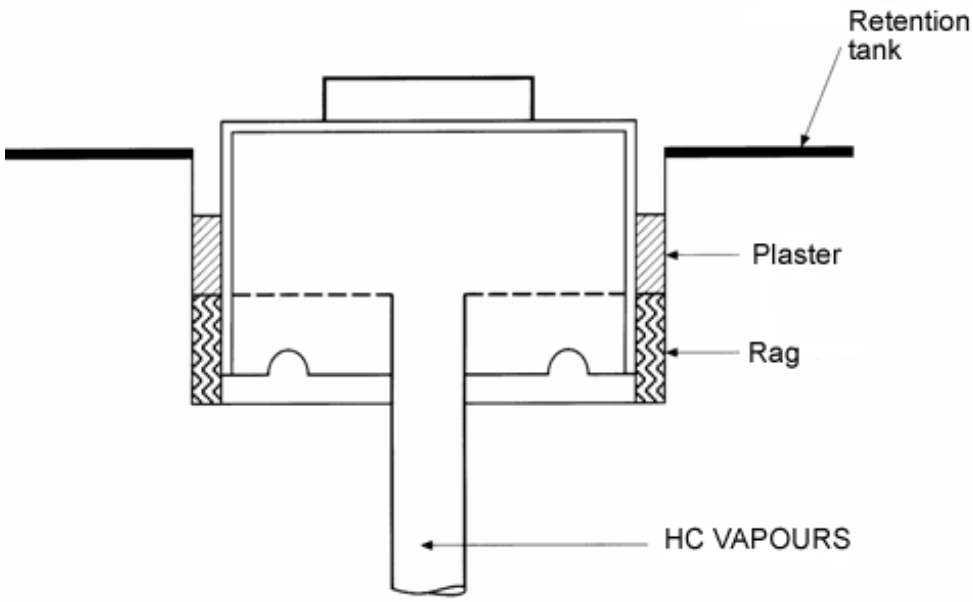


Figure 29: Blanking of a siphoid sight with plaster and cloth before hot works

3.12. AVAILABILITY FOR ENTRY IN A CAPACITY

3.12.1. Working in confined areas

No person may enter a confined space and no work may be carried out in a confined area prior to satisfaction of the following requirements:

- all other options have been eliminated and the grounds for the work in the capacity have been notified,
- all necessary Work Permits have been established and validated, and mention all useful information concerning the response and contingency plan,
- all sources of energy and fluids have been isolated and made safe,
- the atmosphere has been checked, as many times as necessary. Results have been duly recorded,
- a trained agent (or a team) with the appropriate equipment will be near-by and ready to take action,
- unauthorized access will not be possible.

The availability of a capacity for work is subject to very strict rules.

Entering a capacity can be very hazardous if the following rules are not applied: risk of asphyxia, explosion, or fire.

Chronology of availability:

- Establishment of the Availability Procedure (MAD),
  - detailed procedure,
  - identification of procedure stages on PID,
  - listing of bolt running in,
  - plan + numbering of plating/deplating (spading),
  - listing of equipment to be consigned (Mechanical, Instrumentation, Electricity, Production),
  - identification/location of ventilation equipment in the capacity prior to entry,
  - identification/designation/role of actors in and near to the capacity,

- Establishment of the Work Permit,
- Establishment of the Confined Area Work Permit.

Working in confined areas is only authorized if no other method is possible.

In this case, the following requirements apply:

- identify all potential risks, such as the atmosphere in the confined space, any defects in prevention resources (isolation, breathing equipment, etc.),…
- define the necessary prevention resources: place signs at the entrance to the confined space, restrict entry to authorized individuals only, wear the appropriate personal protective equipment, use adequate lighting, always ensure that the atmosphere has been analysed and results recorded, etc.
- establish a Work Permit according to the procedure described in CR EP HSE 036 including a detailed evacuation and rescue plan for emergency situations and the appropriate isolation certificate,
- regularly, and whenever necessary, analyse the atmosphere and record results, place an attendant near to the working area and ensure that emergency equipment is in place and ready-for-use for the entire duration of works. The attendant must have a direct means of communication with the control room and, mainly, with workers inside the confined space.

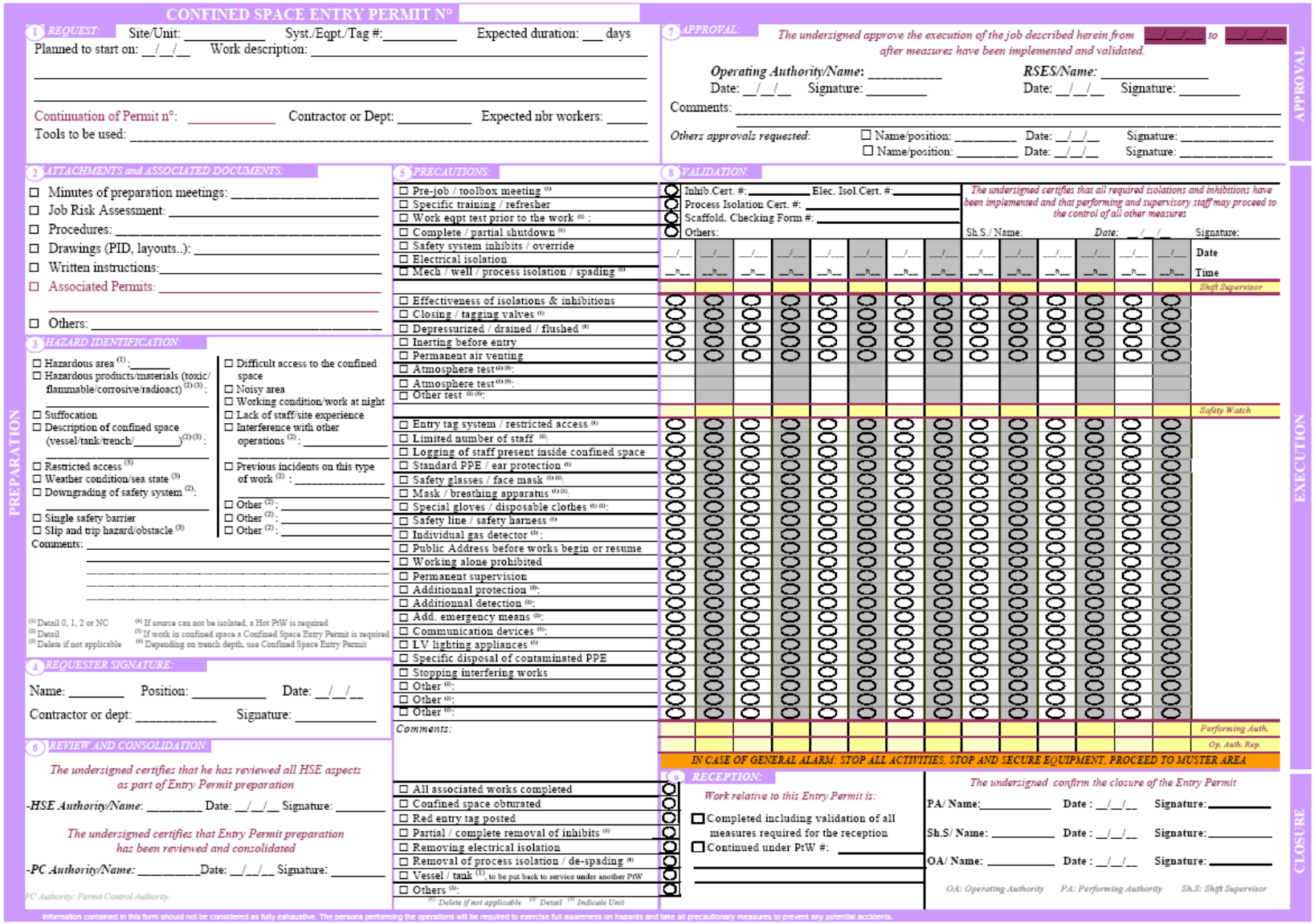


Figure 30: Confined Area Work Permit

3.12.2. Different work phases for a capacity

After having decompressed and drained the capacity and after having isolated all consigned manual and automatic valves, plating will be carried out (these different stages will be defined in an operating procedure validated by the Environment Safety Department).

Should an explosive and/or toxic gaseous phase arise, anti-spark tools and masks (especially if H2S is present) will be used.

Several washing sequences will be applied until clear water appears in the flexible piping connected to the high point (the liquid effluent will be controlled with a mask if H2S is present).

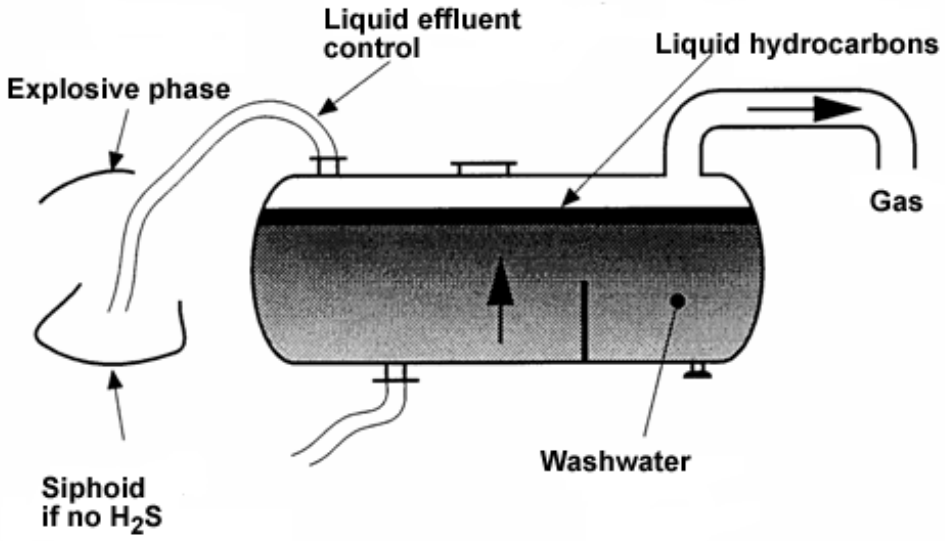


Figure 31: Washing of a capacity

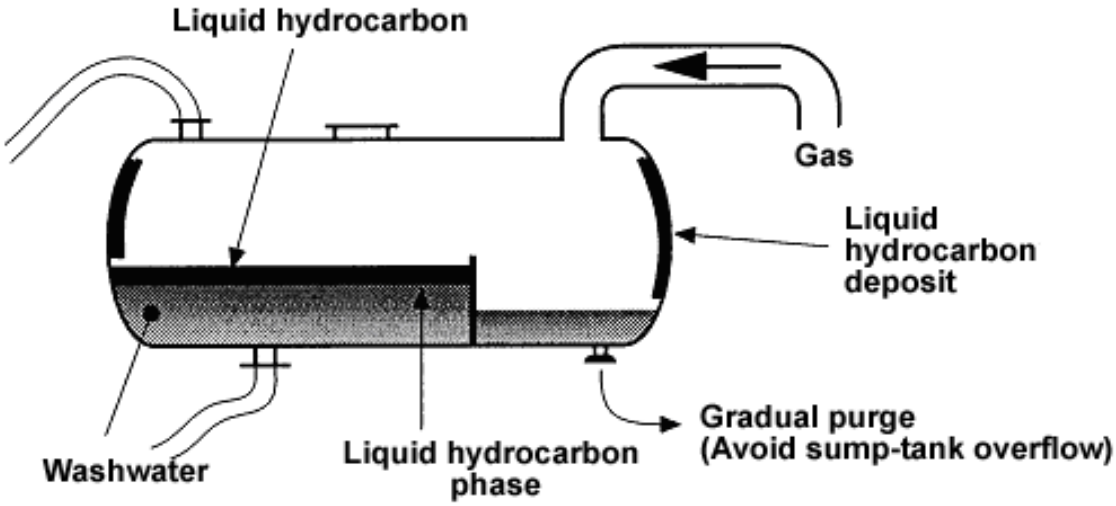


Figure 32: Draining of a capacity

Prior to initial entry, plating or the disconnection of all lines coming into or connecting from the capacity is checked.

The opening will be made with a manhole cover connecting to the potential of the capacity (risk of static electricity).

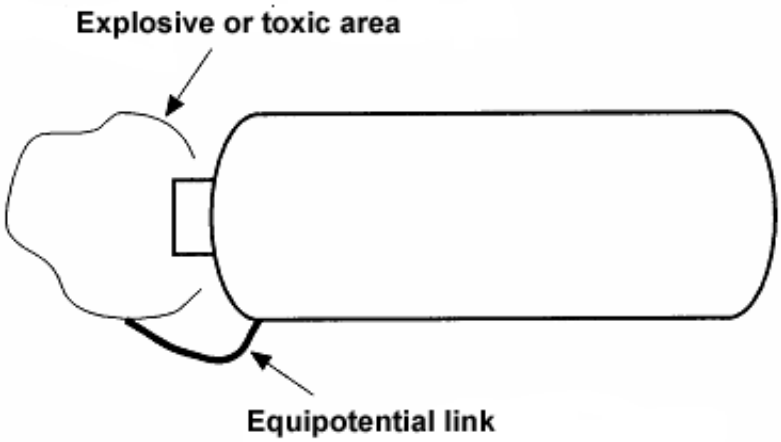
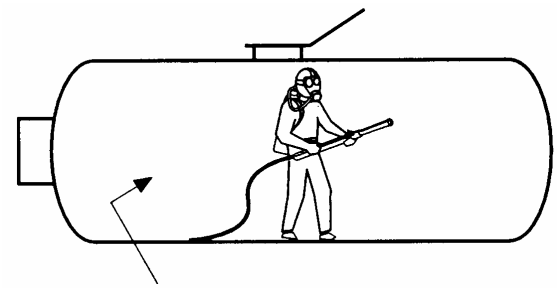


Figure33: Opening in a capacity

Ventilation will be ensured via an extractor which is also connected to the potential of the capacity (risk of static electricity).



Initial entry will be carried out by a safety officer equipped with an independent mask and a fire nozzle connected to the potential for the capacity. This officer will carry out an initial analysis of the atmosphere.

Figure 34: Initial entry

Personnel may enter the capacity with an independent mask or a Hookah rig via the man hole, providing an attendant constantly monitors from the outside (through the same manhole) if:

- LEL < 10%
- H_2S < 50 IPM.

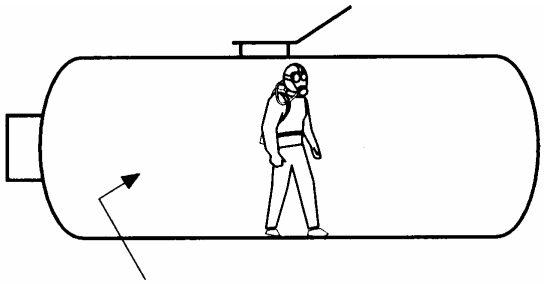
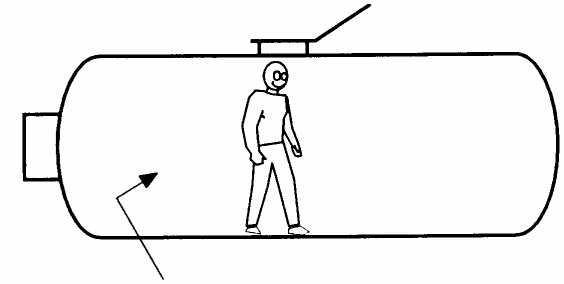


Figure 35: Entry with a mask

Personnel may enter the capacity without a mask if:



- LEL < 0%
- H_2S = 0 IPM; (19 < O_2 < 21)

Figure 36: Entry without a mask

Attention: In case of hot works, LEL must be equal to 0

Attention: A mask must be worn for all cleaning operations and LEL must not exceed 20%

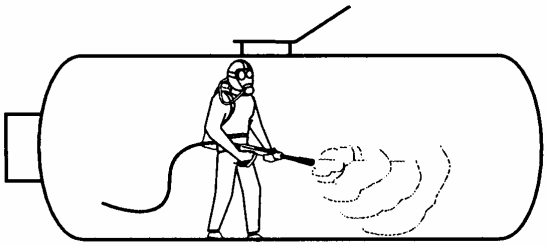
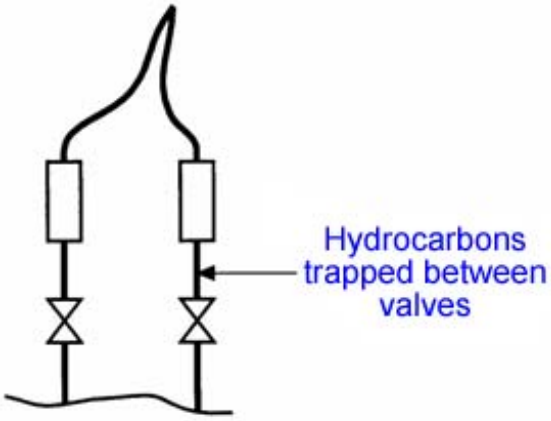


Figure 37: Cleaning operation



Attention: When working on valves, hydrocarbons may be trapped between two sets of valves (Block valve & PSV)

Figure 38: Work on valves

Attention: When working on sight gauges, check that they have been plated to the edge of the capacity (risk of the introduction of hydrocarbons in the capacity).

Attention: Whenever possible, pass the gas piping via a different manhole than the one enabling personnel to access the capacity.

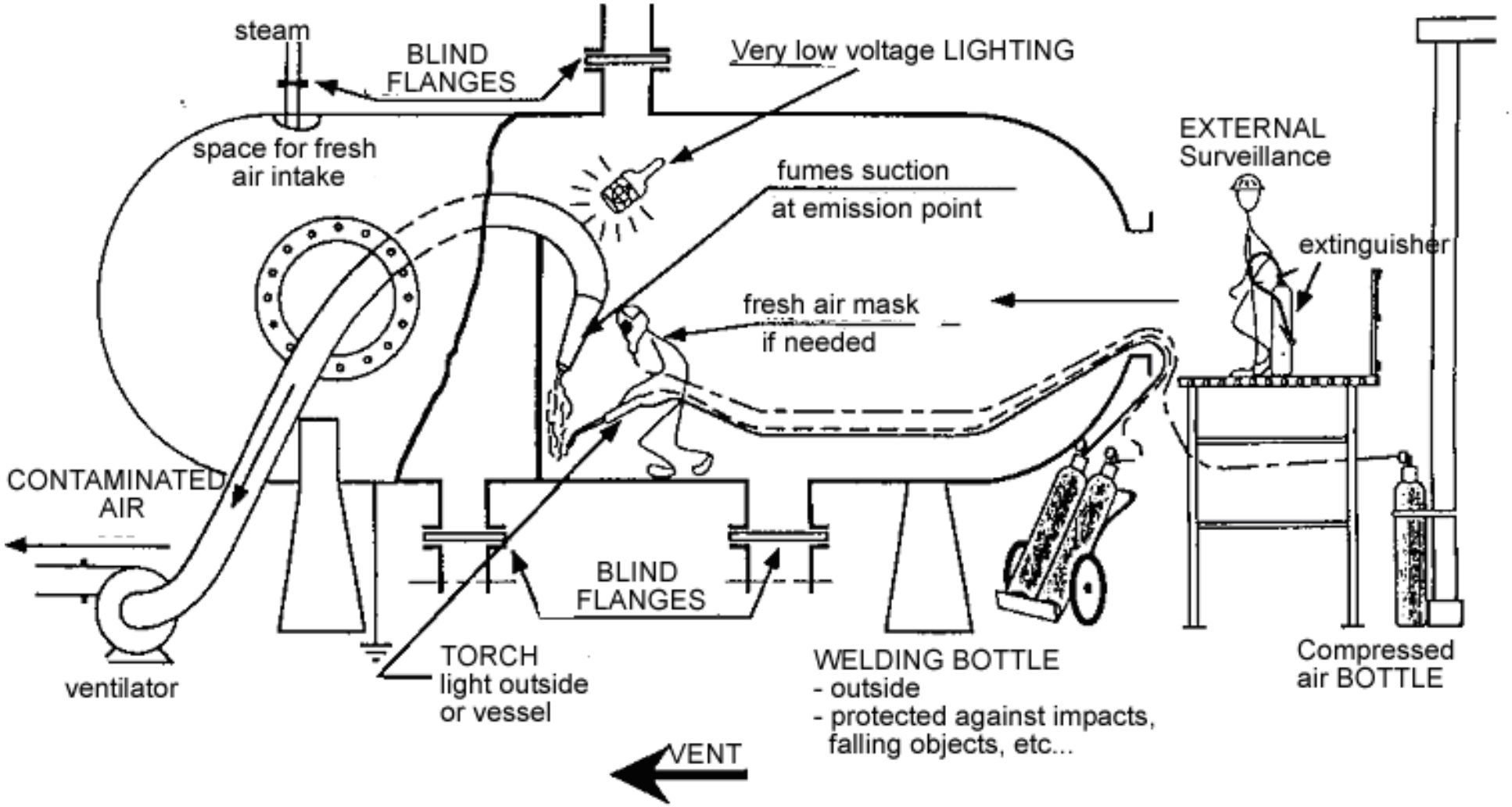


Figure 39: Working in a capacity

Risks	Precautions	Accidents
Hydrocarbon leak	Decompression, isolation and draining of the lines or the capacity, according to the operating procedure validated by the Environment Safety Department. Consignment of pumps and automatic valves. Plating or disconnection of all lines coming in to the capacity. Plating as near to the dimensions of the capacity as possible. Check the LEL regularly and whenever changing shift. Hot work will only be authorized if LEL = 0% and after the cleaning of the capacity (the distillation of hydrocarbon deposits leading to the formation of flammable gas). Bottles of oxyacethylenic gas outside the capacity and always start the blowpipe outside.	Explosion of a gas platform plus condensates following the transmission of gas via the drains when cleaning separators. LOSS PREVENTION: near miss asphyxia by N_2 when working in the capacity. Explosion in a works column, caused by the emission of gas due to the decomposition of sulphide deposits stuck to the wall.
Asphyxia	In the presence of H_2S, degassing of the capacity by scavenging to the flame with purified gas or water. When opening a capacity, use of air extractors equipped with equipotential connections. Analysis of the atmosphere by the Safety service prior to the entry of personnel (0 IPM H_2S max. or use of a mask and 19<0²<21). When working with a mask, calculate the autonomy of air reserves when preparing the site and monitor from outside via the manhole. Entry slip validated by the Safety and Maintenance service.	Asphyxia due to H_2S when working in a capacity.

Risks	Precautions	Accidents
Ignition	In the presence of iron sulphides due to the presence of H_2S in fluid in the capacity: flushing and washing of the walls with a fire nozzle equipped with an equipotential connection. Equipotential connections when opening a manhole, when using the fire nozzle or air extractors. Safety lighting (ADF) Compliance with the hot work procedure (no hot works near to the capacity during opening and degassing). Welding work carried out outside of the capacity. No opening of the capacity or degassing during stormy weather.	Fire caused by pyrophoric sulphides.

Table 8: Inspection and working in a capacity

Risks	Precautions	Accidents
Hydrocarbon leak	Decompression, isolation, draining of lines and capacities. Consignment of pumps and automatic valves. Plating or disconnection of all lines coming in to the capacity. When inerting, the drop in the hydrocarbon content of gases is controlled at the Tankscope by two separate orifices in the cistern, at a distance, located 2m under the deck, at mid-cistern and at the bottom of the cistern. The cistern is considered as scavenged if the values measured at all locations are less than 2%. When opening, personnel may only enter the cistern without a mask if, and only if, LEL < 1 %. The LEL will be permanently monitored by the detection marker. Bottles of oxyacethylenic gas outside the capacity and always start the blowpipe outside.	
Asphyxia	Compliance with the inerting procedure, aeration: ventilation starts at 5% LEL after inerting, triple renewal of the cistern volume, ventilation ducts at the bottom of bunkers. Taking of samples of O_2 > 19%, LEL < 10% and CO < 50 IPM at three different levels in the cistern, and recorded on the entry log. Samples will be taken every 15 min and subsequently every 30 min and at each change of shift. Number of respirators = number of actors (skid-connected Hookah rig at 90m^3 of air) plus stand-by respirators on the deck. Use of inert gas (OMI). Dilution or travel schematic.	Two deaths due to asphyxia in the cistern of a stocker. Unplated valves.

Risks	Precautions	Accidents
Ignition	When opening a panel, it will be connected to the hull via an equipotential connection. No opening or works during a storm if the capacity has not been degassed. No hot work near-by when opening or cleaning the capacity. Washing of walls if a risk of pyrophoric sulphide exists. VLV (Very Low Voltage) and Ex type lighting Lines for the extraction of air with equipotential connections.	
Climbing	Permanent monitoring from the surface by a deck safety office who will record the time and names of the people entering or leaving the cistern. The number of individuals present in the capacity must be restricted. Walkie-talkie link between the cistern and the deck. Marking out the working area. Emergency equipment on the deck at the disposal of trained members of personnel. Idem for the following rescue equipment: - Hookah rig, ARI, Colt, shell type stretcher pneumatic splints, safety stretcher + air winch, -3 safety harnesses, - 3 controls, - roping to reach the most distant location in the cistern, - Ex type flashlights Battery fire nozzles. Near-by extinguishers. Presence of a nurse on the site.	

Risks	Precautions	Accidents
Difficult thermal environment	Use the thermometer to permanently check changes in the ambient temperature (T less than 50°C). Change staff every 15 minutes.	
Electrification	Installing electric equipment.	
Falling from heights	Marking out and permanent surveillance of cistern openings. Marking out of openings corresponding to changes in level on a ladder.	When working in the front peak of Serepca 1, fall in the hole to the second ladder (5.3m).

Table 9: Inspection and working in a stocker cistern

3.12.3. Maintenance and inspection operations

With works, the use of a respirator supplied by air bottles (trolley or rack) is strongly recommended.

The attendant must be trained to be able to:

- Ensure that marking and safety rules are complied with near to the working area,
- Ensure that the air supply is in correct working order,
- Provide assistance to an actor in case of an incident,
- Call for emergency assistance.

The attendant must:

- Be located in an upwind position (wherever possible) from the working area,
- Be able to directly see the working area,
- Be equipped with an ARI in the stand-by position,
- Have a means of radio communication.

Wearing respiratory protection is necessary when opening circuits if the concentration of toxic gas exceeds the TLV for the circuit or if it exceeds the IDLH behind the initial isolation valve. Other parameters must clearly be taken into account.

Disassembled equipment must be washed to avoid any risk of the emission of toxic vapours in the workshop. If this risk cannot be eliminated, special precautions must be considered.

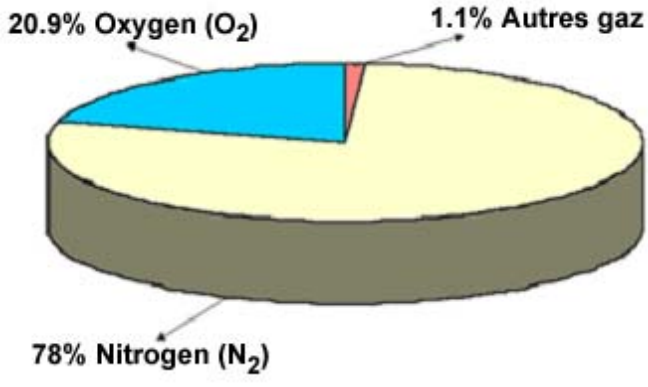
When equipment is open, the installation of one or several mobile gas detection markers enables the monitoring of concentrations in the working area to ensure that levels are acceptable.

Toxic gases may be released, particularly during:

- chemical cleaning. Risks relating to the formation of H_2S during chemical cleaning due to the effect of acids on metal sulphides (iron, etc.) present in circuits or capacities must be accounted for.
- the discharge of water saturated in H_2S,
- the loading of trucks or tankers with crude charged with H_2S,
- the replacement of molecular sieves.

3.12.4. Anoxia risks

The air mainly consists of nitrogen and oxygen in very specific proportions.

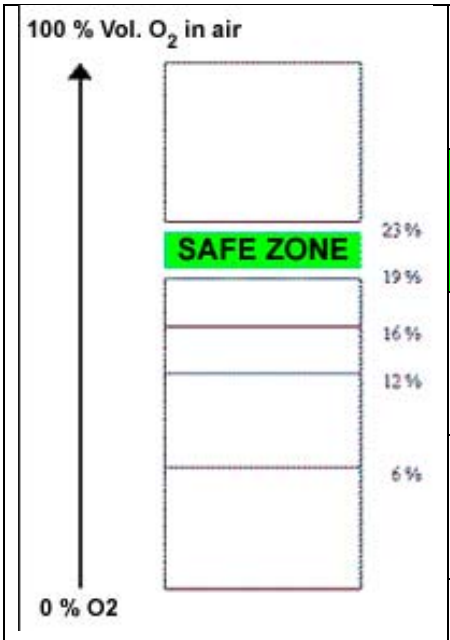


Components	% Vol
Nitrogen	78
Oxygen	20.93
Argon	0.96
Carbon dioxide	0.03
Hydrogen	0.01
Neon	0.0018
Helium	0.0005
Krypton	0.0001
Xenon	0.00001

Figure 40: Composition of the air

The reduction in the percentage of oxygen, due to an increase in the percentage of another component, leads to the risk of ANOXIA

The safety range is including 0_2 between 19 and 23%



% Vol of O_2 in the air	
19-23%	Normal level of O_2
16-19%	Difficulties in breathing, nausea, vomiting, vertigo
12-16%	Loss of consciousness
< 12%	Immediate loss of consciousness + death

Figure 41: Concentration of O_2 in the air

3.12.4.1. Neutral or inert gases

Inert gases do not give forewarning; they act quickly, 2 mouthfuls of oxygen-free atmosphere means almost immediate death.

The following precautions must be taken:

- Always take care if the nitrogen indicator comes on;
- Never use nitrogen instead of compressed air (e.g. for pneumatic tools) and use different connections;
- A cartridge respirator or autoflow will be ineffective as nitrogen is chemically 100% inert;
- Mark out and indicate all equipment using nitrogen;
- The wearing of ARI is recommended during some draining procedures or circuit deplating.

Nitrogen is not a source of concern as it is frequently used, an **unnoticed risk is even more hazardous than an obvious risk.**

Nitrogen cannot be picked up by our senses; the only means of detection is the oxygen meter

Nitrogen is mortal and acts quickly, which increases the inherent hazard. Only use nitrogen if fire risks are high and ensure you avoid systematic use.

After having inerted for nitrogen, ventilate the air whenever possible. The explosimeter cannot detect any residual fuel other than in the presence of oxygen

3.12.5. Pyrophoric iron sulphides

Iron oxides present on the internal surfaces of steel equipment react with the H_2S and the mercaptans contained in the gaseous hydrocarbons and form iron sulphides. These sulphides can spontaneously ignite in contact with the oxygen in the air and cause a fire or explosion, with the emission of SO_2. This risk mainly exists when opening circuits.

3.12.5.1. Equipment opening

Risks: the oxygen in the air will react with sulphides and lead to a significant increase in temperature. If liquid or gaseous hydrocarbons are present, a significant risk of fire or explosion will exist.

Control:

- Nitrogen bleeding of circuits prior to opening,
- Washing with water whenever possible,
- Scavenging with pure nitrogen and controlled oxidation with scavenging with nitrogen-enriched air (5% oxygen, 95% nitrogen), whenever possible.

3.12.5.2. Partial or complete interruption of circuits

Risk: air input via the vents and draw-off points.

Control: various means of isolation to prevent air form entering stopped circuits.

3.12.5.3. Storage of residue

Risk: risk of spontaneous ignition. It must be noted that SO2 (toxic) is emitted in oxidation.

Control:

- Controlled oxidation in a separate area,
- Waste diluted in water.

3.12.5.4. Return of possibly-contaminated equipment to the workshop

Risk: risk of spontaneous ignition in the workshop.

Control: washing of the equipment.

3.13. SAMPLING PRECAUTIONS

Taking samples on oil installations is a daily operation. It can therefore be treated as a "routine" task, and hence lead to catastrophic situations for personnel and equipment. The procedures implemented on sites for this type of operation must therefore be closely applied.

2 types of sampling can be distinguished:

- the standard daily sampling of site effluents (identified, scheduled, with a procedure, carried out by assigned personnel for the installation): this type of sampling may or may not be subject or not to a Work Permit (rather a work slip), depending on the risk level applied, and according to site regulations;
- the specific sampling of some effluents to study purposes (e.g.: sampling of oil to test new chemical products): this type of sampling will require a mandatory Work Permit if a person/team other than assigned personnel for the installation participate in sampling or if special and specific equipment is to be used.

Risks	Precautions
Degassing of hydrocarbons	Optimize the duration/frequency of sampling.
Liquid discharge, splashes	Installation of containers connecting to DO, use of double valves with a reduced section.
Intoxication of the operator	Use of a fresh air mask (trolley or ARI), installation of a "closed" sampling system.
Ignition, Explosion	Prohibit hot works during sampling, use of an equipotential connection.
Blocking of the sampling valve in the open position	Use of a double valve with a reduced section, double valves.
Trigger of ESD due to the detection of gas	Temporary prohibition due to the local detection of gas, optimize the duration/frequency of sampling.
Breakage/leakage of sampling bottles	Use of standard bottles, identified per type of effluent, use of standard impact-resistant transport buckets/cases.

Table 10: (Atmospheric) Sampling risks and precautions

3.14. LIFTING AND HANDLING

No work involving the use of a crane, hoist or other mechanical device, including lift trucks, may start if the following conditions are not simultaneously satisfied:

- the lifting equipment and method have been defined by an authorized individual,
- actors are trained and qualified. Cranes, lifts and trolleys are handled by certified personnel,
- all safety devices on lifting equipment are in correct working order,
- all lifting devices and accessories (slings, shackles, etc.) have been certified as apt for use following inspection,
- a colour marking system or equivalent is used,
- the weight of the load is known and within the limits of maximum capacity,
- all visual accessories have been visually checked prior to use.

3.14.1. Study of risks for lifting operations

Fill in the following table to determine if lifting is CRITICAL.

	YES	NO
This lifting operation has never been executed previously		
A specific procedure is required as no previous procedure with documentation exists and no appropriate general lifting operation procedure exists		
Personnel is not familiar with the equipment/devices to be used		
Lifting exceeds 80% of the crane capacity for the intended distance		
Lifting using more than 85% of the maximum boom length		
The load will travel above a process installation/unprotected machines (the crane will move with the load suspended)		
Lifting above, through or near to active installations (static crane)		
Following lifting, it is more dangerous to recover the load than to place it at its final location		
Lifting of personnel		
Horizontal or vertical lifting in excess of 40 tonnes		

Lifting in excess of 20 tonnes requiring a boom of more than 160' (50 m)		
Lifting using a fixed or lifting fly jib in excess of 10 tonnes (lattice crane) or 5 tonnes (hydraulic crane)		
Lifting from a ship/a barge likely to make large movements during the operation		
Lifting in an area in which environmental conditions play a significant role (substantial tide amplitude, low visibility due to fog, extreme temperatures, etc.)		
Load without specific lifting points or with sharp edges		
Lifting requiring the prior modification of lifting equipment		
Lifting requiring the special configuration of lifting equipment (e.g. tower crane for a track crane or Super Lift configuration)		
Lifting requiring the design or manufacture of special lifting equipment.		
Lifting in excess of 20 tonnes requiring the moving of the load with a track crane or a truck crane		
Use of two cranes or more (multi-crane operation)		
Movement of the crane/lifting near to high voltage electric lines, at a distance less than the recommended safety limit		
Lifting of high value loads		
Lifting of an important component, the loss of which would i) lead to a significant interruption in operations, ii) represent an unacceptable risk of accidents for personnel or tangible damage, ii) cause the discharge of hazardous substances, iv) cause undetected damage leading to safety problems in the future		
The centre of gravity of the load is located over the lifting points, or a high centre of gravity		
The weight of the load is unknown and/or the dimensions or shape of the load are complex		
The load is easy to damage, e.g. by torsion of a long and flat load		
The load must rotate (e.g. 2 cranes lifting a column) or cross supports (e.g. 2 or more cranes moving with a load)		
Lifting in a confined space, or in an area with free height and a restricted hook		
The load has a large surface area which may react with the wind		
The lifting operation is under-water or requires the use of divers		

Table 11: Analysis of risks for lifting operations

If any of the above questions is ticked as YES, this is a CRITICAL LIFTING operation and must be subject to scrutiny prior to execution.

3.14.2. Standard lifting operation plan

LIFTING OPERATION PLAN

Description of Load:	**Weight of Load**:	**Operation Ref No**:
Brief Description of Lifting Operation (including location):		

Possible Considerations (not exhaustive) – tick if relevant and address each point in "step-by-step" plan section below)

1 Weight of load unknown	☐	7 No dedicated rigging	☐	13 Lifting thro' substantial heights	☐
2 High centre of gravity	☐	8 Load has sharp edges	☐	14 Dynamic factors involved	☐
3 Unstable load	☐	9 Extremely heavy load	☐	15 Hazards to personnel	☐
4 Awkward size/shape	☐	10 No lift point above load	☐	16 Communications	☐
5 Fragile load	☐	11 Restricted headroom	☐	17 Tag lines needed	☐
6 No dedicated lift points	☐	12 Confined work area	☐	18 Poor light conditions	☐

Step by step plan of lifting operation (include site clearance if relevant):

(if required use reverse side of this sheet for additional information/sketches etc)

Route to be travelled & laydown area:

1 Has route to be travelled been selected and cleared of obstructions?	☐ yes	☐ n/a
2 Is laydown / landing area(s) adequate in both size and load bearing capacity?	☐ yes	☐ n/a
3 Is suitable packing available for protection of load/slings while landing the load (e.g. timber)?	☐ yes	☐ n/a
4 Have barriers been positioned to prevent access to unauthorised personnel?	☐ yes	☐ n/a
5 Have you confirmed that the laydown area is within the operating radius of the crane?	☐ yes	☐ n/a
6 Will the banksman always be visible to the crane operator or has an alternative method of communication (e.g. radios) been established?	☐ yes	☐ n/a
7 Have weather conditions been considered regarding their effect on the safety of the lifting operation?	☐ yes	☐ n/a

Levels of personnel competence & manning Levels required:	Banksman Slinger ☐ Qty:	Rigger ☐ Qty:
	Technician ☐ Qty:	Support engineering required: Yes ☐ No ☐

Rigging/material list:

Level of Rigging & Lifting Risk Assessment Performed:	Use of Pocket Card ☐ Written assessment ☐	(attach copy to this plan)
Level of Manual Handling Risk Assessment Performed:	Not applicable ☐ Written assessment ☐	Use of Pocket Card ☐ (attach copy to this plan)

Ensure all the lifting equipment and accessories have been checked and deemed safe

Lifting operation planned by:	**Date:**
Job Title:	**Signature:**

Important Note: Before filing update plan with any changes/improvements made during actual operation.

Figure 42: Lifting operation plan

3.14.3. Additional advice for lifting operations

Detailed advice for critical and non-critical lifting operations must be included in the lifting standards/procedures for each subsidiary. The specific points mentioned below can complete or replace the standards included in the rules of procedure, due to the type of operations executed by Total offshore or in inland sea areas.

The load tables and graphs, etc. provided by the crane manufacturer, should be constantly referred to. In particular, if a crane designed for onshore use is used on a floating unit, the manufacturer's appropriate maximum load graph (adapted for sea work) will be enclosed with the lifting plan.

Dynamic amplification factors will be taken into account for all onshore or offshore lifting and mobile track cranes.

All underground obstacles or services must be physically identified on the site, particularly pipes or cables installed offshore or in rivers.

The distribution of the load in view of the ground carrying capacity must be correctly studied and documented in the lifting plan.

The stability of equipment/the barge must be correctly studied and details included in the lifting plan.

Contemplate lifting restrictions due to environmental conditions, etc. Find weather forecasts prior to starting lifting, particularly for lengthy lifting operations or if thunderstorms, violent winds or gusts are announced.

Account for tides if lifting requires the use of a floating barge or a docked barge. If the barge is docked, check that the angle corresponds to the crane manufacturer's recommendations and that structural integrity (due to the concentration of the load) has been checked.

The use of chains (risk of sudden failure) must be avoided as far as possible. Cables are inherently safer.

All other activities executed in the area (including non-critical operations) or near-by, must be suspended during lifting operations.

The lifting venue must be blocked off with a safety line to restrict access by personnel.

3.14.4. SIMOPS - Placing large packages

Risks	Precautions	Accidents
Falling packages	Minimum personnel on the platform. Partially or totally stopped platform depending on the size of the lifting operation. Checking of lifting equipment. Slings: maximum load for adapted use; coupled; less than 20% of legs damaged; presence of cable-grip at ends; colour code for annual inspection; correctly stored. Selection of the lifting equipment and barge. Pre-slinging of packages (slings are regularly inspected). Authorized crane operator and qualified sling operators; radio communication and conventional gestures. Handling responsible is the deck officer. With night work, organize sessions to restrict hazardous operations and install effective lighting. Mark out evacuations.	Fall of the logging cabin subsequent to the breaking of the attachment flat bars on the spreader. Fall raising line, during the lifting of equipment, leading to the rupture of the socket shield: foundry default.
Thumping	Lower waves at the limit calculated on the basis of the crane boom (weather conditions defined by the insurance policy). Ballasting according to the loads to be transferred. Inhibition of CO_2 bottles, automatic extinction restricted to the duration of the placing of a package (risk of impact due to vibration when placing the package).	During a DTM, when transferring a package, the unbalance of the drilling tender will trigger the crane safety system. Due to swinging, the package will regularly hit the platform until the ballasting is re-balanced.

Risks	Precautions	Accidents
Collision	The wetting plan will require the prior approval of the logistic's service of the subsidiary or the competent person contracted by the subsidiary. The wetting plan will only be modified with the written agreement of the subsidiary. The competent Total representative for marine logistics will be on-board for the entire duration of approach handling. Weather conditions (current speed limits and maximum wave heights will be defined by the insurance policy).	The drilling tender hits the platform during the DTM.
Falls by personnel	Safety harness + attached safety strap. (Avoid fall-prevention devices with clearance heights of up to 6 metres). Slings for large packages equipped with guidance lines. During assembly, check that fall-prevention protection is systematically installed. The life line, footbridges and bucket must be systematically used. Experience of personnel. Life jackets for works over water. Rubber speed boat on stand-by for works over water.	Death of a driller subsequent to a fall via an opening in the floor when assembling a derrick set.
Ignition	Compliance with hot work procedures.	

Figure 43: Risks in placing large packages

3.14.5. Verifications to be carried out prior to using cranes

Item	Position	Notes
Table of capacities	Displayed in the crane operator's cabin	
Weight indicator	In correct working order at the last calibration	
Conventional order signs	Displayed	
Rotating machines	Correctly protected Housing in place	
Boom angle and safety stop indicator		
General condition of the boom	No bent or rusted flanges	
Travel stops	Regularity of operating tests on the hoist and boom	
Cables	In good condition, and anti-gyrating	
Hooks	Equipped with a safety pawl Extended with a coupled sling	
Load holding cords	Available, mandatory use	
Personnel transfer buckets	In good condition, procedures known and applied Sling + damper	
Control handles	Clearly identified	
Drip collector		
Crane certification	Recording of the load test	
Communications equipment	Crane operator equipped with radio headphones on the same channel as the ship	
Operating manual	Present in the crane cabin	
Certified crane operator		
Maximum crane capacity	Clearly marked on the crane	

Table 12: Verifications prior to use of the crane

3.14.6. Slings

3.14.6.1. The types of slings

Several types of slings exist:

- Rope slings
- Steel cable slings
- Chain slings (type of material to avoid)
- Flat slings (textile)

The sling type must be chosen according to the following criteria:

- Load weight
- Type of load (wood, metal, etc.)
- General shape of the load (plank, tube, case, etc.)
- Particularities of the load (sharp edges, rough sections, etc.)
- Load temperature

3.14.6.2. Sling control

The following must be checked:

- The maximum lifting load (CMU in French = *Charge Maximale d'Utilisation*) for the sling is compatible with the load lifted
- The sling condition

Cable slings with the following must be destroyed:

- 20% of threads cut through two cable lays
- a broken strand
- substantial crushing
- substantial distortion

- substantial oxidation
- rings in bad condition (faulty splices, cable clamps, sleeves)

Link chain slings with a link defect must be destroyed (crack, permanent distortion, welding default, etc.)

3.14.6.3. Storage of slings

Slings must be stored in a ventilated location, out of sun light and protected from heat.

All contact with acids and chemicals must be avoided.

Slings must be greased prior to storage and wiped down prior to use.

Cables must never be bent with a radius of less than six times their diameter and it is prohibited to tie knots in cables and slings.

They must be protected from sharp edges and crushing.

Some steel cables have a textile core and must be paid attention.

Steel cables must only be handled with gloves

3.14.7. Sling techniques

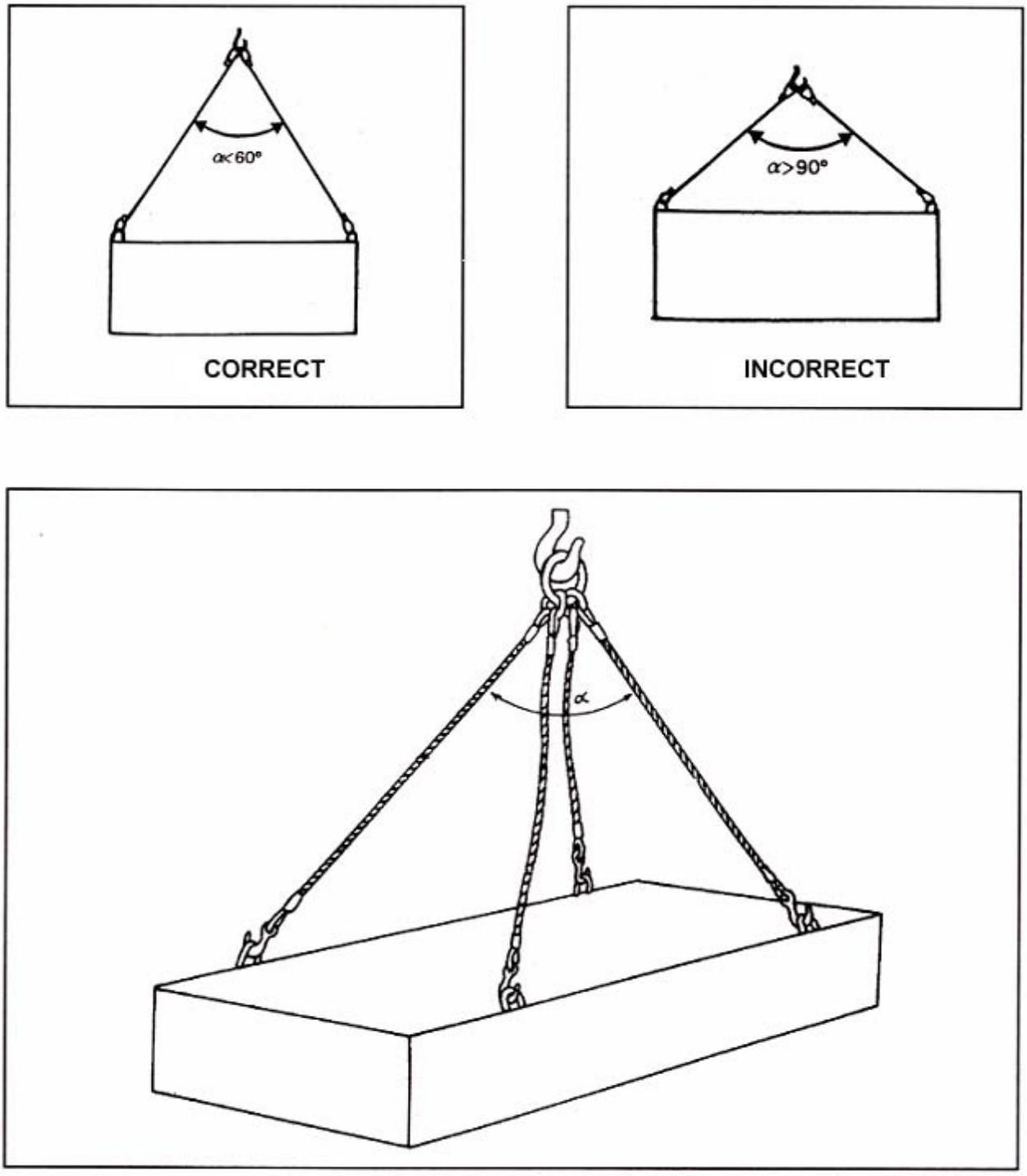


Figure 44: Sling techniques

If legs do not have identical lengths, four-leg slinging must be considered as having two useful legs, as only two legs will be taking the tension.

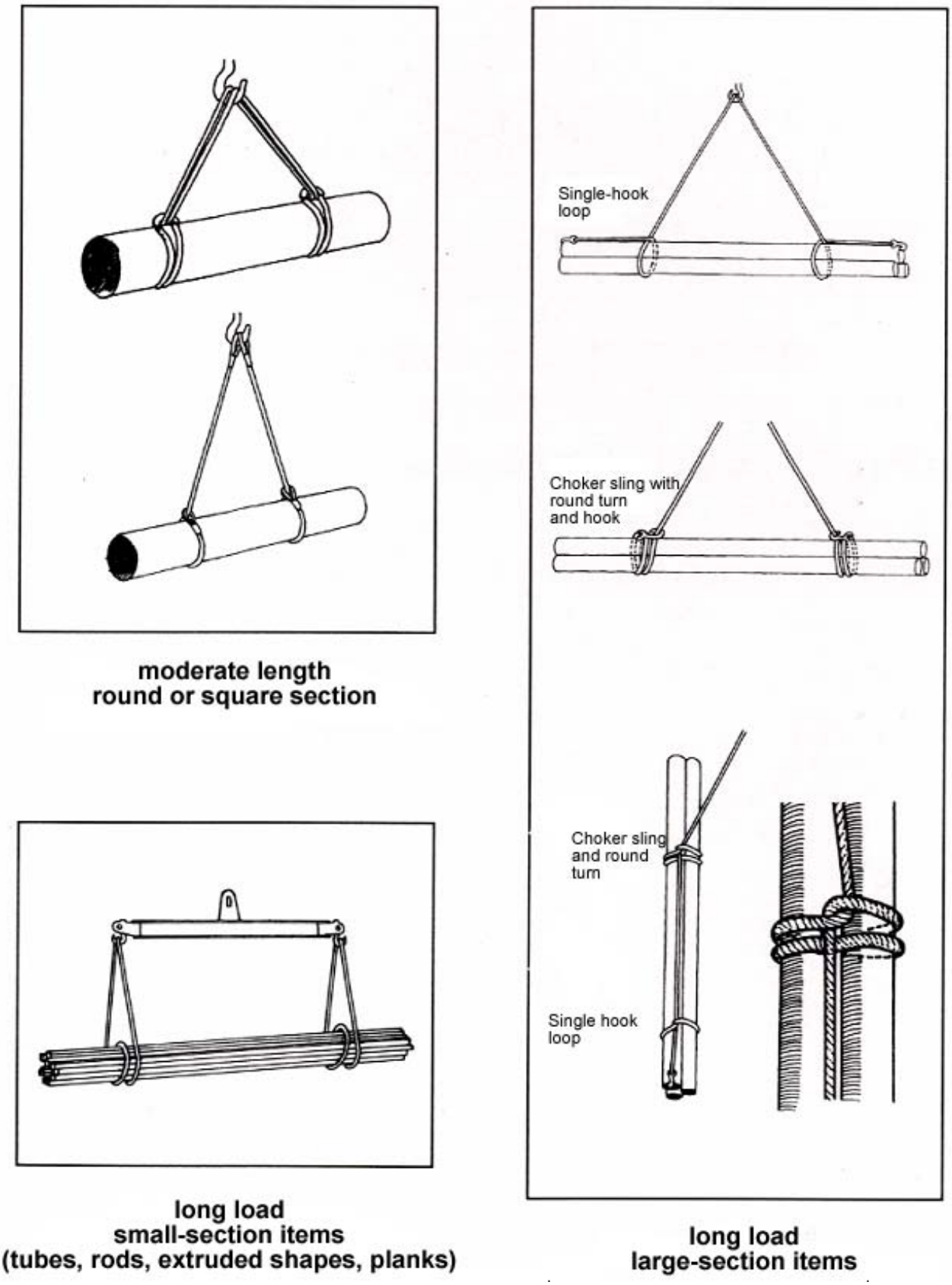


Figure 45: Special slinging

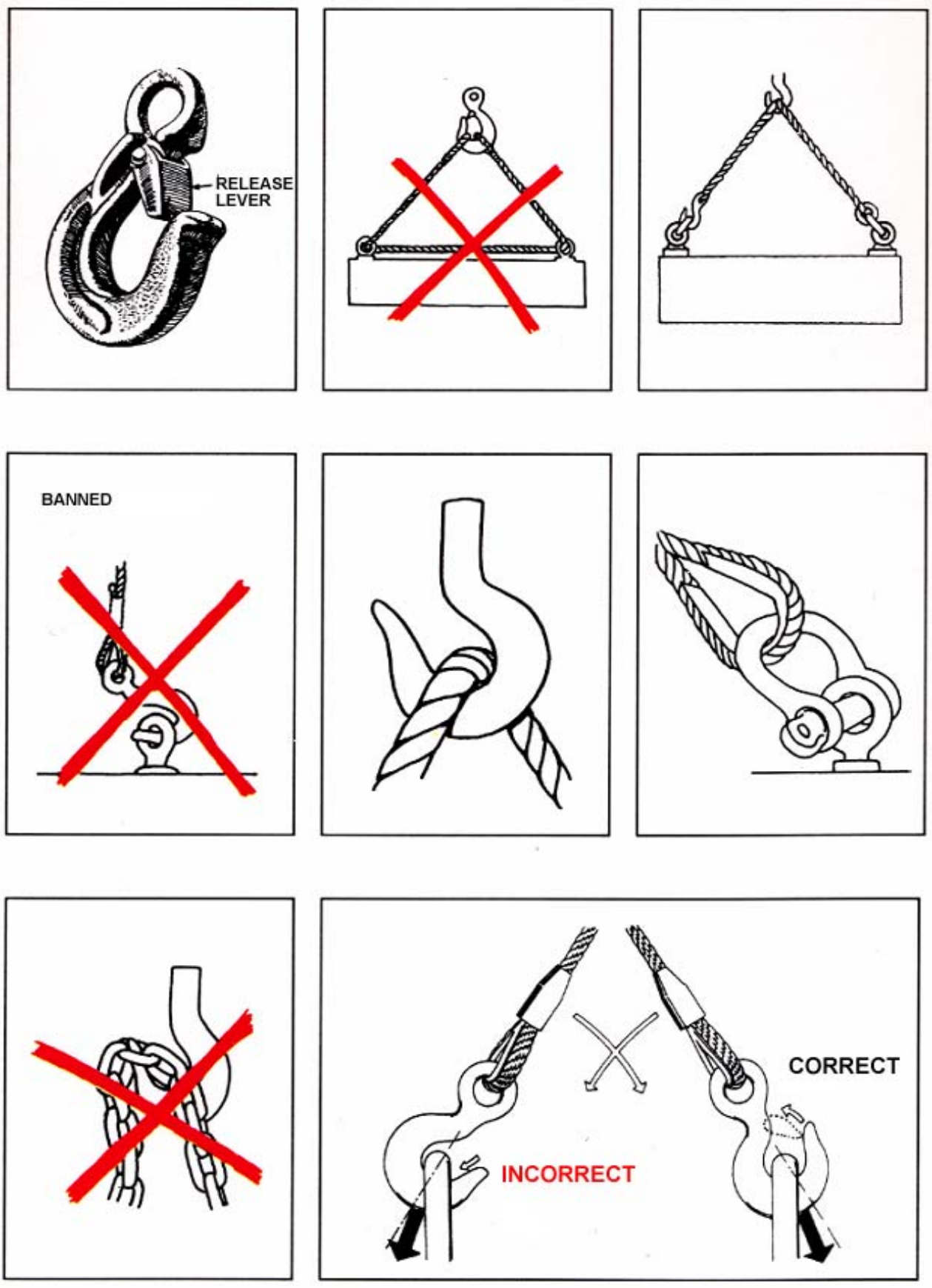


Figure 46: Recommendations

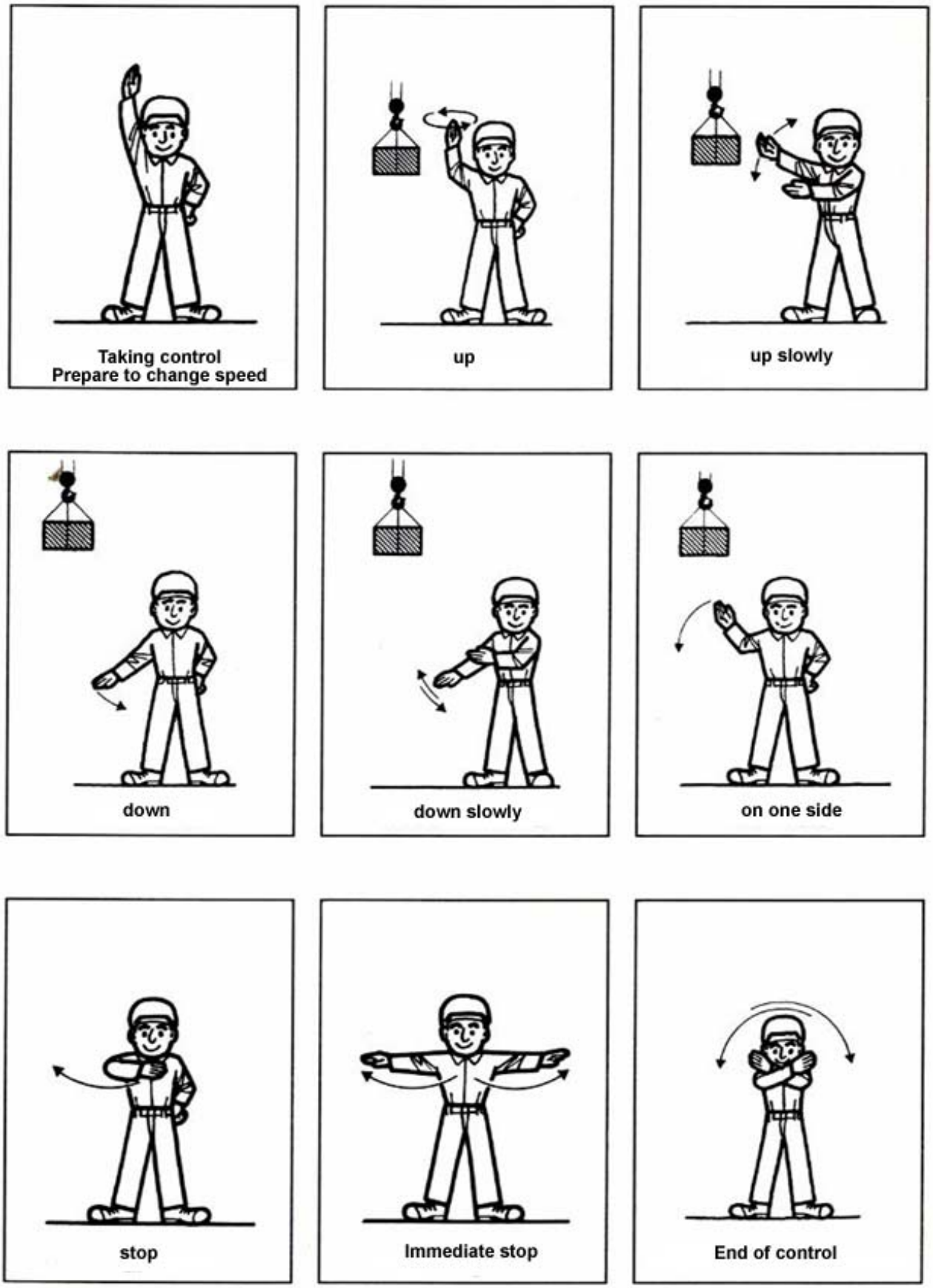


Figure 47: Hand commands

3.15. TRIGGERING SAFETY DEVICES FOR EQUIPMENT

Examples of triggering the safety devices for equipment

3.15.1. Work on systems with an energy supply

Works requiring the isolation of sources of energy (hydraulic, mechanic, electric, process, etc.) will only be undertaken if the following conditions are satisfied:

- prior identification of all tasks requiring this isolation, for a strict check of operations,
- definition of the isolation and discharge method for the energy stored,
- discharge of the energy,
- use of a locking and tagging system with integrated safety devices, i.e., authorization of over-consignment when several separate tasks exist,
- verification of isolation and regular checking of effectiveness,
- recording of the start and end of insulation in a specific log

3.15.2. Availability of a pump for the disassembly of the main pump unit for repairs

Plan the date, the duration and the pump operation

Establishment of the availability procedure (MAD)

- detailed procedure
- identification of procedure stages on PID
- listing of the bolts running in
- plan + numbering of plating/deplating
- listing of equipment to be consigned (Mech, Instrum, Elec, Prod)
- identification/location of lifting/handling equipment
- identification/designation/role of actors

Establishment of the Work Permit

Establishment of the consignment sheet (strict application of the site consignment procedure)

Implementation of the compensatory measures mentioned in the Work Permit

Interruption of the equipment (with the approval of SDC)

Application of the consignment procedure (+ inactivity/equipment operation test)

Application of the MAD procedure (verification of the pressure/fluid/atmosphere check)

Authorization to start disassembling the unit.

3.16. USE OF THE AIR SYSTEM FOR BREATHING

Buddy working is a basic rule for interventions (maintenance, searching for leaks, etc.) requiring the use of a self-contained respirator. While one or several individuals intervene, one person must remain outside of the hazardous area to monitor colleagues and provide assistance if necessary. Means of communication between the actors and the control room must be used.

Activities must be interrupted if the concentration in the air exceeds TLV (10 or 15 IPM H_2S depending on the country) and if all of the individuals in the contaminated area are not wearing respiratory protection.

In accidental situations, it may be necessary to bring additional respiratory protection near to the site.

3.16.1. Breathable air

The ambient air contains approximately 21% of oxygen, 78% of nitrogen and 1% of other gases.
According to EN12021, breathable air must not have either an odour or a significant taste and oxygen content must be 21±1% in terms of volume, with oil content < 0.5 mg/m3, CO2 < 500 IPM, CO < 15 IPM. No free liquid water must be present and the dew point must be sufficiently low to avoid condensation and icing (the dew point must be at least 5°C below the probable lowest temperature).
In the USA, air must comply with the specifications of the Compressed Gas Association (CGA).

Percentage of oxygen	Effects of a lack of oxygen on the body
21%	Normal air content
19%	Tolerance limit
12%–17%	Increased breathing and pulse rate, sweating, slight disturbance of the coordination of movements
10%-14%	Emotional state, difficulty in breathing, unusual fatigue
6%-10%	Nausea, vomiting, possible loss of conscience
< 6%	Convulsion, loss of conscience, heart attack

Table 13: Breathable air

Consumption of air:

- 35 l/minute when resting,
- 50 l/minute when making a moderate physical effort,
- 80-100 l/minute when making a sustained physical effort (usual state when intervening),
- 300 l/minute (peak figure) when making an intense physical effort.

3.16.2. Contaminated air

Air may be contaminated by:

- Gas or vapours with varying degrees of toxicity,
- Solid or liquid aerosols (dust, fog, smoke, etc.),
- Bacteria or viruses.

Maximum professional exposure values (PTLV) and the limit for hazardous effects in terms of life or health (IDLH) are defined for many contaminants.

3.16.3. Respiratory protective equipment

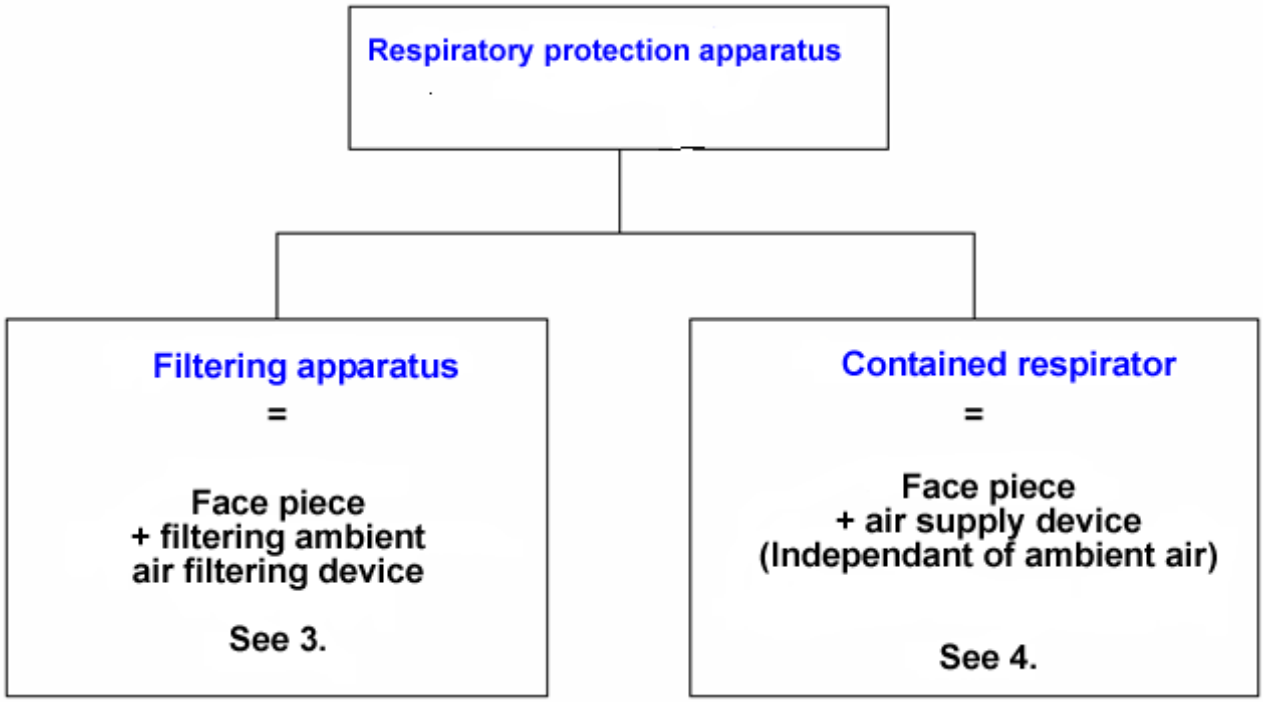


Figure 48: General information on respiratory protective equipment

The global effectiveness of protective respirator depends on:

- the filter device for ambient air or the air supply system,
- entry of ambient air via components, and particularly the correct fitting of the mask on the face. The higher the "total leakage to the mask", the less the user is protected.

3.16.3.1. Classification of respirators according to usage

Respirators can be distinguished according to usage:

- work – See paragraph 6 "Selection of respiratory equipment for work purposes",
- emergency situations (response, evacuation, survival) – See paragraph 7 "Selection of respiratory equipment for emergency situations",

3.16.4. Air-purifying respirators

Air-purifying respirator

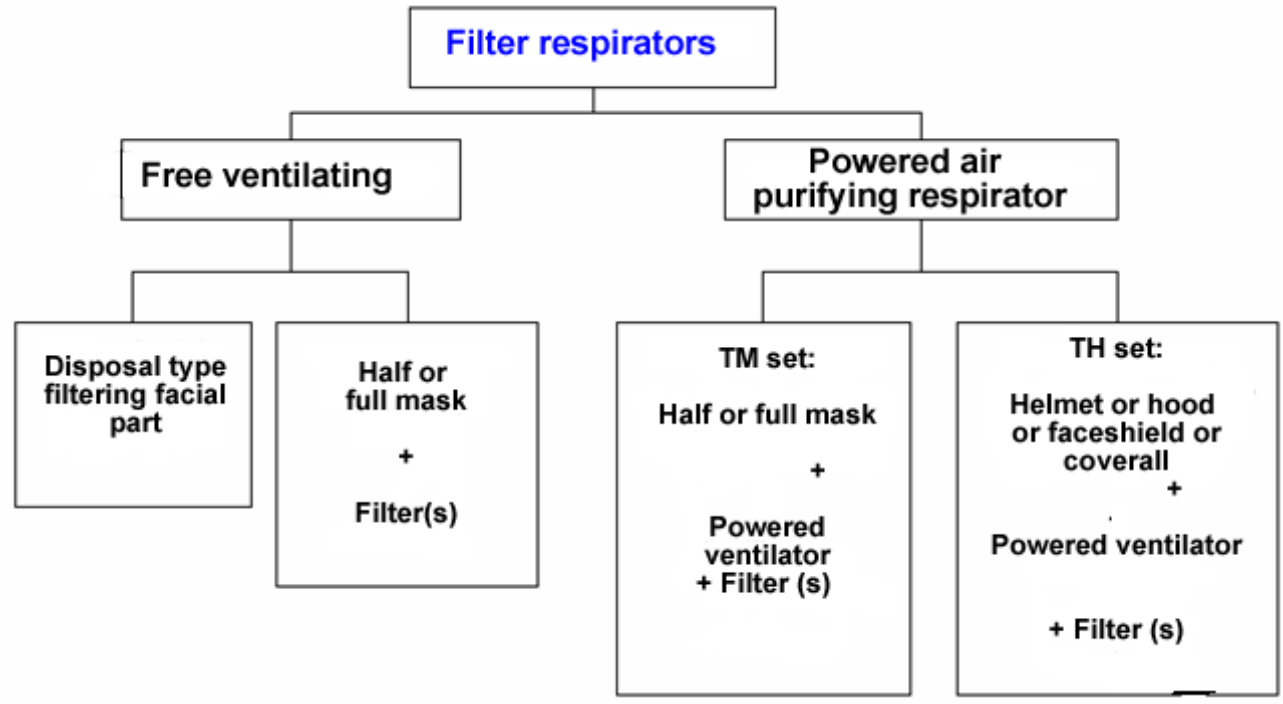


Figure 49: Air-purifying respirators

Reminder: Air purification cannot make the air breathable if the air does not contain enough oxygen or if the air contains one or more non-filterable gases.

An air-purifying respirator is considered a "free ventilation" respirator if the air transiting through the filter is exclusively provided by the breathing of the user, and an "assisted ventilation" respirator if the air is obtained using a powered ventilator.

3.16.4.1. General information on filters

In terms of filters for solid or liquid aerosols (anti-dust), two filter techniques are used:

- mechanical filtering (particles trapped in a mesh),
- electrostatic filtering (electrostatic attraction of the particles which come and "stick" to the fibres).

The two techniques are sometimes combined into one single filter.

A filter for solid or liquid aerosols is the outcome of a compromise between effective filtering and the loss of load when inhaling, as, for a given filter content, the more effective the filter, the harder the breathing for the user.

3.16.4.2. Filters for gas and vapours

Different types of anti-gas filter exist, depending on the type of gas or vapour they are supposed to trap. The filter layer generally consists of active carbon. The operation of these filters depends on 2 principles:

- Physical adsorption. The gas molecules are trapped in the pores of the active carbon.
- A chemical reaction. The active carbon is impregnated with a specific chemical mixture which reacts with the gas molecules to be trapped.

Each type of filter is designated with a marking including one letter accompanied with a strip of a specific colour. Other combined filters also exist to protect against both aerosols and gases and vapours. The air initially crosses the aerosol filter.

Saturation time (or discharge time) is the actual protection time provided by an anti-gas filter. Beyond this time, the filter saturates very rapidly and allows all pollutants to transit.

This can be measured according to a standardized testing procedure.

To give an indication, the minimum discharge time required by European standard EN 141 is of:

Filter class	Concentration of the test gas	Minimum discharge time
B1	1,000 IPM H_2S	40 minutes
B2	5,000 IPM H_2S	40 minutes
B3	10,000 IPM H_2S	60 minutes

Table 14: Minimum discharge time according to EN 141

The relation between discharge time and the concentration of the pollutant can be considered as linear (except for low concentrations where the influence of air humidity becomes more significant). To give an example, a B2 cartridge intended for a duration of at least 40 minutes at a concentration of 5,000 IPM of H_2S will last at least 400 minutes with a concentration of 500 IPM. Saturation will be reached more rapidly if:

- The ambient concentration of gas or vapour is high
- The flow of the filter air is important
- The temperature and the degree of relative humidity are high.

Vibrations can reduce the absorption capacity of a filter by piling, hence modifying the density of the absorbent material.



Figure 50: Masks with filter

Each filter is designated with a marking including one letter accompanied with a strip of a specific colour according to EN 141.

Type	Colour	Field of use
A	Brown	Organic gases and vapours whose boiling point is higher than 65°C (e.g.: benzene)
B	Grey	Inorganic gases and vapours (except carbon monoxide) (e.g.: H2S, mercaptans, chlorine, etc.)
E	Yellow	Sulphur dioxide (SO2) and other gases and acid vapours (HCl, etc.)
K	Green	Ammoniac and organic amine derivatives
HgP3	Red + white	Mercury vapours
NOP3	Blue + white	Nitrogen oxides
AX	Brown	Organic compounds with a low boiling point (less than 65°C)
SX	Purple	Specific compounds designed by the manufacturer

Table 15: Filter markings according to EN 141

A filter may provide protection against several gas groups simultaneously. It will then be designated by adding the corresponding letters and colour strips.

Other combined filters also exist to protect against both aerosols and gases and vapours.

Note: the same filters may be used on free ventilation devices and powered ventilation devices. Filters will then be marked.

The expiry date (year and month) is marked on each filter.

3.16.5. Self-contained respirators

The user is fully isolated from the ambient air. The air inhaled is provided either from an external source (air line or a compressed air bottle), or by recycling exhaled air (closed system).

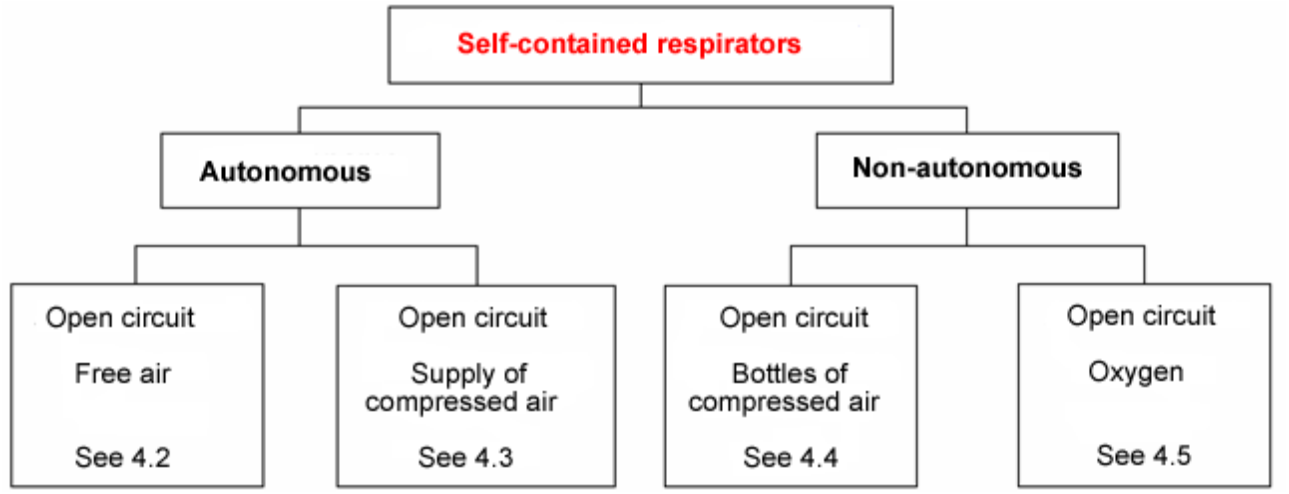


Figure 51: Self-contained respirators

A respirator is considered as "independent" if the source of air is carried with the apparatus.

A self-contained respirator is known as "open system" if the air exhaled is discharged into the surrounding atmosphere via an outlet valve and as "closed system" if the air exhaled is processed and recycled.

A self-contained respirator is known as a "fresh air" apparatus if the user is connected via tubing to a near-by area where the air is not contaminated. These apparatus may be "power-assisted" or "non-assisted.

Fresh air respirators are more adapted to static work with a local risk of pollution. They are little used by E&P.

With **compressed air line respirators** (Supplied Air Respirator), the user is connected to the source of air via a tube connected to a system of breathable air, or a reserve of compressed air.



Figure 52: Compressed air line respirator

These respirators consist of:

- a face mask (full mask, half-mask or hood). Many models exist, each adapted to a specific task (sanding, welding, painting, etc.).
- a "low pressure" respiratory tube,
- a regulator, generally attached to the belt,
- a compressed air supply tube.

According to the air regulation system, these respirators may be

- "continuous flow" if air is continually supplied. The regulator is equipped with a valve to control air flow. This valve is equipped with a safety which ensures a minimum flow of approximately 100 L/minute, even when closed.
- "at request" if they include a device restricting the flow of air to the quantity required for each inhalation.
- "at request, positive pressure" if they include a device ensuring slight overpressure in the mask during inhalation and exhalation phases.

The directives of the manufacturer must be applied with regards air flow, the diameter and the length of ducts. The entire system must be supplied by the same manufacturer.

Hoods must not be used in confined spaces or in atmospheres representing an immediate hazard for life or health (> IDLH limit). In these conditions, only the full mask with an at request, positive pressure valve, combined with an emergency reserve of compressed air (see photo), may be used.

The autonomy of the auxiliary bottle must be adequate to allow the person to leave the contaminated area should air supply stop.

3.16.5.1. Breathable air system

Some sites are equipped with a **specific system for breathable air** maintained at 4-8 bars.

A second compressor or "buffer" bottles must be able to maintain system pressure should the primary compressor fail.

It may be necessary to install various devices such as water or oil traps, or air heating/cooling systems on the supply line.

Connections to the breathable air system must be specific and incompatible with the connections to other utilities. Isolation valves must be locked open.

3.16.5.2. High capacity bottles or air frame

Units must comply with applicable regulations for pressurized respirators.

A filter skid can trap solid particles, water or oil prior to supplying the mask. The filter device is often combined with the distributor in order to supply several items of equipment simultaneously.

Units are generally equipped with an integrated lifting system.

3.16.5.3. Verifications prior to use

Check that the entire system is in good condition (hose, connections, mask, etc.),

Ensure that the hose is protected from external mechanical interference and can move freely with the user,

Ensure that the valves operate correctly,

Ensure that the air pressure is correct,

Adjust the air flow.

3.16.6. Open-system breathing respirators (ARI)

According to the air regulation system, these respirators may be

- "at request" if they include a device restricting the flow of air to the quantity required for each inhalation.
- "at request, positive pressure" if they include the same device and a slight overpressure is maintained.

ARI are generally equipped either with a 9l bottle, or 2 6l bottles of compressed air.



Figure 53: Self-contained respirator

The compressed air is expanded in 2 stages. A regulator expands the pressure in the bottle to an average pressure level (approximately 10 bars). The air will then transit via the "request valve" attached to the mask, which will adjust flow at the request of the user.

With positive pressure respirators, the valve will maintain a slight overpressure (a few millibars) in the mask.

Exhaled air will exit from the mask via 1 or 2 exhalation valves.

The respirator is equipped with a pressure gauge indicating pressure of remaining air.

ARI are equipped with a whistle, which will be armed when opening the bottle and triggered if residual pressure drops under 50 bars.

Weight of a fully equipped ARI: 15 8 kg

ARI must be stored in a dry location. Any trace of humidity on the valves could modify operation. Unused respirators must be stored separately and marked / tagged as such.

Verifications prior to use

- Check the general good condition of the mask,
- Check that the whistle is armed when the bottle is opened,
- Check the mask seals. Respirators are over-pressurized, the user simply needs to block breathing and any leak will be picked up by the micro-regulator which will provide air.
- Check the bottle pressure on the pressure gauge.

3.16.6.1. Compressed air bottles

Materials	Volume	Pressure	Empty weight
Steel	6 litres	300 bar	8 kg
	9 litres	200 bar	10.8 kg
Aluminium composite coated with carbon fibre	6 litres	300 bar	4 kg–4.5 kg
	9 litres	200 bar	4.5 kg
		300 bar	5.8 kg

Table 16: General information on compressed air bottles

3.16.6.2. Bottle autonomy

Autonomy depends on the capacity of the bottle (volume and pressure) and the air consumption of the user.

Autonomy is calculated on the basis of a simplified formula

Max. autonomy = P x V / Q

Where P is the pressure in the bottle (as shown on the pressure gauge) in bars),
Where V is the geometric volume of the bottle in litres,
Where Q is the air consumption of the user in litres per minute.

To give an example, with a 9-litre bottle, inflated to 300 bars, a user consuming 50 L/minute (moderate effort) will have a theoretical autonomy of 54 minutes. If consumption rises to 90 L/minute (sustained effort), the theoretical autonomy drops to 30 minutes.

3.16.6.3. Selection of a respiratory equipment for the job

The selection of protective respiratory equipment requires knowledge of the hazards and the level of exposure for those involved and the knowledge of how to use the equipment.

Criteria for the selection of equipment:

- Field of vision/eye protection. The field of vision of full masks is restricted. However, they provide optimum eye protection.
- Resistance to inhaling. Over-pressurized respirators are the most comfortable.
- Freedom of movement. ARI are awkward to carry, however they do not have air pipes.
- Degree of protection. Protection is improved with positive pressure respirators (especially for those having beard).
- Autonomy. The duration of use must account for the time required to enter and leave the contaminated area.
- Physical effort (increase in work load for the user). Effort will be substantially higher with an ARI.
- Perception of the environment (difficulty in communication, sensory drawbacks).
- Handling of the equipment. Avoid having different models of equipment on a site.

Air-purifying respirators or self-contained respirators?

A self-contained respirator will be selected for the job if at least one of the following conditions exists:

- Oxygen concentration is less than 19% (19.5% according to US standards),
- There is no filter cartridge suitable for the toxic gas potentially present in the atmosphere,
- The concentration of aerosols or gas represents an immediate hazard for life or health (concentration > IDLH limit),
- The performances of air-purifying respirators are not adequate to provide effective protection (concentration > 40 times the VME),
- The contaminants are not identified. →See paragraph on "Self-contained respirator".

In all other cases, select an air-purifying respirator appropriate to the job and the contaminants present in the atmosphere.

3.16.6.4. Maintenance and inspection operations

With works, the use of a respirator supplied by air bottles (trolley or rack) is strongly recommended. The attendant must be trained to be able to:

- Ensure that signs and safety rules are complied with near to the working area,
- Ensure that the air supply is in correct working order,
- Provide assistance to an actor in case of an incident,
- Call for emergency assistance.

The attendant must:

- Be located in an upwind position from the working area,
- Be able to directly see the working area,
- Be equipped with an ARI in the stand-by position,
- Have a means of radio communication.

Air cylinders have to be inspected once a year by a third party.

3.17. RISKS OF HYDROCARBON TRAPPING

The opening of equipment or lines carrying hydrocarbons always represents a risk.

One risk which often causes accidents is the presence of hydrocarbons trapped in the equipment or the line to be opened.

Even after the full evacuation of hydrocarbons via draining, repeated flushing and inerting in total compliance with all availability regulations, hydrocarbons can still be trapped:

- in high points (gas),
- in low points (liquids),
- in the substances in the equipment or line (e.g.: a pipe carrying hydrocarbons including H2S may continue to degas for a lengthy period after having been decompressed, drained, and flushed with nitrogen).

It is therefore essential to integrate this parameter in the definition of the availability (MAD) procedure and systematically analyse the system to identify traps and ensure that the means of evacuating all hydrocarbons are present.

3.18. DOUBLE BLOCK & BLEED

Some maintenance operations or short-term jobs require the availability of equipment without actually requiring the flow of fluids to be stopped, or the depressurization of a large section of the installation, or plating (e.g.: replacement of a pump impeller, replacement of a section of corroded line, etc.).

For this type of availability, it is strongly recommended, whenever possible, to ensure the installation of 2 safety barriers (2 closed and consigned valves) upstream and downstream from the equipment in order to guarantee the blocking of pressurized fluids. Rules require this "double block" plus the possibility for decompression ("bleed") between the 2 valves to be used to control any leakage from valves if the pressure upstream/downstream from the block is (judged) too excessive or if the fluid is a toxic gas whose partial pressure exceeds 1 bar g.

Example of a double block & bleed:

In order to be able to disassemble the LV5003A without need for decompression/ the drainage of large sections of piping, the 2 upstream valves and the 2 downstream valves can be closed and padlocked and the unit can be bled via the 2 3/4" valves.

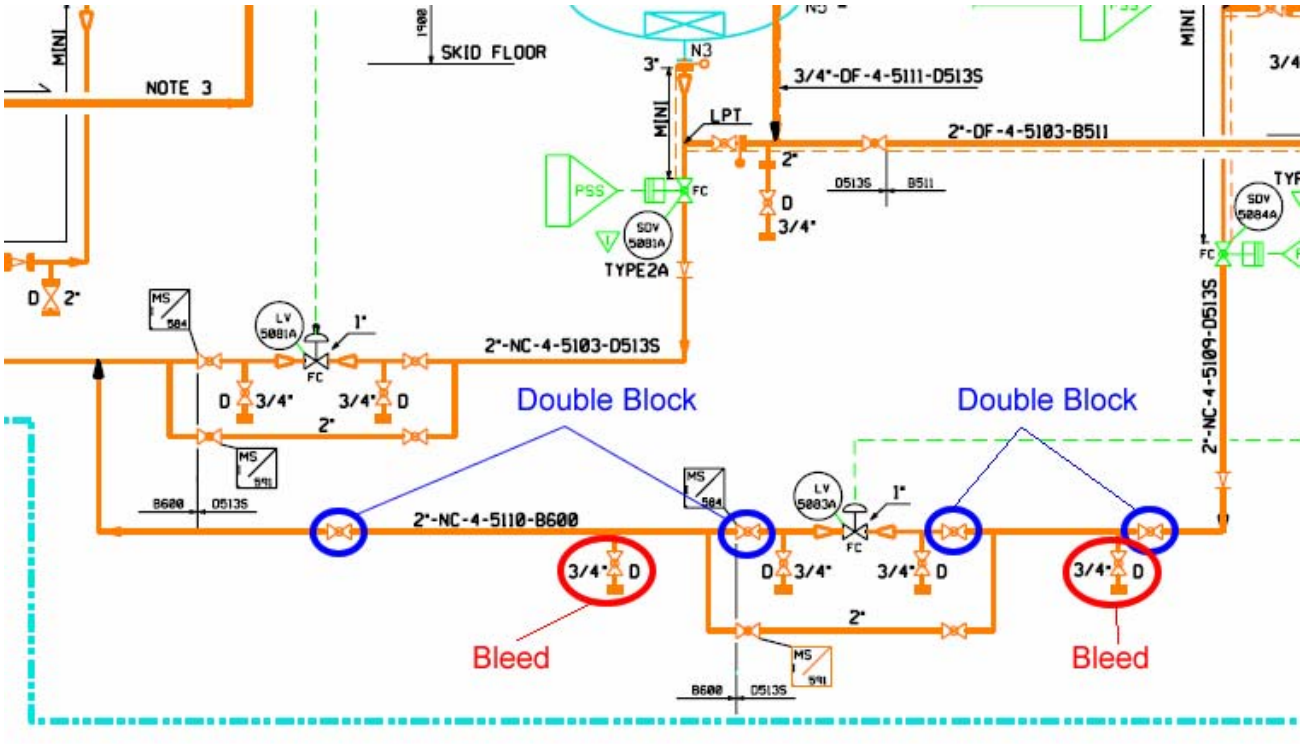


Figure 54: Example of a "Double block & bleed"

3.19. SANDING AND PAINTING

Risks	Precautions	Accidents
Hydrocarbon leak	Prior to sanding and pickling operations using needles on lines and/or capacities, with the approval of the inspection/corrosion service. Restrict the quantity of solvent and paint on platforms to the strict minimum (use a floating base or, if this is not possible, define a storage area which is limited in terms of area and duration). Evacuate empty containers having held flammable liquids.	Rupture of the sight glass on a gas compressor suction drum subsequent to surface preparation works prior to painting
Ignition	For large sites, request that heat engine units be stored on a floating base. Sanding = hot work (projection of sand on metal). Static electricity: equipotential connection between the sanding machine and the platform. If sanding/paint works are executed on a floating storage unit, only authorize wet sanding.	
Tangible non-availability of the platform.	Protection of siphoids. Protection of fixed measuring instruments on platforms. Protection of air suctions for technical rooms. Protection of rotating machinery. Obstruction of lift pump risers or fire pumps by site sand: following the completion of the site, test a fire unit for one hour to check that its suction riser is not obstructed. Do not store site equipment on/under the fall-prevention boards. Suspension of scaffolding on the platform structure and not on the fall-prevention boards.	Fall of barrels stored on the helideck nets during sanding/painting works. In addition, the scaffolding was suspended from the nets and not from the platform structures.

Risks	Precautions	Accidents
Fall	For this risk type, see "SIMOPS Interventions on existing installations".	
Rupture due to overpressure in the sanding machine	Presence and inspection of overpressure limiters in capacities (valves). Only personnel required for the job is present.	Sanding machine worker received face injuries due to the opening of the overpressure valve on the sanding machine.

Table 17: Sanding and painting risks

3.20. RULES FOR THE USE OF FLANGES

The use or replacement of flanges (e.g.: when disassembling or opening equipment or lines) must comply with the following rules:

- strict compliance with the series (pressure),
- strict compliance with dimensions.
- strict compliance with the seal which goes with the flange (material, quality, dimensions, etc)

Note: The details of series, and flange characteristics are itemized in the PIPING course.

3.21. USE OF HOSES

The use or replacement of hoses during operations requires authorization. Site regulations define the operations requiring the use of hoses and the scope for the use of the former.

The use or replacement of hoses must absolutely comply with the following:

- strict compliance with the service pressure of the hose,
- strict compliance with the connection dimensions for the hose,
- strict compliance with the fluid types authorized for the hose,
- strict compliance with the maximum dates for use of the hose,
- strict compliance with storage rules for hoses.

Careful attention must be paid to the use of hoses carrying fluids between the supply ships and offshore sites. This use must be subject to a procedure.

Even a slightly damaged hose must be cut, disposed of and replaced with another of the same type.

3.22. LIFEBOAT, ENTRY RULES

The entry of personnel into lifeboats for maintenance operations is subject to a strict site procedure. Accidents have happened due to non-compliance with basic safety instructions (e.g.: accidental launching of a lifeboat during a preventive maintenance operation with personnel on-board).

Entry into a lifeboat for maintenance/inspection operations will only be authorized if the lashings are attached and secured.

The lashings will be detached following the completion of the operation and when all personnel has left the lifeboat.

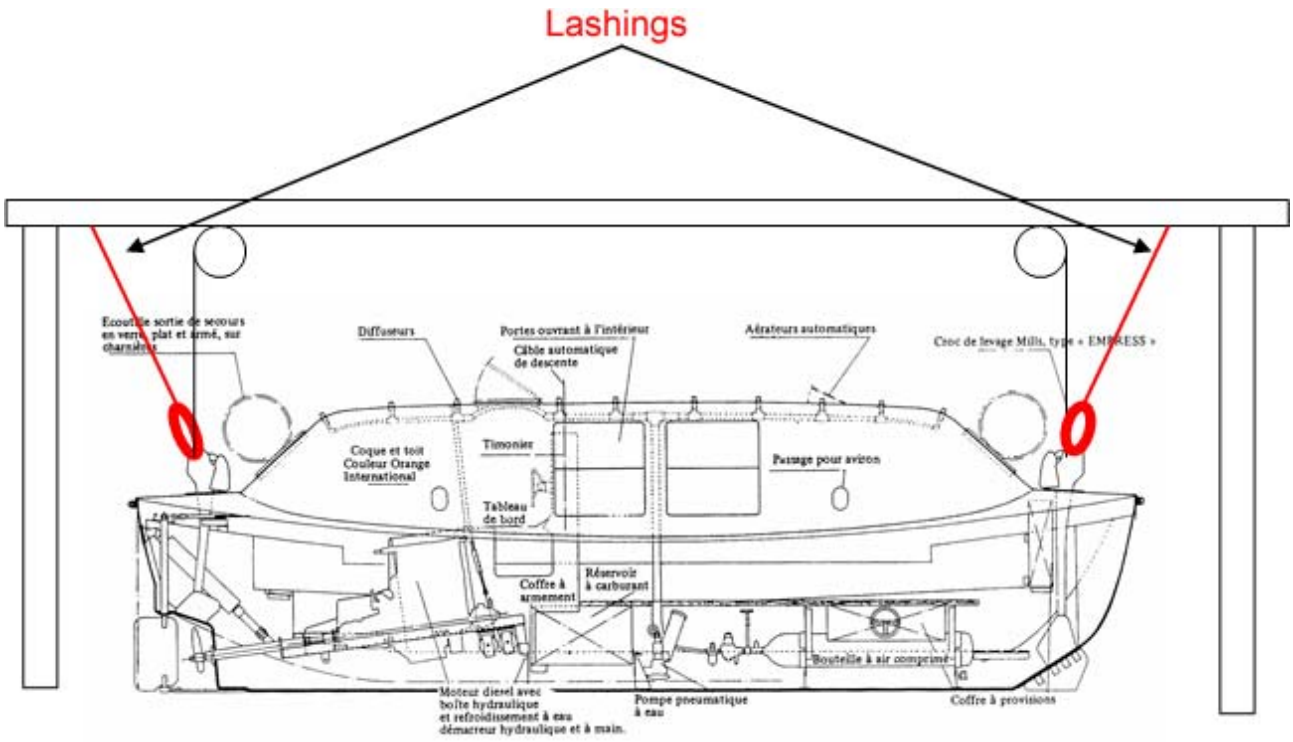


Figure 55: Descriptive schematic of a watercraft secured by lashings

3.23. RISKS OF USING INAPPROPRIATE OR RE-USED JOINTS

Independently to the types of joints used, they must be replaced with new joints with identical properties and the "old" joints will be cut in two to ensure that re-use is not possible. This is to avoid creating a hazard situation (leakage) which could be catastrophic.

Each type of joint has its own properties and its field of use, and cannot therefore be replaced by another type.

Specific attention will be paid when replacing RTJ/RX/BX type ring joints, their profiles are similar and it is easy to make a mistake if not careful. This is particularly the case for RTJ and RX ring joints. RX joints are often used for assembling wellheads; therefore it is essential not to mix the joints up when procuring a workover.

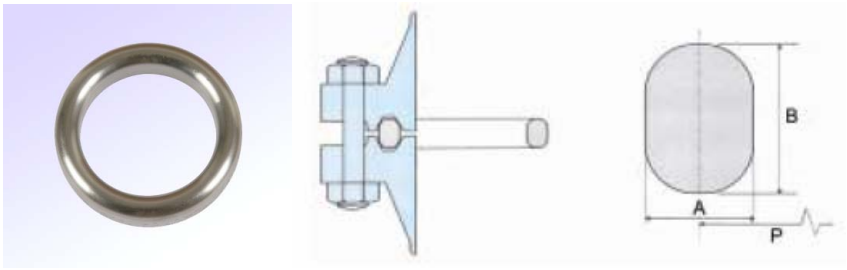


Figure 56 : Seal type RTJ Oval Stainless Steel

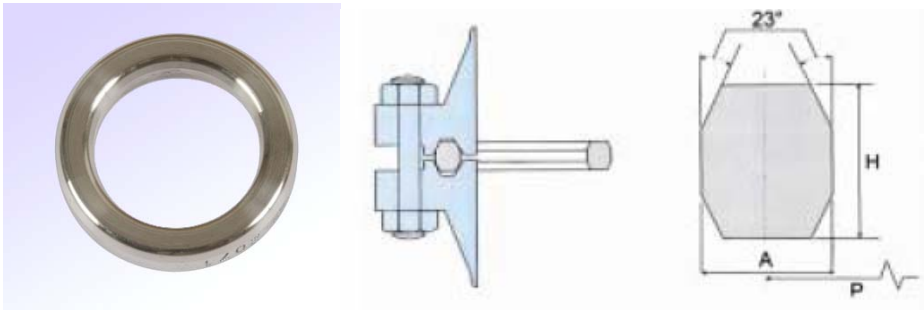


Figure 57 : Seal type RTJ octagonal Stainless Steel

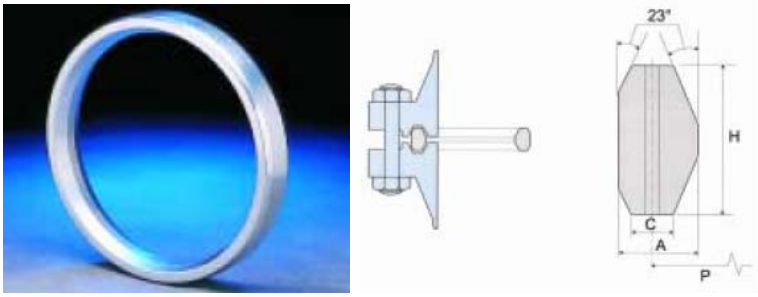


Figure 58 : Seal type RX Stainless Steel

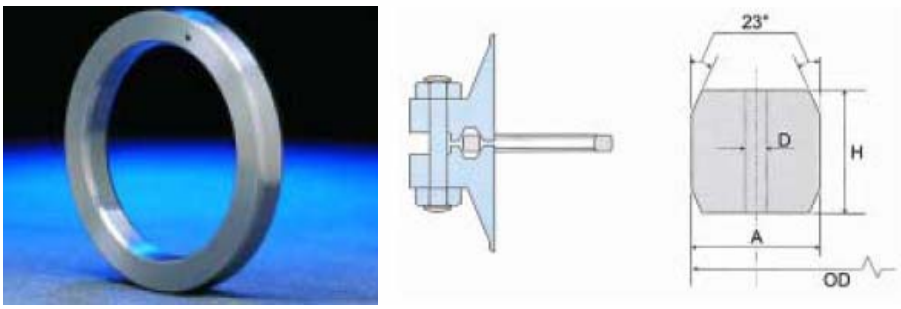


Figure 59 : Seal type BX Stainless Steel

Specific attention will also be paid when replacing GRAYLOCK and TECLOCK type joints, their profiles are similar and it is easy to make a mistake if not careful.

These joints are often used for assembling flowlines; therefore it is essential to not mix the joints up when procuring a flowline replacement/installation.

3.24. REROUTING/ TEMPORARY LINE INSTALLATION

3.24.1. Modification of installations

Any modification to existing installations is subject to the modifications management procedure, which is specifically based on the following principles:

- The need for the modification is assessed via a specific study. If necessary, the regulatory process (HSE study to be carried out, Work Permit to be obtained, etc.) is applied, all modifications are subject to a complete risk assessment process including the identification and analysis of all potential impact during or after the implementation of modifications. In both cases, it must be ensured that risks are at an acceptable level.

- The study to be carried out is entrusted to a designated individual from the main profession concerned. All professions concerned will be consulted for each modification,

- The updating of related documents (e.g.: HSE folder for the installation, pertinent technical documents, etc.) is systematically considered as part of the modification process. Emergency and priority modifications, particularly modifications enabling the resolution of HSE problems and the improvement of the HSE situation for installations, may be processed rapidly, providing the approval process is fully applied, specific meetings with all members of personnel concerned are organized to speed up the process, and no part of the modification process is neglected.

3.24.2. Temporary installations

Construction works in temporary installations, and any modification to these installations, are subject to the same level of preparation and surveillance of installations. The duration of use of temporary installations is closely monitored. A maximum date for use will be systematically defined and complied with. In addition, temporary installations are used for as short a period as possible and are disassembled/taken out of service as rapidly as possible. They must be declared as compensatory measures for downgraded situations and managed as such.

3.25. OPEN DRAIN/ CLOSED DRAIN INTERCONNECTION

Drainage systems carry a large number of different effluents. Improper segregation of drainage systems is one of the main causes of accidents in the oil and gas installations. Segregation shall therefore be incorporated as early as possible in the design of drainage systems.

Closed drains shall always be segregated from open drains so as to prevent pressure-driven gas from the closed drains to come back up the plant via the open drains system.

The piping systems collecting closed drains and open drains shall be independent (no connections at all, even for maintenance purposes).

The closed drain drum shall receive no effluents coming directly from the open drains. And the closed drain drum shall not discharge into an atmospheric enclosure (tank, drum) receiving open drain effluents.

3.25.1. Definitions

Eh	D1 - Oily effluents	They include liquid hydrocarbons or waters which may be contaminated by oil products or derivatives.
Cp	D2 - Main collectors	They carry all of the effluents drained by the secondary collectors on the different decks to the caisson sump, the drain tank or the water processing unit, if any.
Cs	D3 - Secondary collectors	Piping receiving several connections from installations or devices located on the different decks.
Rt	D4 – Connections	Narrow piping connecting specific collection points such as the drum bleeding, pumps, levels, etc.
Ch	D5 - Hydraulic guards	Liquid height blocking the gas transiting from downstream to upstream without preventing the evacuation of liquid. This hydraulic guard will prevent the propagation of an explosion or fire in the sewerage.
Vp	D6 - Primary ventilation	The vertical extension of the main fall collector, including the free air outlet, equipped with a flame-stopper, must appear: - A minimum of 3 metres above the piping racks - At a minimum of 10 metres (horizontal projection) from any potential source of bare flame and the suction of ventilators or other machines aspiring air (compressor, blowers, booster, etc.) - At a minimum of 5 metres from platforms and circulation areas (ladder, stairs, footbridge, etc.) - At a vent placed against the wall of a building overhanging the upper edge by at least 1 metre.

Et	D7 - Vents	Vent tubing will be installed according to the flow layout of STD TUY.B01.506.P drawing. If installation conditions so allow, vent tubing may be connected to the primary ventilation collector, in order to avoid multiplying final flame-stoppers and therefore multiplying hazardous areas. This tubing may be large enough to evacuate maximum gas flow for the system section, or the devices protected. The tubing run must ensure that no low point exists and that the outlet of the vent equipped with a flame-stopper (should this not be the same as the Vp) is installed according to the same conditions as the final primary ventilation.
Rs	D8 - Siphoid sights	Small cell equipped with a siphon, placed at the entry to a drain pipe, in a specific deck or in the pans collecting drips or run-off water. The hydraulic guard for the siphon prevents gas from travelling from the tubing to the atmosphere.

Table 18: Definitions Closed Drain – Open Drain

3.25.2. General

With a traditional platform, the main sources of discharge considered as drainage are:

- Rain and washing water
- Water draw-off from separators
- Manual bleeding of capacities (oil or water)
- Pressurized unit bleeding.

Depending on the fluids to be collected, two collection systems will be installed on the platform for the fluids to be eliminated.

Open system: to the caisson sump

This system collects unpressurized fluids with no or little hydrocarbon content.

Closed system: to the drain tank

This system collects fluids with hydrocarbon content, generally unpressurized, and including little or no gas.

NB: Water outlets for processing units (desalter, decanters, coalescing separators) generally go directly to the sea via a discharge tube, possibly equipped with a dissolver. But only if those waters are “clean”, water outlet of a separator is treated before being disposed off.

3.25.3. Design of the open system

Effluents collected by the open system are carried to the caisson sump which will eliminate all traces of hydrocarbons contained in the effluents prior to discharge at sea.

3.25.3.1. The caisson sump

The caisson sump is an integral part of the structure and must have a maximum diameter which is compatible with construction limitations. Depth depends on the local climate (waves, tides) and may vary between 10 and 30 metres as compared with sea level. The caisson sump has two inlets.
The lower inlet (approximately 6m for installations in the Gulf of Guinea) receives the unpressurized discharge waters containing little or no hydrocarbons.

3.25.3.2. Rain and washing water

Rain and washing water is collected on the decks or in collection pans under the capacities. The diameters of the connections to collector pans are determined on the basis of site rain levels and must not be less than 3" (plan for short lengths between flanges in view of disassembly) while collectors generally have a diameter of 4". Each pan is equipped with a bucket-type siphon for which the height of the water holder must be sufficient to avoid the degassing of the caisson sump. In addition, an adequate slope must exist between the pans and the collector. Siphons must be able to be effectively blocked to prevent return flow, particularly in case of works or interventions.

3.25.3.3. Other discharge

All other sources of drips or leaks generally collected in the pans or funnels (pump glands, product tanks), and generally speaking, all systematically unpressurized drainage using solely gravity-based flow, will be connected to this system.

Processed water discharge will occasionally be connected if flow is high.

The upper inlet (approximately - 4m for installations in the Gulf of Guinea) receives water containing low quantities of hydrocarbons, i.e. mainly manual bleeding and the overflow for the drain tank.

3.25.3.4. Caisson sump equipment

The caisson sump is a vertical decanter constructed with the equipment described below.

Unpressurized water runs through this unit and the line must be equipped with a standpipe to avoid the transmission of pulsations due to wave movements to the siphons.

The upper inlet is for pressurized water: the conduct is simply sloped downwards.

3.25.3.5. Degassing

A flame-stopper will be positioned on the caisson sump vent.

3.25.3.6. Recovery of hydrocarbons

The suction of the hydrocarbon recovery pump is immerged approximately 1m below sea level or the lowest possible level.

The level can generally be visually controlled via the quality of the effluent pumped, but also by controlling the interface level if a guard is installed high enough to ensure that measurements are not affected by sea movements.

Drained hydrocarbons can be recovered either by a gas pump or by an electric pump.

Note: It is beneficial to be able to adjust the height of oil recovery in the caisson sump in order to avoid recycling sea water.

3.25.4. Design of the closed system

The closed system carries liquid hydrocarbons (oils), generally unpressurized and therefore without a significant quantity of gas. These are collected at the drain tank.

3.25.4.1. Drain tank

The drain tank includes a horizontal tank which may include two compartments placed at low points to allow for gravity-based flow.
Its capacity is at least equal to the liquid volume at the lowest point of the largest capacity on the platform, and the supply tube is sloped downwards (arriving at a low point).

The tank is equipped with a level regulator which controls a recovery pump for drips:

- From the oil outflow collector, i.e. upstream from the final stage of the three-phase separation of the production chain (this will generally be the atmospheric separator). The tank is equipped with a high level alarm (LAH), and a low level safety (LSH) which stop/start production.

- A flame-stopper located on the capacity will be installed on the vent conduct not connected to the flare, but to a vent line leading to a non-hazardous area, e.g. a

flare-boom beam located a minimum of 10-15 metres from the edge of the platform or to the right of the support tripod for footbridge flares.

- The manual bleeding system for the drain tank and the overflow will be connected to the caisson sump via a common pipe.

3.25.4.2. Effluents collected

As a general rule, oil effluents containing little gas and coming from previously decompressed lines or capacities are collected at the drain tank, e.g.:

- Manual bleeding of separators (previously decompressed at the flare)
- Line bleeding
- Scraper guards (except HP gas circuit)
- The flare scrubber bleeding (for low flow).

Note 1: Sight glasses are collected at the drain tank due to the low quantities of liquid and the need for cleaning.

Note 2: With platforms with no flare or flare scrubber, some decompression lines (e.g.: choke holders and manifold well lines) may be collected in the drain tank. In this case, the position of the vent must be carefully considered.

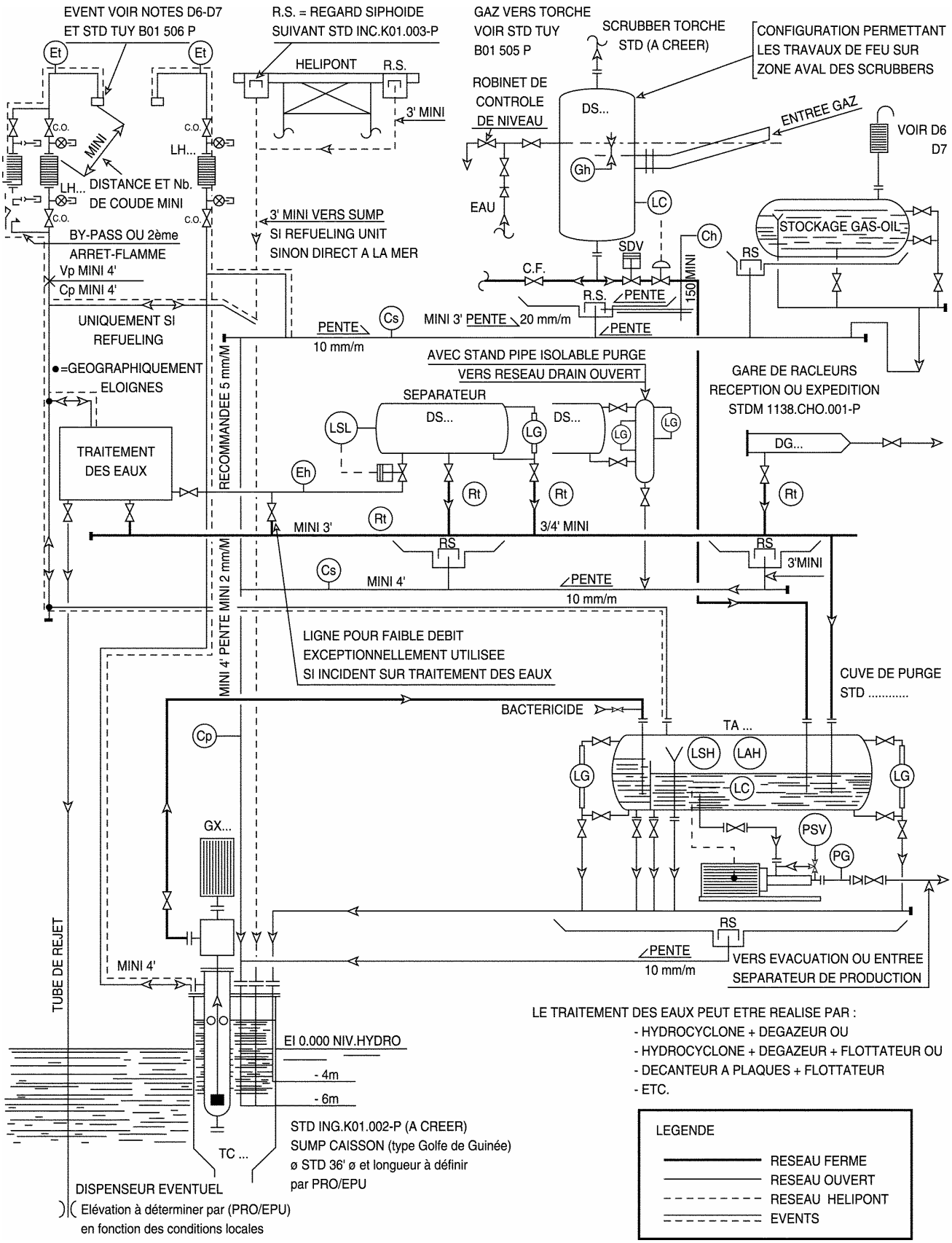


Figure 60: Open drains and closed drains

3.26. USE OF TRANSPORT VEHICLES

Speeding is not acceptable, even for service reasons. The following precautions apply to all vehicles:

- drivers will comply with all regulations, all rules and the applicable driving policy,
- speed limits will be strictly met,
- vehicles will be inspected, authorized and approved for the intended use,
- all individuals driving a company vehicle will have received training in preventive driving, and a refresher course if required,
- one seat will be assigned per passenger, and the safety belt will remain fastened for the entire duration of the journey,
- passengers and goods will not be carried in the same compartment,
- mobile telephones or walkie-talkies may not be used while driving,

Specific precautions will be taken to account for hazard situations or hazardous driving conditions.

4. GLOSSARY

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